



## High-performance 6-axis IMU (inertial measurement unit) with sensor fusion, ASC (adaptive self-configuration), EIS (electronic image stabilization)



**LGA-14L**  
(2.5 x 3.0 x 0.83 mm) typ.

| Product status |
|----------------|
| Custom data    |

| Product summary        |                                  |
|------------------------|----------------------------------|
| Order code             | LSM6DSVETR                       |
| Temperature range [°C] | -40 to +85                       |
| Package                | LGA-14L<br>(2.5 x 3.0 x 0.83 mm) |
| Packing                | Tape and reel                    |

| Product resources                             |
|-----------------------------------------------|
| <a href="#">TN0018</a> (design and soldering) |

### Features

- Dual-channel architecture for UI and EIS data processing
- "Always-on" experience with low power consumption for both accelerometer and gyroscope
- Smart FIFO up to 4.5 KB
- Android compliant
- $\pm 2/\pm 4/\pm 8/\pm 16$  g full scale
- $\pm 125/\pm 250/\pm 500/\pm 1000/\pm 2000/\pm 4000$  dps full scale
- SPI / I<sup>2</sup>C & MIPI I3C<sup>®</sup> v1.1 serial interface with main processor data synchronization
- EIS dedicated channel on primary interface with dedicated filtering
- Advanced pedometer, step detector, and step counter
- Significant motion detection, tilt detection
- Standard interrupts: free-fall, wake-up, 6D/4D orientation, click and double click
- Programmable finite state machine for accelerometer, gyroscope, and external sensor data processing with high rate @ 960 Hz
- Embedded adaptive self-configuration (ASC)
- Embedded sensor fusion low-power algorithm
- Embedded temperature sensor
- Independent I/O supply
  - I<sup>2</sup>C voltage range: 1.62 V to 3.6 V
  - SPI / MIPI I3C<sup>®</sup> extended voltage range: 1.08 V to 3.6 V
- Power consumption: 0.65 mA in combo high-performance mode
- Compact footprint: 2.5 mm x 3 mm x 0.83 mm
- ECOPACK and RoHS compliant

### Applications

- Motion tracking and gesture detection
- [Wearables](#)
- Indoor navigation
- [IoT and connected devices](#)
- Smartphones and handheld devices
- EIS for camera applications
- Vibration monitoring and compensation

### Description

The LSM6DSVE is a high-end, low-noise, low-power 6-axis small IMU, featuring a 3-axis digital accelerometer and a 3-axis digital gyroscope, with a dual-channel architecture for processing acceleration and angular rate data on two separate channels (user interface and EIS) with dedicated configuration, processing, and filtering.



The device enables processes in edge computing, leveraging embedded advanced dedicated features such as a sensor fusion low-power (SFLP) algorithm and a finite state machine (FSM) for configurable motion tracking.

The LSM6DSVE supports the adaptive self-configuration (ASC) feature, which allows the FSM to automatically reconfigure the device in real time based on the detection of a specific motion pattern, without any intervention from the host processor.

ST confidential  
DRAFT



## 1 Overview

The LSM6DSVE is a system-in-package featuring a high-performance 3-axis digital accelerometer and 3-axis digital gyroscope.

The LSM6DSVE delivers best-in-class motion sensing that can detect orientation and gestures in order to empower application developers and consumers with features and capabilities that are more sophisticated than simply orienting their devices to portrait and landscape mode.

The event-detection interrupts enable efficient and reliable motion tracking and context awareness, implementing hardware recognition of free-fall events, 6D orientation, click and double-click sensing, activity or inactivity, stationary/motion detection and wake-up events. Finite state machine processing allows moving some algorithms from the application processor to the LSM6DSVE sensor, enabling consistent reduction of power consumption.

The LSM6DSVE supports the main OS requirements, offering real, virtual, and batch mode sensors. In addition, the LSM6DSVE can efficiently run the sensor-related features specified in Android, saving power and enabling faster reaction time. In particular, the LSM6DSVE has been designed to implement hardware features such as significant motion detection, stationary/motion detection, tilt, pedometer functions, timestamping and to support the data acquisition of external sensors.

The LSM6DSVE offers hardware flexibility to connect the pins with different mode connections to external sensors to expand functionalities such as adding a sensor hub.

The LSM6DSVE offers advanced design flexibility for EIS applications. Both channels have a dedicated processing path with independent filtering and enhanced EIS channel gyroscope data are read over the primary interfaces I<sup>2</sup>C / MIPI I3C<sup>®</sup> v1.1 / SPI.

Channel 1 has been designed for user-interface data processing for motion tracking. Data are available on the primary output of I<sup>2</sup>C / SPI / I3C<sup>®</sup> for the accelerometer and gyroscope with independent ODR and FS.

Channel 2 has been design for enhanced EIS. Data are available in freerun mode in the output registers or in FIFO with dedicated tag and timestamp.

Up to 4.5 KB of FIFO with compression and dynamic allocation of significant data (that is, external sensors, timestamp, and so forth) allows overall power saving of the system.

The LSM6DSVE embeds a sensor fusion low-power (SFLP) algorithm able to provide a 6-axis (accelerometer + gyroscope) game rotation vector represented as a quaternion. The X, Y, Z quaternion components are stored in FIFO.

Like the entire portfolio of MEMS sensor modules, the LSM6DSVE leverages the robust and mature in-house manufacturing processes already used for the production of micromachined accelerometers and gyroscopes. The various sensing elements are manufactured using specialized micromachining processes, while the IC interfaces are developed using CMOS technology that allows the design of a dedicated circuit, which is trimmed to better match the characteristics of the sensing element.

The LSM6DSVE embeds advanced dedicated features like a finite state machine and data filtering for EIS, and motion processing.

The LSM6DSVE is available in a small plastic, land grid array (LGA) package of 2.5 x 3.0 x 0.83 mm to address ultracompact solutions.



## 2 Embedded low-power features

The LSM6DSVE has been designed to be fully compliant with Android, featuring the following on-chip functions:

- 4.5 KB FIFO data buffering, data can be compressed two or three times
  - 100% efficiency with flexible configurations and partitioning
  - Possibility to store timestamp
- Event-detection interrupts (fully configurable)
  - Free-fall
  - Wake-up
  - 6D orientation
  - Click and double-click sensing
  - Activity/inactivity recognition
  - Stationary/motion detection
- Specific IP blocks (called "embedded functions") with negligible power consumption and high performance
  - Pedometer functions: step detector and step counters
  - Tilt
  - Significant motion detection
  - Finite state machine (FSM)
  - Adaptive self-configuration (ASC)
  - Embedded sensor fusion low-power (SFLP) algorithm
- Sensor hub
  - Up to 6 total sensors: 2 internal (accelerometer and gyroscope) and 4 external sensors

### 2.1 Pedometer functions: step detector and step counters

The LSM6DSVE embeds an advanced pedometer with an algorithm running in an ultralow-power domain in order to ensure extensive battery life in battery-constrained applications.

Leveraging enhanced configurability, the advanced embedded pedometer is suitable for a large range of applications from mobile to wearable devices.

The algorithm processes and analyzes the accelerometer waveform in order to count the user's steps during walking and running activities.

The pedometer works at 30 Hz and it is not affected by the selected device power mode (ultralow-power, low-power, high-performance), thus guaranteeing an ultralow-power experience and extreme flexibility in conjunction with other device functionalities.

The pedometer output can be batched in the device's FIFO buffer, in order to decrease overall system power consumption.

ST freely provides the support and the tools for easily configuring the device and tuning the algorithm configuration for a best-in-class user experience.

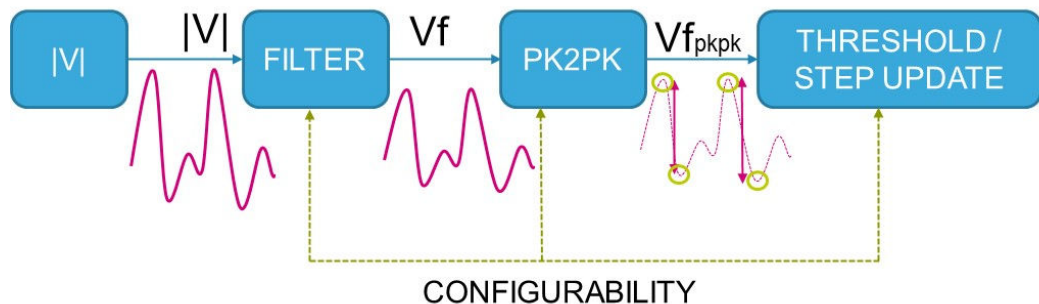


## 2.2 Pedometer algorithm

The pedometer algorithm is composed of a cascade of four stages:

1. Computation of the acceleration magnitude signal in order to detect the signal independently from device orientation
2. FIR filter to extract relevant frequency components and to smooth the signal by cutting off high frequencies
3. Peak detector to find the maximum and minimum of the waveform and compute the peak-to-peak value
4. Step count: if the peak-to-peak value is greater than the settled threshold, a step is counted

Figure 1. Four-stage pedometer algorithm



The LSM6DSVE embeds a dynamic internal threshold for step detection that is updated after each peak-to-peak evaluation: the internal threshold is increased with a configurable speed if a step is detected or decreased with a configurable speed if a step is not detected.

This approach ensures high accuracy when the user starts to walk and a false peak rejection when the user is walking or running.

An internal configurable debounce algorithm can be also set to filter false walks: indeed, an accelerometer pattern is recognized as a walk or run only if a minimum number of steps are counted.

The LSM6DSVE has been designed to reject a false-positive signal inside the algorithm core.

On top of the mechanisms detailed above, the LSM6DSVE allows enabling a dedicated false-positive rejection block to further boost pedometer accuracy.

## 2.3 Tilt detection

The tilt function helps to detect activity change and has been implemented in hardware using only the accelerometer to achieve targets of both ultralow power consumption and robustness during the short duration of dynamic accelerations.

The tilt function is based on a trigger of an event each time the device's tilt changes and can be used with different scenarios, for example:

- Triggers when a phone is in a front pants pocket and the user goes from sitting to standing or standing to sitting
- Does not trigger when a phone is in a front pants pocket and the user is walking, running, or going upstairs

## 2.4 Significant motion detection

The significant motion detection (SMD) function generates an interrupt when a 'significant motion', that could be due to a change in user location, is detected. In the LSM6DSVE device this function has been implemented in hardware using only the accelerometer.

SMD functionality can be used in location-based applications in order to receive a notification indicating when the user is changing location.



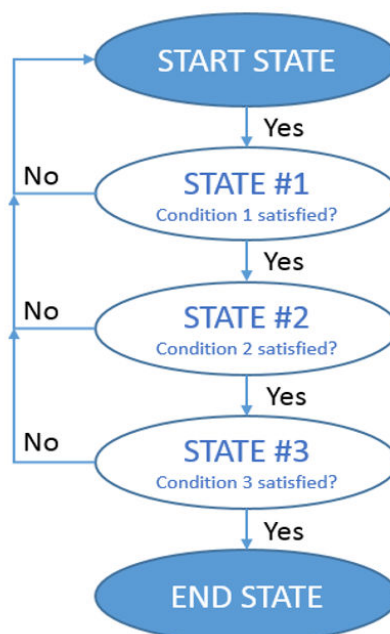
## 2.5 Finite state machine

The LSM6DSVE can be configured to generate interrupt signals activated by user-defined motion patterns. To do this, up to 8 embedded finite state machines can be programmed independently for motion detection such as glance gestures, absolute wrist tilt, shake and double-shake detection.

### Definition of finite state machine

A state machine is a mathematical abstraction used to design logic connections. It is a behavioral model composed of a finite number of states and transitions between states, similar to a flow chart in which one can inspect the way logic runs when certain conditions are met. The state machine begins with a start state, goes to different states through transitions dependent on the inputs, and can finally end in a specific state (called stop state). The current state is determined by the past states of the system. The following figure shows a generic state machine.

Figure 2. Generic state machine



### Finite state machine in the LSM6DSVE

The LSM6DSVE works as a combo accelerometer-gyroscope sensor, generating acceleration and angular rate output data. It is also possible to connect an external sensor like a magnetometer or pressure sensor by using the sensor hub feature (mode 2). These data can be used as input of up to 8 programs in the embedded finite state machine (Figure 3. State machine in the LSM6DSVE).

All 8 finite state machines are independent: each one has its dedicated memory area and it is independently executed. An interrupt is generated when the end state is reached or when some specific command is performed.

Figure 3. State machine in the LSM6DSVE





## 2.6 Adaptive self-configuration (ASC)

The LSM6DSVE supports the adaptive self-configuration (ASC) feature, which allows the FSM to automatically reconfigure the device in real time based on the detection of a specific motion pattern, without any intervention from the host processor. The FSM can write a subset of the device registers using the SETR command, which allows indicating the register address and the new value to be written in such a register. The access to these device registers is mutually exclusive to the host.

## 2.7 Sensor fusion low power

A sensor fusion low-power (SFLP) block is available in the LSM6DSVE for generating the following data based on the accelerometer and gyroscope data processing:

- Game rotation vector, which provides a quaternion representing the attitude of the device
- Gravity vector, which provides a three-dimensional vector representing the direction of gravity
- Gyroscope bias, which provides a three-dimensional vector representing the gyroscope bias

The SFLP block is enabled by setting the SFLP\_GAME\_EN bit to 1 of the EMB\_FUNC\_EN\_A (04h) embedded functions register.

The SFLP block can be reinitialized by setting the SFLP\_GAME\_INIT bit to 1 of the EMB\_FUNC\_INIT\_A (66h) embedded functions register.

**Table 1. Sensor fusion performance**

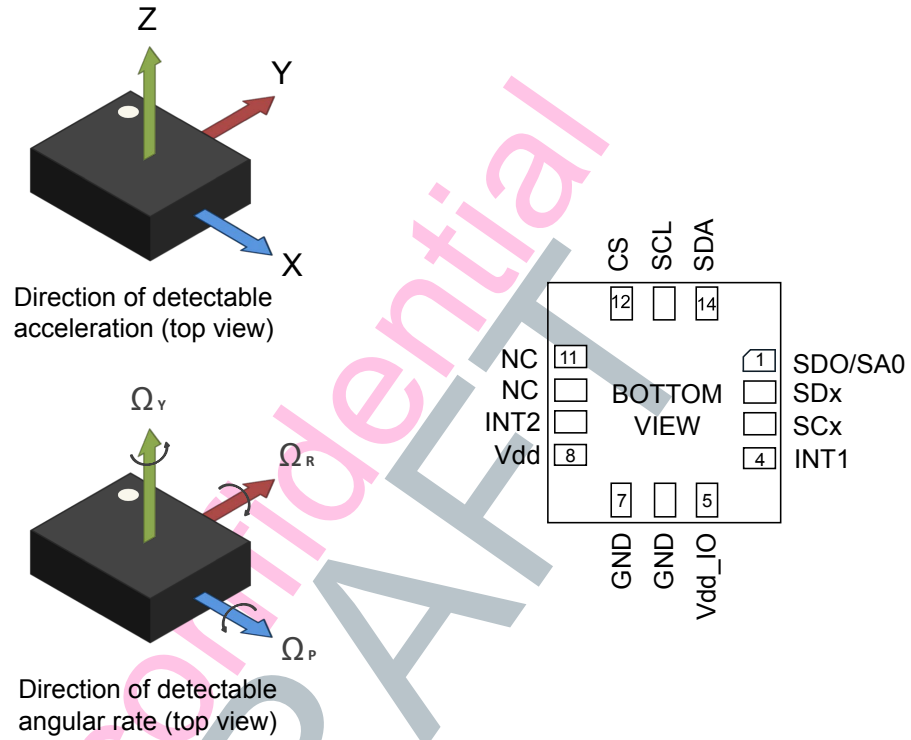
| Parameter                      |               | Value                      |
|--------------------------------|---------------|----------------------------|
| Static accuracy                | heading / yaw | 0.5 deg. / 5 minutes       |
|                                | pitch         | 1.5 deg.                   |
|                                | roll          | 1.5 deg.                   |
| Low dynamic accuracy           | heading / yaw | 0.7 deg. / 5 minutes       |
|                                | pitch         | 0.5 deg.                   |
|                                | roll          | 0.5 deg.                   |
| High dynamic accuracy          | heading / yaw | 5.9 deg. / 5 minutes       |
|                                | pitch         | 1.6 deg.                   |
|                                | roll          | 1.2 deg.                   |
| Calibration time               |               | 0.8 seconds <sup>(1)</sup> |
| Orientation stabilization time |               | 0.7 seconds                |

1. Time required to reach steady state



### 3 Pin description

Figure 4. Pin connections





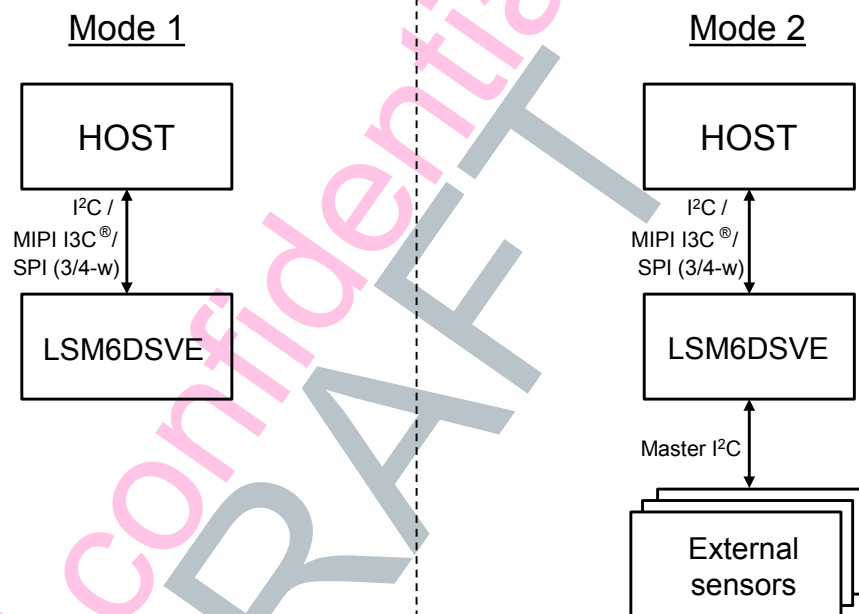


### 3.1 Pin connections

The LSM6DSVE offers flexibility to connect the pins in order to have two different mode connections and functionalities. In detail:

- **Mode 1:** I<sup>2</sup>C / MIPI I3C<sup>®</sup> slave interface or SPI (3- and 4-wire) serial interface is available.
- **Mode 2:** I<sup>2</sup>C / MIPI I3C<sup>®</sup> slave interface or SPI (3- and 4-wire) serial interface and master I<sup>2</sup>C interface for external sensor connections are available.

Figure 5. LSM6DSVE connection modes





In the following table, each mode is described for the pin connections and function.

**Table 2. Pin description**

| Pin# | Name                  | Mode 1 function                                                                                                                                                                                                                        | Mode 2 function                                                                                                                                                                                                                        |
|------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | SDO/SA0               | SPI 4-wire interface serial data output (SDO)<br>I <sup>2</sup> C least significant bit of the device address (SA0)                                                                                                                    | SPI 4-wire interface serial data output (SDO)<br>I <sup>2</sup> C least significant bit of the device address (SA0)                                                                                                                    |
| 2    | SDx                   | Connect to Vdd_IO or GND                                                                                                                                                                                                               | I <sup>2</sup> C master serial data (MSDA)                                                                                                                                                                                             |
| 3    | SCx                   | Connect to Vdd_IO or GND                                                                                                                                                                                                               | I <sup>2</sup> C master serial clock (MSCL)                                                                                                                                                                                            |
| 4    | INT1                  | Programmable interrupt in I <sup>2</sup> C and SPI                                                                                                                                                                                     |                                                                                                                                                                                                                                        |
| 5    | Vdd_IO <sup>(1)</sup> | Power supply for I/O pins                                                                                                                                                                                                              |                                                                                                                                                                                                                                        |
| 6    | GND                   | 0 V supply                                                                                                                                                                                                                             |                                                                                                                                                                                                                                        |
| 7    | GND                   | 0 V supply                                                                                                                                                                                                                             |                                                                                                                                                                                                                                        |
| 8    | Vdd <sup>(1)</sup>    | Power supply                                                                                                                                                                                                                           |                                                                                                                                                                                                                                        |
| 9    | INT2                  | Programmable interrupt 2 (INT2) / Data enabled (DEN)                                                                                                                                                                                   | Programmable interrupt 2 (INT2) / Data enabled (DEN) /<br>I <sup>2</sup> C master external synchronization signal (MDRDY)                                                                                                              |
| 10   | NC                    | Connect to Vdd_IO or leave unconnected <sup>(2)</sup>                                                                                                                                                                                  | Connect to Vdd_IO or leave unconnected <sup>(2)</sup>                                                                                                                                                                                  |
| 11   | NC                    | Connect to Vdd_IO or leave unconnected <sup>(2)</sup>                                                                                                                                                                                  | Connect to Vdd_IO or leave unconnected <sup>(2)</sup>                                                                                                                                                                                  |
| 12   | CS                    | I <sup>2</sup> C / MIPI I3C <sup>®</sup> / SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C / MIPI I3C <sup>®</sup> communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C / MIPI I3C <sup>®</sup> disabled) | I <sup>2</sup> C / MIPI I3C <sup>®</sup> / SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C / MIPI I3C <sup>®</sup> communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C / MIPI I3C <sup>®</sup> disabled) |
| 13   | SCL                   | I <sup>2</sup> C / MIPI I3C <sup>®</sup> serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                                                                             | I <sup>2</sup> C / MIPI I3C <sup>®</sup> serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                                                                             |
| 14   | SDA                   | I <sup>2</sup> C / MIPI I3C <sup>®</sup> serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                                                                                 | I <sup>2</sup> C / MIPI I3C <sup>®</sup> serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                                                                                 |

1. Recommended 100 nF filter capacitor.

2. Leave pin electrically unconnected and soldered to PCB.



## 4 Module specifications

### 4.1 Mechanical characteristics

@Vdd = 1.8 V, T = 25°C, unless otherwise noted.

**Table 3. Mechanical characteristics**

| Symbol   | Parameter                                                                  | Test conditions     | Min. | Typ. <sup>(1)</sup> | Max. | Unit     |
|----------|----------------------------------------------------------------------------|---------------------|------|---------------------|------|----------|
| LA_FS    | Linear acceleration measurement range                                      |                     |      | ±2                  |      | g        |
|          |                                                                            |                     |      | ±4                  |      |          |
|          |                                                                            |                     |      | ±8                  |      |          |
|          |                                                                            |                     |      | ±16                 |      |          |
| G_FS     | Angular rate measurement range                                             |                     |      | ±125                |      | dps      |
|          |                                                                            |                     |      | ±250                |      |          |
|          |                                                                            |                     |      | ±500                |      |          |
|          |                                                                            |                     |      | ±1000               |      |          |
|          |                                                                            |                     |      | ±2000               |      |          |
| LA_So    | Linear acceleration sensitivity                                            | FS = ±2 g           |      | 0.061               |      | mg/LSB   |
|          |                                                                            | FS = ±4 g           |      | 0.122               |      |          |
|          |                                                                            | FS = ±8 g           |      | 0.244               |      |          |
|          |                                                                            | FS = ±16 g          |      | 0.488               |      |          |
| G_So     | Angular rate sensitivity <sup>(2)</sup>                                    | FS = ±125 dps       |      | 4.375               |      | mdps/LSB |
|          |                                                                            | FS = ±250 dps       |      | 8.75                |      |          |
|          |                                                                            | FS = ±500 dps       |      | 17.50               |      |          |
|          |                                                                            | FS = ±1000 dps      |      | 35                  |      |          |
|          |                                                                            | FS = ±2000 dps      |      | 70                  |      |          |
|          |                                                                            | FS = ±4000 dps      |      | 140                 |      |          |
| G_So%    | Sensitivity tolerance <sup>(2)</sup>                                       | at component level  |      | ±1                  |      | %        |
| LA_SoDr  | Linear acceleration sensitivity change vs. temperature <sup>(3)</sup>      | from -40°C to +85°C |      | ±0.01               |      | %/°C     |
| G_SoDr   | Angular rate sensitivity change vs. temperature <sup>(3)</sup>             | from -40°C to +85°C |      | ±0.007              |      | %/°C     |
| LA_TyOff | Linear acceleration zero-g level offset accuracy <sup>(4)</sup>            |                     |      | ±20                 |      | mg       |
| G_TyOff  | Angular rate zero-rate level <sup>(4)</sup>                                |                     |      | ±1                  |      | dps      |
| LA_OffDr | Linear acceleration zero-g level change vs. temperature <sup>(3)</sup>     |                     |      | ±0.07               |      | mg/°C    |
| G_OffDr  | Angular rate typical zero-rate level change vs. temperature <sup>(3)</sup> |                     |      | ±0.008              |      | dps/°C   |
| Rn       | Rate noise density in high-performance mode <sup>(5)</sup>                 |                     |      | 3.4                 |      | mdps/√Hz |
| RnRMS    | Gyroscope RMS noise in low-power mode <sup>(6)</sup>                       |                     |      | 70                  |      | mdps RMS |
| An       | Acceleration noise density in high-performance mode <sup>(7)</sup>         | FS = ±2 g - ±16 g   |      | 60                  |      | μg/√Hz   |
|          | Acceleration noise density in normal mode <sup>(8)(9)</sup>                | FS = ±2 g - ±16 g   |      | 100                 |      |          |
| RMS      | Accelerometer RMS noise in low-power mode                                  | LPM1                |      | 2.3                 |      | mg RMS   |
|          |                                                                            | LPM2                |      | 1.8                 |      |          |
|          |                                                                            | LPM3                |      | 1.2                 |      |          |



| Symbol | Parameter                                                                             | Test conditions | Min. | Typ. <sup>(1)</sup>   | Max. | Unit |
|--------|---------------------------------------------------------------------------------------|-----------------|------|-----------------------|------|------|
| LA_ODR | Linear acceleration output data rate                                                  |                 |      | 1.875 <sup>(10)</sup> |      | Hz   |
|        |                                                                                       |                 |      | 7.5                   |      |      |
|        |                                                                                       |                 |      | 15                    |      |      |
|        |                                                                                       |                 |      | 30                    |      |      |
|        |                                                                                       |                 |      | 60                    |      |      |
|        |                                                                                       |                 |      | 120                   |      |      |
|        |                                                                                       |                 |      | 240                   |      |      |
|        |                                                                                       |                 |      | 480                   |      |      |
|        |                                                                                       |                 |      | 960                   |      |      |
|        |                                                                                       |                 |      | 1.92 k                |      |      |
|        |                                                                                       |                 |      | 3.84 k                |      |      |
|        |                                                                                       |                 |      | 7.68 k                |      |      |
| G_ODR  | Angular rate output data rate                                                         |                 |      | 7.5                   |      | Hz   |
|        |                                                                                       |                 |      | 15                    |      |      |
|        |                                                                                       |                 |      | 30                    |      |      |
|        |                                                                                       |                 |      | 60                    |      |      |
|        |                                                                                       |                 |      | 120                   |      |      |
|        |                                                                                       |                 |      | 240                   |      |      |
|        |                                                                                       |                 |      | 480                   |      |      |
|        |                                                                                       |                 |      | 960                   |      |      |
|        |                                                                                       |                 |      | 1.92 k                |      |      |
|        |                                                                                       |                 |      | 3.84 k                |      |      |
|        |                                                                                       |                 |      | 7.68 k                |      |      |
| HAODR  | ODR variation over temperature and supply range in high-accuracy mode <sup>(11)</sup> | Gyro on         |      |                       | ±1   | %    |
|        |                                                                                       | Gyro off        |      |                       | ±3   |      |
| Vst    | Linear acceleration self-test output change <sup>(12)(13)(14)</sup>                   |                 | 50   |                       | 1700 | mg   |
|        | Angular rate self-test output change <sup>(15)(16)</sup>                              | FS = ±250 dps   | 20   |                       | 80   | dps  |
|        |                                                                                       | FS = ±2000 dps  | 150  |                       | 700  | dps  |
| Top    | Operating temperature range                                                           |                 | -40  |                       | +85  | °C   |

- Typical specifications are *not* guaranteed.
- Sensitivity tolerance for FS up to ±2000 dps.
- Measurements are performed in a uniform temperature setup and they are based on characterization data in a limited number of samples. Not measured during final test for production.
- Value after calibration.
- Gyroscope rate noise density in high-performance mode is independent of the ODR and FS setting up to ±2000 dps.
- Gyroscope RMS noise in low-power mode is independent of the ODR and FS setting up to ±2000 dps.
- Accelerometer noise density in high-performance mode is independent of the selected ODR and FS. Valid when XL\_DualC\_EN = 0 in register CTRL8 (17h).
- Accelerometer noise density in normal mode is independent of the ODR and FS setting. Valid when XL\_DualC\_EN = 0 in register CTRL8 (17h).
- Noise RMS related to BW = ODR/2.
- This ODR is available when the accelerometer is in low-power mode.
- Values specified by design.
- The sign of the linear acceleration self-test output change is defined by the ST\_XL\_[1:0] bits in a dedicated register for all axes.
- The linear acceleration self-test output change is defined with the device in stationary condition as the absolute value of: OUTPUT[LSb] (self-test enabled) - OUTPUT[LSb] (self-test disabled). 1LSb = 0.061 mg at ±2 g full scale.
- Accelerometer self-test limits are full-scale independent.
- The sign of the angular rate self-test output change is defined by the ST\_G\_[1:0] bits in a dedicated register for all axes.
- The angular rate self-test output change is defined with the device in stationary condition as the absolute value of: OUTPUT[LSb] (self-test enabled) - OUTPUT[LSb] (self-test disabled). 1LSb = 70 mdps at ±2000 dps full scale.



## 4.2 Electrical characteristics

@Vdd = 1.8 V, T = 25°C, unless otherwise noted.

**Table 4. Electrical characteristics**

| Symbol     | Parameter                                                           | Test conditions                             | Min.         | Typ. <sup>(1)</sup> | Max.         | Unit |
|------------|---------------------------------------------------------------------|---------------------------------------------|--------------|---------------------|--------------|------|
| Vdd        | Supply voltage                                                      |                                             | 1.71         | 1.8                 | 3.6          | V    |
| Vdd_IO     | Power supply for I/O                                                |                                             | 1.08         |                     | 3.6          | V    |
| IddHP      | Gyroscope and accelerometer supply current in high-performance mode |                                             |              | 0.65                |              | mA   |
| LA_IddHP   | Accelerometer supply current in high-performance mode               |                                             |              | 190                 |              | μA   |
| LA_IddNM   | Accelerometer supply current in normal mode                         |                                             |              | 100                 |              | μA   |
| LA_IddLPM2 | Accelerometer supply current in low-power mode (LPM2)               | ODR = 60 Hz                                 |              | 20                  |              | μA   |
|            |                                                                     | ODR = 1.875 Hz                              |              | 4.2                 |              |      |
| LA_IddLPM1 | Accelerometer supply current in low-power mode (LPM1)               | ODR = 60 Hz                                 |              | 17                  |              | μA   |
|            |                                                                     | ODR = 1.875 Hz                              |              | 4.0                 |              |      |
| IddPD      | Gyroscope and accelerometer supply current during power-down        |                                             |              | 2.6                 |              | μA   |
| Ton        | Turn-on time - gyroscope                                            |                                             |              | 40                  |              | ms   |
| VIH        | Digital high-level input voltage                                    | Vdd_IO = 1.8 V (typ)                        | 0.7 * Vdd_IO |                     |              | V    |
|            |                                                                     | Vdd_IO = 1.2 V (typ)                        | 0.8 * Vdd_IO |                     |              |      |
| VIL        | Digital low-level input voltage                                     | Vdd_IO = 1.8 V (typ)                        |              |                     | 0.3 * Vdd_IO | V    |
|            |                                                                     | Vdd_IO = 1.2 V (typ)                        |              |                     | 0.2 * Vdd_IO |      |
| VOH        | High-level output voltage                                           | Vdd_IO = 1.8 V<br>IOH = 4 mA <sup>(2)</sup> | Vdd_IO - 0.2 |                     |              | V    |
|            |                                                                     | Vdd_IO = 1.2 V<br>IOH = 2 mA <sup>(2)</sup> | Vdd_IO - 0.2 |                     |              |      |
| VOL        | Low-level output voltage                                            | Vdd_IO = 1.8 V<br>IOL = 4 mA <sup>(2)</sup> |              |                     | 0.2          | V    |
|            |                                                                     | Vdd_IO = 1.2 V<br>IOL = 2 mA <sup>(2)</sup> |              |                     | 0.2          |      |
| Top        | Operating temperature range                                         |                                             | -40          |                     | +85          | °C   |

1. Typical specifications are not guaranteed.

2. IOL/IOH are the driving capabilities, that is, the DC currents that can be sourced/sunk by the digital pin to guarantee the correct digital output voltage levels VOH and VOL.



### 4.3 Temperature sensor characteristics

@Vdd = 1.8 V, T = 25°C unless otherwise noted.

**Table 5. Temperature sensor characteristics**

| Symbol              | Parameter                                     | Test condition | Min. | Typ. <sup>(1)</sup> | Max. | Unit   |
|---------------------|-----------------------------------------------|----------------|------|---------------------|------|--------|
| TODR <sup>(2)</sup> | Temperature refresh rate                      |                |      | 60                  |      | Hz     |
| Toff                | Temperature offset <sup>(3)</sup>             |                | -15  |                     | +15  | °C     |
| TSen                | Temperature sensitivity                       |                |      | 256                 |      | LSB/°C |
| TST                 | Temperature stabilization time <sup>(4)</sup> |                |      |                     | 500  | µs     |
| T_ADC_res           | Temperature ADC resolution                    |                |      | 16                  |      | bit    |
| Top                 | Operating temperature range                   |                | -40  |                     | +85  | °C     |

1. Typical specifications are not guaranteed
2. When the accelerometer is in low-power mode and the gyroscope part is turned off, the TODR value is equal to the accelerometer ODR.
3. The output of the temperature sensor is 0 LSB (typ.) at 25°C.
4. Time from power ON to valid data based on characterization data.



## 4.4 Communication interface characteristics

### 4.4.1 SPI - serial peripheral interface

Subject to general operating conditions for Vdd and Top. @Vdd\_IO = 1.8 V, T = 25°C unless otherwise noted.

**Table 6. SPI slave timing values (Vdd\_IO at 1.8 V)**

| Symbol                  | Parameter               | Value <sup>(1)</sup> |     |     | Unit |
|-------------------------|-------------------------|----------------------|-----|-----|------|
|                         |                         | Min                  | Typ | Max |      |
| f <sub>c</sub> (SPC)    | SPI clock frequency     |                      |     | 10  | MHz  |
| t <sub>c</sub> (SPC)    | SPI clock period        | 100                  |     |     | ns   |
| t <sub>high</sub> (SPC) | SPI clock high period   | 45                   |     |     |      |
| t <sub>low</sub> (SPC)  | SPI clock low period    | 45                   |     |     |      |
| t <sub>su</sub> (CS)    | CS setup time (mode 3)  | 5                    |     |     |      |
|                         | CS setup time (mode 0)  | 20                   |     |     |      |
| t <sub>h</sub> (CS)     | CS hold time (mode 3)   | 20                   |     |     |      |
|                         | CS hold time (mode 0)   | 20                   |     |     |      |
| t <sub>su</sub> (SI)    | SDI input setup time    | 5                    |     |     |      |
| t <sub>h</sub> (SI)     | SDI input hold time     | 15                   |     |     |      |
| t <sub>v</sub> (SO)     | SDO valid output time   |                      | 15  | 25  |      |
| t <sub>dis</sub> (SO)   | SDO output disable time |                      |     | 50  |      |
| C <sub>load</sub>       | Bus capacitance         |                      |     | 100 | pF   |

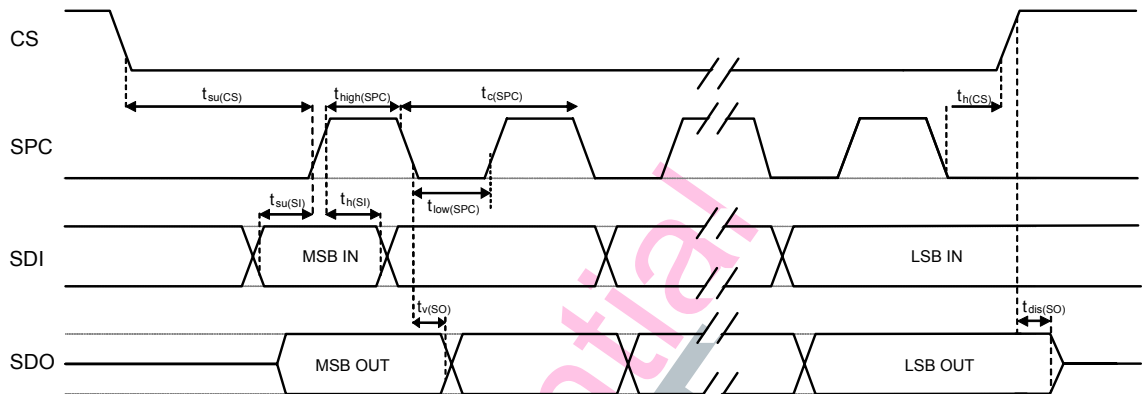
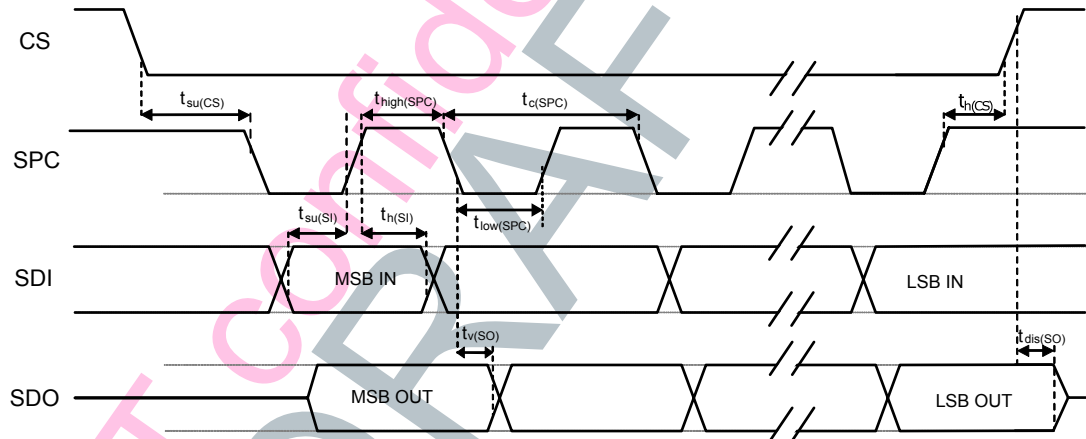
1. Values are evaluated at 10 MHz clock frequency for SPI with both 4 and 3 wires, based on characterization results, not tested in production.

Subject to general operating conditions for Vdd and Top. @Vdd\_IO = 1.2 V, T = 25°C unless otherwise noted.

**Table 7. SPI slave timing values (Vdd\_IO at 1.2 V)**

| Symbol                  | Parameter               | Value <sup>(1)</sup> |     |     | Unit |
|-------------------------|-------------------------|----------------------|-----|-----|------|
|                         |                         | Min                  | Typ | Max |      |
| f <sub>c</sub> (SPC)    | SPI clock frequency     |                      |     | 8   | MHz  |
| t <sub>c</sub> (SPC)    | SPI clock period        | 125                  |     |     | ns   |
| t <sub>high</sub> (SPC) | SPI clock high period   | 62.5                 |     |     |      |
| t <sub>low</sub> (SPC)  | SPI clock low period    | 62.5                 |     |     |      |
| t <sub>su</sub> (CS)    | CS setup time (mode 3)  | 30                   |     |     |      |
|                         | CS setup time (mode 0)  | 30                   |     |     |      |
| t <sub>h</sub> (CS)     | CS hold time (mode 3)   | 30                   |     |     |      |
|                         | CS hold time (mode 0)   | 30                   |     |     |      |
| t <sub>su</sub> (SI)    | SDI input setup time    | 30                   |     |     |      |
| t <sub>h</sub> (SI)     | SDI input hold time     | 30                   |     |     |      |
| t <sub>v</sub> (SO)     | SDO valid output time   |                      |     | 50  |      |
| t <sub>dis</sub> (SO)   | SDO output disable time |                      |     | 50  |      |
| C <sub>load</sub>       | Bus capacitance         |                      |     | 50  | pF   |

1. Values are evaluated at 8 MHz clock frequency for SPI with both 4 and 3 wires, based on characterization results, not tested in production.


**Figure 6. SPI slave timing in mode 0**

**Figure 7. SPI slave timing in mode 3**


**Note:** Measurement points are done at  $0.3 \cdot V_{dd\_IO}$  and  $0.7 \cdot V_{dd\_IO}$  for both input and output ports.





#### 4.4.2 I<sup>2</sup>C - inter-IC control interface

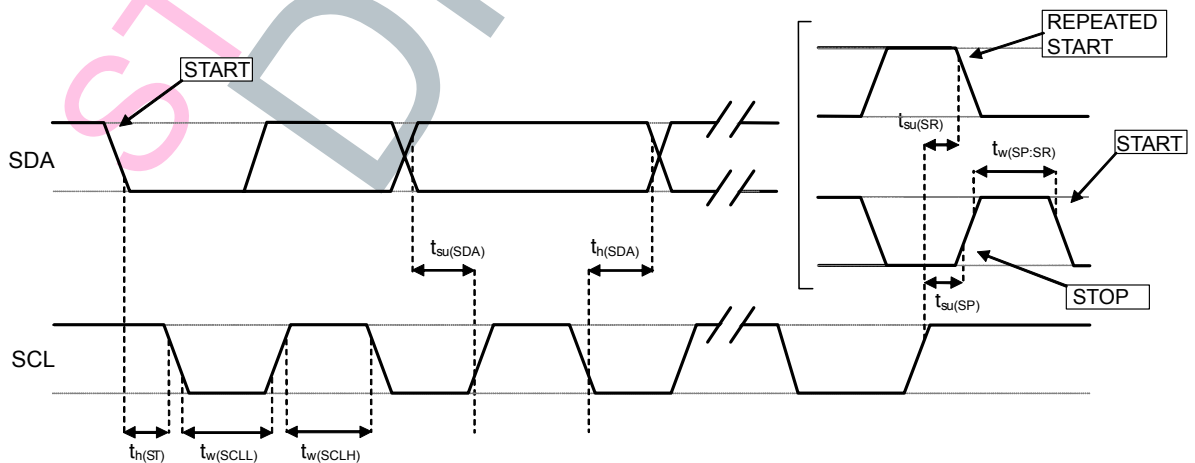
Subject to general operating conditions for V<sub>dd</sub> and T<sub>op</sub>. @V<sub>dd\_IO</sub> = 1.8 V, T = 25°C unless otherwise noted.

**Table 8. I<sup>2</sup>C slave timing values**

| Symbol                | Parameter                                      | I <sup>2</sup> C fast mode <sup>(1)(2)</sup> |     | I <sup>2</sup> C fast mode plus <sup>(1)(2)</sup> |                    | Unit |
|-----------------------|------------------------------------------------|----------------------------------------------|-----|---------------------------------------------------|--------------------|------|
|                       |                                                | Min                                          | Max | Min                                               | Max                |      |
| f <sub>(SCL)</sub>    | SCL clock frequency                            | 0                                            | 400 | 0                                                 | 1000               | kHz  |
| t <sub>w(SCLL)</sub>  | SCL clock low period                           | 1.3                                          |     | 0.5                                               |                    | μs   |
| t <sub>w(SCLH)</sub>  | SCL clock high period                          | 0.6                                          |     | 0.26                                              |                    |      |
| t <sub>su(SDA)</sub>  | SDA setup time                                 | 100                                          |     | 50                                                |                    | ns   |
| t <sub>h(SDA)</sub>   | SDA data hold time                             | 0                                            | 0.9 | 0                                                 |                    | μs   |
| t <sub>h(ST)</sub>    | START/REPEATED START condition hold time       | 0.6                                          |     | 0.26                                              |                    |      |
| t <sub>su(SR)</sub>   | REPEATED START condition setup time            | 0.6                                          |     | 0.26                                              |                    |      |
| t <sub>su(SP)</sub>   | STOP condition setup time                      | 0.6                                          |     | 0.26                                              |                    |      |
| t <sub>w(SP:SR)</sub> | Bus free time between STOP and START condition | 1.3                                          |     | 0.5                                               |                    |      |
|                       | Data valid time                                |                                              | 0.9 |                                                   | 0.45               |      |
|                       | Data valid acknowledge time                    |                                              | 0.9 |                                                   | 0.45               |      |
| C <sub>B</sub>        | Capacitive load for each bus line              |                                              | 400 |                                                   | 550 <sup>(3)</sup> | pF   |

1. Data based on standard I<sup>2</sup>C protocol requirement, not tested in production.
2. Data for I<sup>2</sup>C fast mode and I<sup>2</sup>C fast mode plus have been validated by characterization, not tested in production.
3. Maximum capacitive load if V<sub>dd\_IO</sub> ≥ 1.62 V. If V<sub>dd\_IO</sub> < 1.62 V, the maximum load must be reduced to 400 pF.

**Figure 8. I<sup>2</sup>C slave timing diagram**



**Note:** Measurement points are done at 0.3·V<sub>dd\_IO</sub> and 0.7·V<sub>dd\_IO</sub> for both ports.



## 4.5 Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

**Table 9. Absolute maximum ratings**

| Symbol           | Ratings                                                                           | Maximum value       | Unit     |
|------------------|-----------------------------------------------------------------------------------|---------------------|----------|
| Vdd              | Supply voltage                                                                    | -0.3 to +4.8        | V        |
| T <sub>STG</sub> | Storage temperature range                                                         | -40 to +125         | °C       |
| Sg               | Acceleration <i>g</i> for 0.2 ms                                                  | 20,000              | <i>g</i> |
| ESD              | Electrostatic discharge protection (HBM)                                          | 2                   | kV       |
| Vin              | Input voltage on any control pin<br>(including CS, SCL/SPC, SDA/SDI/SDO, SDO/SA0) | -0.3 to Vdd_IO +0.3 | V        |

**Note:** Supply voltage on any pin should never exceed 4.8 V.



This device is sensitive to mechanical shock, improper handling can cause permanent damage to the part.



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.



## 4.6 Terminology

### 4.6.1 Sensitivity

Linear acceleration sensitivity can be determined, for example, by applying 1 g acceleration to the device. Because the sensor can measure DC accelerations, this can be done easily by pointing the selected axis towards the ground, noting the output value, rotating the sensor 180 degrees (pointing towards the sky) and noting the output value again. By doing so,  $\pm 1$  g acceleration is applied to the sensor. Subtracting the larger output value from the smaller one, and dividing the result by 2, leads to the actual sensitivity of the sensor. This value changes very little over temperature and over time. The sensitivity tolerance describes the range of sensitivities of a large number of sensors (see [Table 3](#)).

An angular rate gyroscope is a device that produces a positive-going digital output for counterclockwise rotation around the axis considered. Sensitivity describes the gain of the sensor and can be determined by applying a defined angular velocity to it. This value changes very little over temperature and time (see [Table 3](#)).

### 4.6.2 Zero-g and zero-rate level

Linear acceleration zero-g level offset (TyOff) describes the deviation of an actual output signal from the ideal output signal if no acceleration is present. A sensor in a steady state on a horizontal surface measures 0 g on both the X-axis and Y-axis, whereas the Z-axis measures 1 g. Ideally, the output is in the middle of the dynamic range of the sensor (content of OUT registers 00h, data expressed as two's complement number). A deviation from the ideal value in this case is called zero-g offset.

Offset is to some extent a result of stress to the MEMS sensor and therefore the offset can slightly change after mounting the sensor onto a printed circuit board or exposing it to extensive mechanical stress. Offset changes little over temperature, see "Linear acceleration zero-g level change vs. temperature" in [Table 3](#). The zero-g level tolerance (TyOff) describes the standard deviation of the range of zero-g levels of a group of sensors.

Zero-rate level describes the actual output signal if there is no angular rate present. The zero-rate level of precise MEMS sensors is, to some extent, a result of stress to the sensor and therefore the zero-rate level can slightly change after mounting the sensor onto a printed circuit board or after exposing it to extensive mechanical stress. This value changes very little over temperature and time (see [Table 3](#)).



## 5 Digital interfaces

### 5.1 I<sup>2</sup>C/SPI interface

The registers embedded inside the LSM6DSVE may be accessed through both the I<sup>2</sup>C and SPI serial interfaces. The latter may be software configured to operate either in 3-wire or 4-wire interface mode. The device is compatible with SPI modes 0 and 3.

The serial interfaces are mapped to the same pins. To select/exploit the I<sup>2</sup>C interface, the CS line must be tied high (that is, connected to Vdd\_IO).

**Table 10. Serial interface pin description**

| Pin name    | Pin description                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CS          | Enables SPI<br>I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) |
| SCL/SPC     | I <sup>2</sup> C serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                                          |
| SDA/SDI/SDO | I <sup>2</sup> C serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                                              |
| SDO/SA0     | SPI serial data output (SDO)<br>I <sup>2</sup> C less significant bit of the device address                                                                                 |

#### 5.1.1 I<sup>2</sup>C serial interface

The LSM6DSVE I<sup>2</sup>C is a bus slave. The I<sup>2</sup>C is employed to write the data to the registers, whose content can also be read back.

The relevant I<sup>2</sup>C terminology is provided in the table below.

**Table 11. I<sup>2</sup>C terminology**

| Term        | Description                                                                              |
|-------------|------------------------------------------------------------------------------------------|
| Transmitter | The device that sends data to the bus                                                    |
| Receiver    | The device that receives data from the bus                                               |
| Master      | The device that initiates a transfer, generates clock signals, and terminates a transfer |
| Slave       | The device addressed by the master                                                       |

There are two signals associated with the I<sup>2</sup>C bus: the serial clock line (SCL) and the serial data line (SDA). The latter is a bidirectional line used for sending and receiving the data to/from the interface. Both the lines must be connected to Vdd\_IO through external pull-up resistors. When the bus is free, both the lines are high.

The I<sup>2</sup>C interface supports fast mode (400 kHz) I<sup>2</sup>C standards as well as fast mode plus (1000 kHz).

In order to disable the I<sup>2</sup>C block, I2C\_I3C\_disable = 1 must be written in IF\_CFG (03h).



### 5.1.2 I<sup>2</sup>C operation

The transaction on the bus is started through a start (ST) signal. A start condition is defined as a high to low transition on the data line while the SCL line is held high. After this has been transmitted by the master, the bus is considered busy. The next byte of data transmitted after the start condition contains the address of the slave in the first 7 bits and the eighth bit tells whether the master is receiving data from the slave or transmitting data to the slave. When an address is sent, each device in the system compares the first seven bits after a start condition with its address. If they match, the device considers itself addressed by the master.

The slave address (SAD) associated to the LSM6DSVE is 110101xb. The SDO/SA0 pin can be used to modify the less significant bit of the device address. If the SDO/SA0 pin is connected to the supply voltage, LSb is 1 (address 1101011b); else if the SDO/SA0 pin is connected to ground, the LSb value is 0 (address 1101010b). This solution permits to connect and address two different inertial modules to the same I<sup>2</sup>C bus.

Data transfer with acknowledge is mandatory. The transmitter must release the SDA line during the acknowledge pulse. The receiver must then pull the data line low so that it remains stable low during the high period of the acknowledge clock pulse. A receiver that has been addressed is obliged to generate an acknowledge after each byte of data received.

The I<sup>2</sup>C embedded inside the LSM6DSVE behaves like a slave device and the following protocol must be adhered to. After the start condition (ST) a slave address is sent, once a slave acknowledge (SAK) has been returned, an 8-bit subaddress (SUB) is transmitted. The increment of the address is configured by the CTRL3 (12h) (IF\_INC).

The slave address is completed with a read/write bit. If the bit is 1 (read), a repeated start (SR) condition must be issued after the two subaddress bytes; if the bit is 0 (write) the master transmits to the slave with direction unchanged. Table 12 explains how the SAD+read/write bit pattern is composed, listing all the possible configurations.

**Table 12. SAD+read/write patterns**

| Command | SAD[6:1] | SAD[0] = SA0 | R/W | SAD+R/W        |
|---------|----------|--------------|-----|----------------|
| Read    | 110101   | 0            | 1   | 11010101 (D5h) |
| Write   | 110101   | 0            | 0   | 11010100 (D4h) |
| Read    | 110101   | 1            | 1   | 11010111 (D7h) |
| Write   | 110101   | 1            | 0   | 11010110 (D6h) |

**Table 13. Transfer when master is writing one byte to slave**

|        |    |         |     |     |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |    |

**Table 14. Transfer when master is writing multiple bytes to slave**

|        |    |         |     |     |     |      |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |      | SAK |    |

**Table 15. Transfer when master is receiving (reading) one byte of data from slave**

|        |    |         |     |     |     |    |         |     |      |      |    |
|--------|----|---------|-----|-----|-----|----|---------|-----|------|------|----|
| Master | ST | SAD + W |     | SUB |     | SR | SAD + R |     |      | NMAK | SP |
| Slave  |    |         | SAK |     | SAK |    |         | SAK | DATA |      |    |

**Table 16. Transfer when master is receiving (reading) multiple bytes of data from slave**

|        |    |           |     |     |     |    |           |     |      |  |      |  |      |    |
|--------|----|-----------|-----|-----|-----|----|-----------|-----|------|--|------|--|------|----|
| Master | ST | SAD+<br>W |     | SUB |     | SR | SAD+<br>R |     | MAK  |  | MAK  |  | NMAK | SP |
| Slave  |    |           | SAK |     | SAK |    |           | SAK | DATA |  | DATA |  | DATA |    |



Data are transmitted in byte format (DATA). Each data transfer contains 8 bits. The number of bytes transferred per transfer is unlimited. Data is transferred with the most significant bit (MSb) first. If a slave receiver does not acknowledge the slave address (that is, it is not able to receive because it is performing some real-time function) the data line must be left high by the slave. The master can then abort the transfer. A low to high transition on the SDA line while the SCL line is high is defined as a stop condition. Each data transfer must be terminated by the generation of a stop (SP) condition.

In the presented communication format, MAK is master acknowledge and NMAK is no master acknowledge.

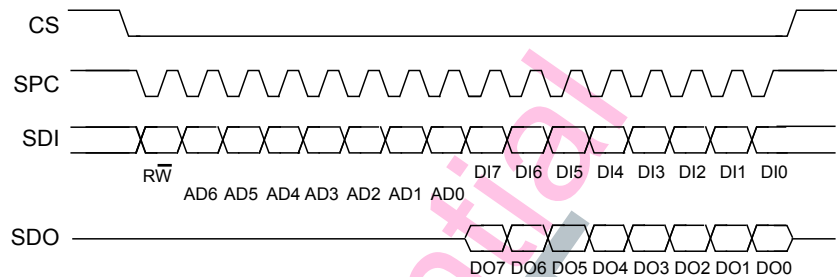
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### 5.1.3 SPI bus interface

The SPI on the LSM6DSVE is a bus slave which allows writing and reading the registers of the device.

**Figure 9. Read and write protocol (in mode 3)**



**CS** enables the serial port and it is controlled by the SPI master. It goes low at the start of the transmission and goes back high at the end. **SPC** is the serial port clock and it is controlled by the SPI master. It is stopped high when **CS** is high (no transmission). **SDI** and **SDO** are, respectively, the serial port data input and output. Those lines are driven at the falling edge of **SPC** and should be captured at the rising edge of **SPC**.

Both the read register and write register commands are completed in 16 clock pulses or in multiples of 8 in case of multiple read/write bytes. Bit duration is the time between two falling edges of **SPC**. The first bit (bit 0) starts at the first falling edge of **SPC** after the falling edge of **CS** while the last bit (bit 15, bit 23, ...) starts at the last falling edge of **SPC** just before the rising edge of **CS**.

**bit 0:**  $\overline{RW}$  bit. When 0, the data  $DI(7:0)$  is written into the device. When 1, the data  $DO(7:0)$  from the device is read. In latter case, the chip drives **SDO** at the start of bit 8.

**bit 1-7:** address  $AD(6:0)$ . This is the address field of the indexed register.

**bit 8-15:** data  $DI(7:0)$  (write mode). This is the data that is written into the device (MSb first).

**bit 8-15:** data  $DO(7:0)$  (read mode). This is the data that is read from the device (MSb first).

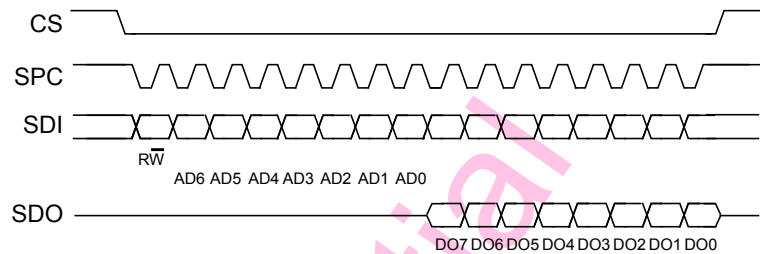
In multiple read/write commands further blocks of 8 clock periods are added. When the **CTRL3 (12h) (IF\_INC)** bit is 0, the address used to read/write data remains the same for every block. When the **CTRL3 (12h) (IF\_INC)** bit is 1, the address used to read/write data is increased at every block.

The function and the behavior of **SDI** and **SDO** remain unchanged.



### 5.1.3.1 SPI read

**Figure 10. SPI read protocol (in mode 3)**



The SPI read command is performed with 16 clock pulses. A multiple byte read command is performed by adding blocks of 8 clock pulses to the previous one.

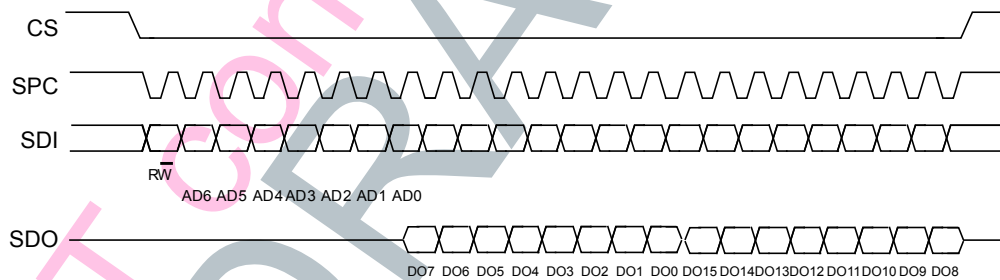
**bit 0:** READ bit. The value is 1.

**bit 1-7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DO(7:0) (read mode). This is the data that is read from the device (MSb first).

**bit 16-...:** data DO(...-8). Further data in multiple byte reads.

**Figure 11. Multiple byte SPI read protocol (2-byte example) (in mode 3)**

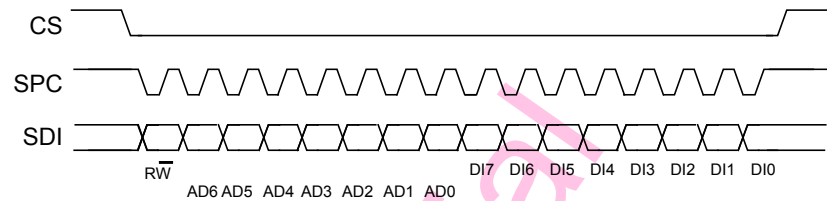






### 5.1.3.2 SPI write

**Figure 12. SPI write protocol (in mode 3)**



The SPI write command is performed with 16 clock pulses. A multiple byte write command is performed by adding blocks of 8 clock pulses to the previous one.

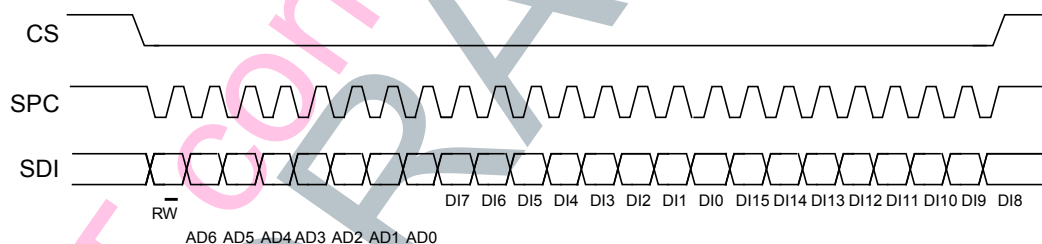
**bit 0:** WRITE bit. The value is 0.

**bit 1 -7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DI(7:0) (write mode). This is the data that is written inside the device (MSb first).

**bit 16-...** : data DI(...-8). Further data in multiple byte writes.

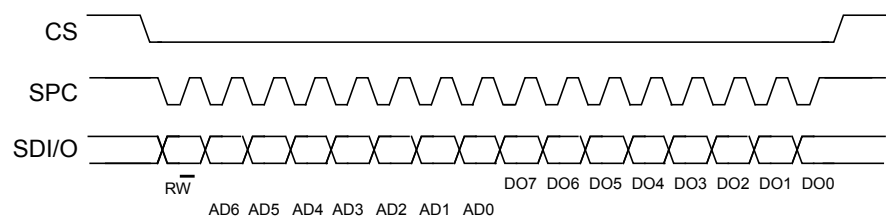
**Figure 13. Multiple byte SPI write protocol (2-byte example) (in mode 3)**



### 5.1.3.3 SPI read in 3-wire mode

3-wire mode is entered by setting the **IF\_CFG (03h) (SIM)** bit equal to 1 (SPI serial interface mode selection).

**Figure 14. SPI read protocol in 3-wire mode (in mode 3)**



The SPI read command is performed with 16 clock pulses:

**bit 0:** READ bit. The value is 1.

**bit 1-7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DO(7:0) (read mode). This is the data that is read from the device (MSb first).

A multiple read command is also available in 3-wire mode.



## 5.2 MIPI I3C<sup>®</sup> interface

### 5.2.1 MIPI I3C<sup>®</sup> slave interface

The LSM6DSVE interface includes an MIPI I3C<sup>®</sup> SDR only slave interface (compliant with release 1.1 of the specification) with MIPI I3C<sup>®</sup> SDR embedded features:

- CCC command
- Direct CCC communication (SET and GET)
- Broadcast CCC communication
- Private communications
- Private read and write for single byte
- Multiple read and write
- In-band interrupt request
- Slave reset pattern
- Group address
- Full range Vdd\_IO support
- Asynchronous modes 0 and 1
- Synchronous mode
- Error detection and recovery methods (S0-S6)

In order to disable the MIPI I3C<sup>®</sup> block, I2C\_I3C\_disable = 1 must be written in IF\_CFG (03h).



### 5.2.2 MIPI I3C® CCC supported commands

The list of MIPI I3C® CCC commands supported by the device is detailed in the following table.

**Table 17. MIPI I3C® CCC commands**

| Command   | Command code | Default                                      | Description                                                                |
|-----------|--------------|----------------------------------------------|----------------------------------------------------------------------------|
| ENTDAA    | 0x07         |                                              | DAA procedure                                                              |
| SETDASA   | 0x87         |                                              | Assign dynamic address using static address 0x6B/0x6A depending on SDO pin |
| ENEC      | 0x80 / 0x00  |                                              | Slave activity control (direct and broadcast)                              |
| DISEC     | 0x81 / 0x01  |                                              | Slave activity control (direct and broadcast)                              |
| ENTAS0    | 0x82 / 0x02  |                                              | Enter activity state (direct and broadcast)                                |
| SETXTIME  | 0x98 / 0x28  |                                              | Timing information exchange                                                |
| GETXTIME  | 0x99         | 0x07<br>0x00<br>0x05<br>0x92                 | Timing information exchange                                                |
| RSTDAA    | 0x06         |                                              | Reset the assigned dynamic address (broadcast only)                        |
| SETMWL    | 0x89 / 0x08  |                                              | Define maximum write length during private write (direct and broadcast)    |
| SETMRL    | 0x8A / 0x09  |                                              | Define maximum read length during private read (direct and broadcast)      |
| SETNEWDA  | 0x88         |                                              | Change dynamic address                                                     |
| GETMWL    | 0x8B         | 0x00<br>0x08<br>(2 byte)                     | Get maximum write length during private write                              |
| GETMRL    | 0x8C         | 0x00<br>0x10<br>0x0A<br>(3 byte)             | Get maximum read length during private read                                |
| GETPID    | 0x8D         | 0x02<br>0x08<br>0x00<br>0x70<br>0x92<br>0x0B | SDO = 1                                                                    |
|           |              | 0x02<br>0x08<br>0x00<br>0x70<br>0x12<br>0x0B | SDO = 0                                                                    |
| GETBCR    | 0x8E         | 0x07<br>(1 byte)                             | Bus characteristics register                                               |
| GETDCR    | 0x8F         | 0x44 default                                 | MIPI I3C® device characteristics register                                  |
| GETSTATUS | 0x90         | 0x00<br>0x00<br>(2 byte)                     | Status register                                                            |
| GETMXDS   | 0x94         | 0x08<br>0x60                                 | Return max write and read speed                                            |



| Command | Command code | Default                      | Description                                                                   |
|---------|--------------|------------------------------|-------------------------------------------------------------------------------|
| GETCAPS | 0x95         | 0x00<br>0x11<br>0x18<br>0x00 | Provide information about device capabilities and supported extended features |
| SETGRPA | 0x9B         |                              | Group address assignment command                                              |
| RSTGRPA | 0x2C / 0x9C  |                              | Reset the group address                                                       |
| RSTACT  | 0x9A / 0x2A  |                              | Configure slave reset action                                                  |

### 5.2.3 Overview of antispikes filter management

The device acts as a standard I<sup>2</sup>C target as long as it has an I<sup>2</sup>C static address. The device is capable of detecting and disabling the I<sup>2</sup>C antispikes filter after detecting the broadcast address (7'h7E/W). In order to guarantee proper behavior of the device, the I<sup>3</sup>C master must emit the first START, 7'h7E/W at open-drain speed using I<sup>2</sup>C fast mode plus reference timing.

After detecting the broadcast address, the device can receive the I<sup>3</sup>C dynamic address following the I<sup>3</sup>C push-pull timing. If the device is not assigned a dynamic address, then the device continues to operate as an I<sup>2</sup>C device with no antispikes filter. For the case in which the host decides to keep the device as I<sup>2</sup>C with antispikes filter, there is a configuration required to keep the antispikes filter active. This configuration is done by writing the ASF\_CTRL bit to 1 in the IF\_CFG (03h) register. This configuration forces the antispikes filter to always be turned on instead of being managed by the communication on the bus.



### 5.3 Master I<sup>2</sup>C interface

If the LSM6DSVE is configured in mode 2, a master I<sup>2</sup>C line is available. The master serial interface is mapped to the following dedicated pins.

Table 18. Master I<sup>2</sup>C pin details

| Pin name | Pin description                                         |
|----------|---------------------------------------------------------|
| MSCL     | I <sup>2</sup> C master serial clock                    |
| MSDA     | I <sup>2</sup> C master serial data                     |
| MDRDY    | I <sup>2</sup> C master external synchronization signal |



## 6 Functionality

### 6.1 Operating modes

In the LSM6DSVE, the accelerometer and the gyroscope can be turned on/off independently of each other and are allowed to have different ODRs and power modes.

The LSM6DSVE has three operating modes available:

- Only accelerometer active and gyroscope in power-down
- Only gyroscope active and accelerometer in power-down
- Both accelerometer and gyroscope sensors active with independent ODR and power mode

The accelerometer is activated from power-down by writing ODR\_XL[3:0] in [CTRL1 \(10h\)](#) while the gyroscope is activated from power-down by writing ODR\_G[3:0] in [CTRL2 \(11h\)](#). For combo mode the ODRs are totally independent.

### 6.2 Accelerometer power modes

In the LSM6DSVE, the accelerometer can be configured in five different operating modes: power-down mode, low-power mode (1, 2, 3), normal mode, high-performance mode, and high-accuracy ODR mode.

The operating mode selected depends on the value of the OP\_MODE\_XL[2:0] bits in [CTRL1 \(10h\)](#).

If the value of the OP\_MODE\_XL[2:0] bits is 000 (default), high-performance mode is valid for all ODRs (from 7.5 Hz up to 7.68 kHz).

Normal mode is available for ODR values from 7.5 Hz to 1.92 kHz and it is enabled by setting the OP\_MODE\_XL[2:0] bits to 111.

In high-performance mode and in normal mode, the analog antialiasing filter is active.

Low-power mode is available for lower ODRs (1.875 Hz, 15 Hz, 30 Hz, 60 Hz, 120 Hz, 240 Hz). The three low-power modes are enabled by setting OP\_MODE\_XL[2:0] to 100 (LPM1), 101 (LPM2), 110 (LPM3).

High-accuracy ODR mode is available for ODR values from 15 Hz up to 7.68 kHz and it is enabled by setting the OP\_MODE\_XL[2:0] bits to 001. Refer to [Section 6.5: High-accuracy ODR mode](#) for more details.

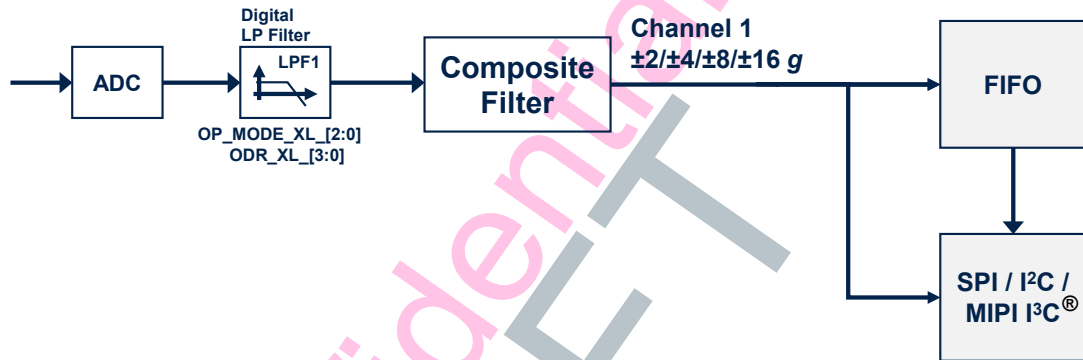
The embedded functions based on accelerometer data (free-fall, 6D/4D, tap/double-tap, wake-up, activity/inactivity, stationary/motion, step counter, step detection, significant motion, tilt) and the FIFO batching functionality are supported in all modes.



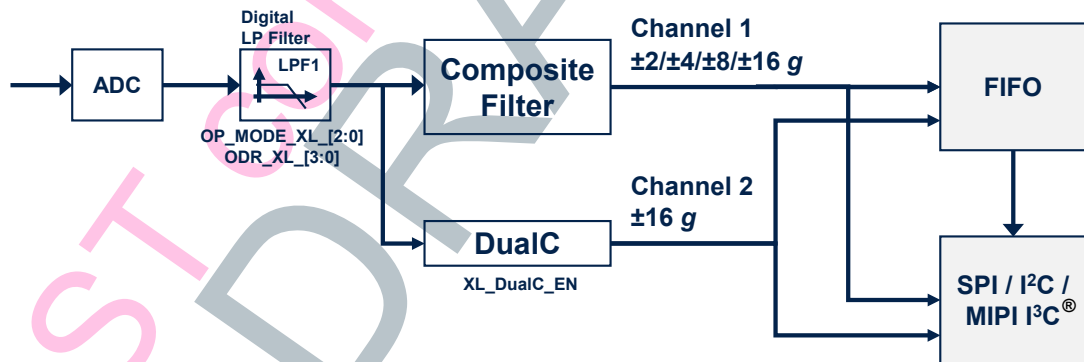
### 6.3 Accelerometer dual-channel mode

The LSM6DSVE accelerometer block has a dual-channel architecture able to work with two different full scales simultaneously. By default, the device operates in single-channel mode supporting FS scale values from  $\pm 2\text{ g}$  through  $\pm 16\text{ g}$  and different power modes, as described in [Section 6.2: Accelerometer power modes](#). The block diagrams in the following figures show the configuration of acceleration data processing in the two different modes.

**Figure 15. Single-channel mode (XL\_DualC\_EN = 0)**



**Figure 16. Dual-channel mode (XL\_DualC\_EN = 1)**



The dual-channel functionality can be enabled/disabled by configuring the bit XL\_DualC\_EN to 1 (enable) or to 0 (disable) in [CTRL8 \(17h\)](#).

Referring to [Figure 16. Dual-channel mode \(XL\\_DualC\\_EN = 1\)](#), when the dual-channel mode has been activated:

1. Channel 1 supports user-selectable full-scale acceleration range of  $\pm 2/\pm 4/\pm 8/\pm 16\text{ g}$  based on the value of the FS\_XL[1:0] bits in the [CTRL8 \(17h\)](#) register.
2. Channel 2 full scale is set to  $\pm 16\text{ g}$ . Acceleration data are available in the output registers from UI\_OUTX\_L\_A\_DualC (34h) and UI\_OUTX\_H\_A\_DualC (35h) through UI\_OUTZ\_L\_A\_DualC (38h) and UI\_OUTZ\_H\_A\_DualC (39h).



## 6.4 Gyroscope power modes

In the LSM6DSVE, the gyroscope can be configured in five different operating modes: power-down mode, sleep mode, low-power mode, high-performance mode and high-accuracy ODR mode.

The operating mode selected depends on the value of the OP\_MODE\_G\_[2:0] bits in CTRL2 (11h).

If the value of the OP\_MODE\_G\_[2:0] bits is 000 (default), high-performance mode is valid for all ODRs (from 7.5 Hz up to 7.68 kHz).

Low-power mode is available for lower ODRs (7.5 Hz, 15 Hz, 30 Hz, 60 Hz, 120 Hz, 240 Hz) and it is enabled by setting the the OP\_MODE\_G\_[2:0] bits to 101.

High-accuracy ODR mode is available for ODR values from 15 Hz up to 7.68 kHz and it is enabled by setting the OP\_MODE\_G\_[2:0] bits to 001. Refer to Section 6.5: High-accuracy ODR mode for more details.

## 6.5 High-accuracy ODR mode

High-accuracy ODR (HAODR) mode can be enabled to reduce the part-to-part output data rate variation. It supports accelerometer only, gyroscope only, and combo (accelerometer and gyroscope) modes. When this mode is used for one sensor (accelerometer or gyroscope), the other sensor also has to be configured in high-accuracy ODR (HAODR) mode.

The main high-accuracy ODR features are:

- Noise level is aligned with high-performance mode
- The power consumption in HAODR mode depends on the selected ODR and on the enabled sensors. As an example, at ODR = 960 Hz in combo mode (accelerometer + gyroscope) it increases 20  $\mu$ A (typical) compared to high-performance mode.
- The UI channel bandwidth can be selected through the gyroscope LPF1 and accelerometer HPF/LPF2 filters.
- When HAODR mode is enabled, it is applied to the UI accelerometer, UI gyroscope, EIS gyroscope, and temperature.

*Note:* HAODR mode has to be enabled / disabled when the device is in power-down mode.

When HAODR mode is enabled, two different sets of ODRs are supported based on the configuration of the HAODR\_SEL\_[1:0] bitfield in the HAODR\_CFG (62h) register, as shown in the table below.

*Note:* High-accuracy ODR mode is not compatible with the activity/inactivity functionality (motion/stationary can be used).

**Table 19. Accelerometer and gyroscope ODR selection in high-accuracy ODR mode**

| ODR_XL_[3:0]<br>ODR_G_[3:0] | ODR [Hz]<br>HAODR_SEL_[1:0] = 00 | ODR [Hz]<br>HAODR_SEL_[1:0] = 01 | ODR [Hz]<br>HAODR_SEL_[1:0] = 10 |
|-----------------------------|----------------------------------|----------------------------------|----------------------------------|
| 0000                        | Power-down                       | Power-down                       | Power-down                       |
| 0001                        | Reserved                         | Reserved                         | Reserved                         |
| 0010                        | Reserved                         | Reserved                         | Reserved                         |
| 0011                        | 15                               | 15.625                           | 12.5                             |
| 0100                        | 30                               | 31.25                            | 25                               |
| 0101                        | 60                               | 62.5                             | 50                               |
| 0110                        | 120                              | 125                              | 100                              |
| 0111                        | 240                              | 250                              | 200                              |
| 1000                        | 480                              | 500                              | 400                              |
| 1001                        | 960                              | 1000                             | 800                              |
| 1010                        | 1920                             | 2000                             | 1600                             |
| 1011                        | 3840                             | 4000                             | 3200                             |
| 1100                        | 7680                             | 8000                             | 6400                             |
| Others                      | Reserved                         | Reserved                         | Reserved                         |





## 6.6 ODR-triggered mode

When ODR-triggered mode is enabled, a reference signal must be provided to the INT2 pin, and the device then automatically aligns (in frequency and phase) the data generation to the edges of the reference signal.

It supports accelerometer only, gyroscope only, and combo (accelerometer and gyroscope) modes. When both the accelerometer and gyroscope are enabled, the user must configure the same ODR on both the accelerometer and gyroscope. It is not possible to select different ODRs for the accelerometer and gyroscope; if different output data rate values are set, the ODR configured for the gyroscope data is also applied to the accelerometer data.

The full-scale configurations are totally independent between the accelerometer and gyroscope and they can be set in any combination.

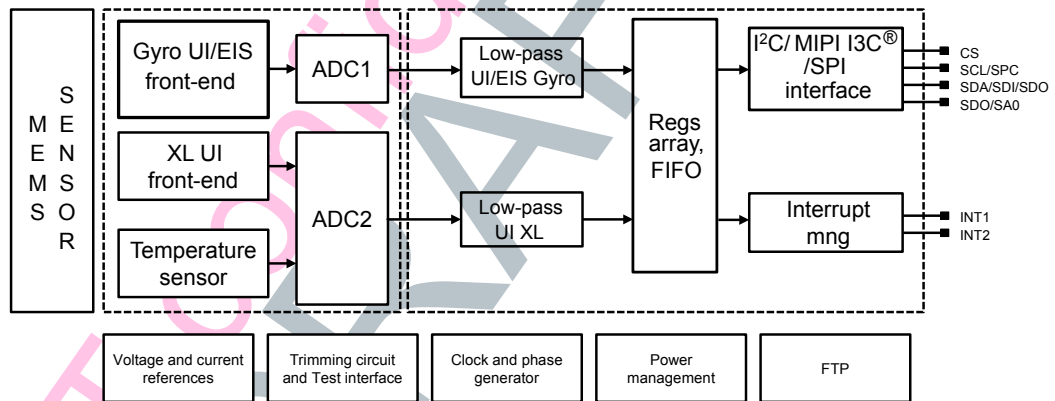
*Note:* ODR-triggered mode has to be enabled / disabled when the device is in power-down mode.

*Note:* When ODR-triggered mode is enabled, the 1100 configuration of the ODR\_XL\_[3:0] bits in register CTRL1 (10h) and the 1100 configuration of the ODR\_G\_[3:0] bits in register CTRL2 (11h) cannot be used.

*Note:* ODR-triggered mode is not compatible with the EIS functionality.

## 6.7 Block diagram of filters

Figure 17. Block diagram of filters





### 6.7.1 Block diagrams of the accelerometer filters

In the LSM6DSVE, the filtering chain for the accelerometer part is composed of the following:

- Digital filter (LPF1)
- Composite filter

Details of the block diagram appear in the following figure.

Figure 18. Accelerometer UI chain

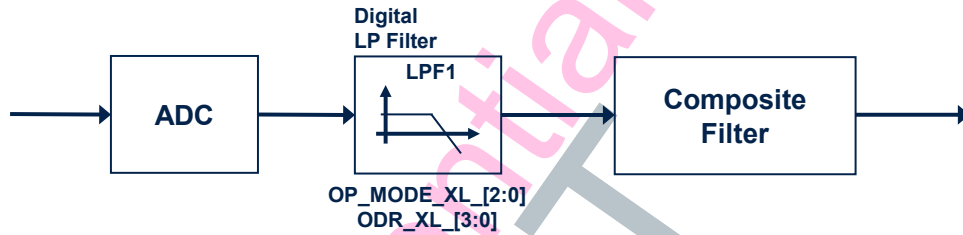
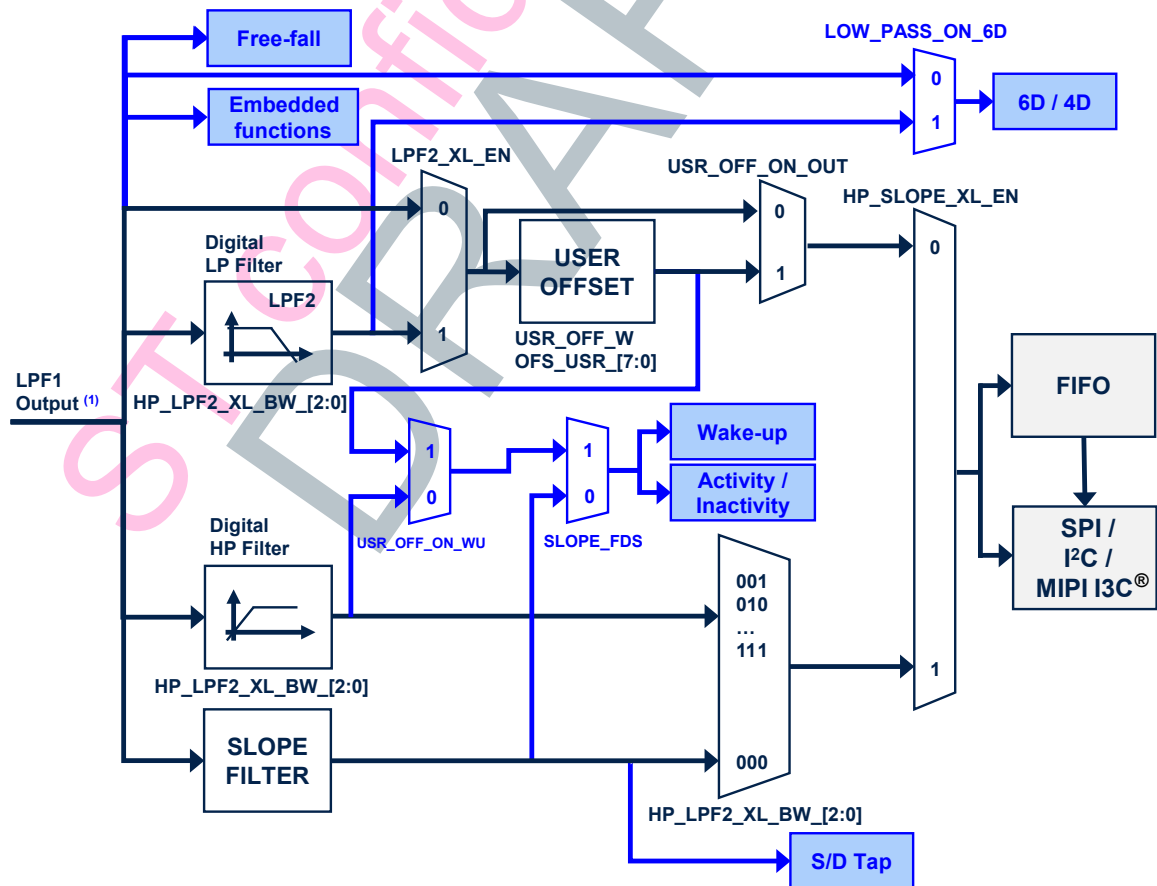


Figure 19. Accelerometer composite filter



1. The cutoff value of the LPF1 output is  $ODR/2$  when the accelerometer is in high-performance mode, high-accuracy ODR mode, or normal mode. This value is equal to 2300 Hz when the accelerometer is in low-power mode 1 (2 mean), 912 Hz in low-power mode 2 (4 mean) or 431 Hz in low-power mode 3 (8 mean).

**Note:** Embedded functions include the finite state machine, pedometer, step detector and step counter, significant motion detection, and tilt functions.

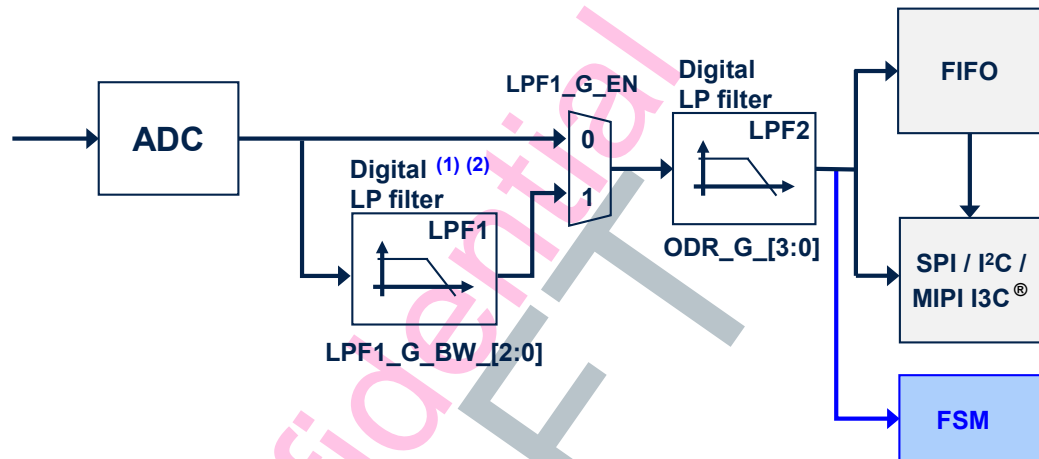


### 6.7.2 Block diagrams of the gyroscope filters

In the LSM6DSVE, the gyroscope filtering chain depends on the mode configuration:

- Mode 1 (for user interface (UI) and electronic image stabilization (EIS) functionality through the primary interface) and mode 2

**Figure 20. Gyroscope digital chain - mode 1 (UI/EIS) and mode 2**



1. When the gyroscope EIS chain is enabled, the LPF1 filter is not available in the gyroscope UI chain. It is recommended to avoid using the LPF1 filter in the gyroscope UI chain when the gyroscope EIS is used.
2. The LPF1 filter is available in high-performance mode only. If the gyroscope is configured in low-power mode, the LPF1 filter is bypassed.

In this configuration, the gyroscope ODR is selectable from 7.5 Hz up to 7.68 kHz. A low-pass filter (LPF1) is available, for more details about the filter characteristics see [Table 62. Gyroscope LPF1 + LPF2 bandwidth selection](#).

The digital LPF2 filter's cutoff frequency depends on the selected gyroscope ODR, as indicated in the following table.

**Table 20. Gyroscope LPF2 bandwidth selection**

| Gyroscope ODR [Hz] | LPF2 cutoff [Hz] |
|--------------------|------------------|
| 7.5                | 3.4              |
| 15                 | 6.6              |
| 30                 | 13.0             |
| 60                 | 24.6             |
| 120                | 49.4             |
| 240                | 96               |
| 480                | 187              |
| 960                | 342              |
| 1.92 kHz           | 491              |
| 3.84 kHz           | 528              |
| 7.68 kHz           | 537              |

**Note:** Data can be acquired from the output registers and FIFO over the primary I<sup>2</sup>C/MIPI I3C<sup>®</sup>/SPI interface.



## 6.8 Enhanced EIS

The LSM6DSVE offers advanced design flexibility for EIS applications: enhanced EIS functionality has a dedicated channel and processing with independent filtering.

Enhanced EIS main features:

- Enhanced EIS channel gyroscope data can be read over the primary interfaces through I<sup>2</sup>C / MIPI I3C<sup>®</sup> / SPI.
- EIS data are available in free-run mode in the output registers (UI\_OUTX\_L\_G\_EIS (2Eh) and UI\_OUTX\_H\_G\_EIS (2Fh) through UI\_OUTZ\_L\_G\_EIS (32h) and UI\_OUTZ\_H\_G\_EIS (33h)) or in FIFO (by setting the G\_EIS\_FIFO\_EN bit to 1 in the FIFO\_CTRL4 (0Ah) register) with dedicated TAG and timestamp configurable using FIFO\_CTRL4 (0Ah).

**Figure 21. Gyroscope enhanced EIS and UI block diagram**



When enhanced EIS mode is activated through the ODR\_EIS\_[1:0] bits in the CTRL\_EIS (6Bh) register:

- Gyroscope UI can be configured only in power-down mode or high-performance mode.
- Gyroscope EIS full scale can be selected by using the FS\_G\_EIS\_[2:0] bits in the CTRL\_EIS (6Bh) register.
- Gyroscope EIS data rate selectable at 1.92 kHz or 960 Hz configurable through the ODR\_G\_EIS\_[1:0] bits in the CTRL\_EIS (6Bh) register.
- LPF\_EIS low-pass filter (refer to Figure 21) bandwidth selection can be configured through the LPF\_G\_EIS\_BW bit in the CTRL\_EIS (6Bh) register.



## 6.9 FIFO

The presence of a FIFO allows consistent power saving for the system since the host processor does not need continuously poll data from the sensor, but it can wake up only when needed and burst the significant data out from the FIFO.

The LSM6DSVE embeds 1.5 KB of data in FIFO (up to 4.5 KB with the compression feature enabled) to store the following data:

- Gyroscope
- Accelerometer
- External sensors (up to four)
- Step counter
- Timestamp
- Temperature
- SFLP output data (quaternion, gyroscope bias, gravity vector)

The applications have maximum flexibility in choosing the rate of batching for physical sensors with FIFO-dedicated configurations: accelerometer, gyroscope, and temperature sensor batch rates can be selected by the user. Writing external sensors in the FIFO can be triggered depending on the sensor hub rate chosen or by an external sensor interrupt. The step counter can be stored in FIFO with an associated timestamp each time a step is detected. Results from SFLP (quaternion, gyroscope bias, gravity vector) can be stored in the FIFO at the rate selected for the algorithm. Moreover, MLC features, filters, and results can be enabled to be stored in the FIFO. It is possible to select decimation for timestamp batching in FIFO with a factor of 1, 8, or 32.

The reconstruction of a FIFO stream is a simple task thanks to the FIFO\_DATA\_OUT\_TAG byte that allows recognizing the meaning of a word in FIFO.

FIFO allows correct reconstruction of the timestamp information for each sensor stored in FIFO. If a change in the ODR or BDR (batch data rate) configuration is performed, the application can correctly reconstruct the timestamp and know exactly when the change was applied without disabling FIFO batching. FIFO stores information of the new configuration and timestamp in which the change was applied in the device.

Finally, FIFO embeds a compression algorithm that the user can enable in order to have up to 4.5 KB data stored in FIFO and take advantage of interface communication length for FIFO flushing and communication power consumption.

The programmable FIFO watermark threshold can be set using the WTM[7:0] bits in the FIFO\_CTRL1 (07h) register. To monitor the FIFO status, dedicated registers (FIFO\_STATUS1 (1Bh), FIFO\_STATUS2 (1Ch)) can be read to detect FIFO overrun events, FIFO full status, FIFO empty status, FIFO watermark status, and the number of unread samples stored in the FIFO. To generate dedicated interrupts on the INT1 and INT2 pins of these status events, the configuration can be set in INT1\_CTRL (0Dh) and INT2\_CTRL (0Eh).

The FIFO buffer can be configured according to seven different modes:

- Bypass mode
- FIFO mode
- Continuous mode
- Continuous-to-FIFO mode
- ContinuousWTM-to-full mode
- Bypass-to-continuous mode
- Bypass-to-FIFO mode

Each mode is selected by the FIFO\_MODE\_[2:0] bits in the FIFO\_CTRL4 (0Ah) register.

### 6.9.1 Bypass mode

In bypass mode (FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0] = 000), the FIFO is not operational and it remains empty. Bypass mode is also used to reset the FIFO when in FIFO mode.



### 6.9.2 FIFO mode

In FIFO mode (FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0] = 001) data from the output channels are stored in the FIFO until it is full.

To reset FIFO content, bypass mode should be selected by writing FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0]) to 000. After this reset command, it is possible to restart FIFO mode by writing FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0]) to 001.

The FIFO buffer memorizes up to 4.5 KB of data (with compression enabled) but the depth of the FIFO can be resized by setting the WTM[7:0] bits in FIFO\_CTRL1 (07h). If the STOP\_ON\_WTM bit in FIFO\_CTRL2 (08h) is set to 1, FIFO depth is limited up to the WTM[7:0] bits in the FIFO\_CTRL1 (07h) register.

### 6.9.3 Continuous mode

Continuous mode (FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0] = 110) provides a continuous FIFO update: as new data arrives, the older data is discarded.

A FIFO threshold flag FIFO\_STATUS2 (1Ch)(FIFO\_WTM\_IA) is asserted when the number of unread samples in FIFO is greater than or equal to FIFO\_CTRL1 (07h) (WTM[7:0]).

It is possible to route the FIFO\_WTM\_IA flag to the INT1 pin by writing in register INT1\_CTRL (0Dh) (INT1\_FIFO\_TH) = 1 or to the INT2 pin by writing in register INT2\_CTRL (0Eh) (INT2\_FIFO\_TH) = 1.

A full-flag interrupt can be enabled, INT1\_CTRL (0Dh) (INT1\_FIFO\_FULL) = 1 or INT2\_CTRL (0Eh) (INT2\_FIFO\_FULL) = 1, in order to indicate FIFO saturation and eventually read its content all at once.

If an overrun occurs, at least one of the oldest samples in FIFO has been overwritten and the FIFO\_OVR\_IA flag in FIFO\_STATUS2 (1Ch) is asserted.

In order to empty the FIFO before it is full, it is also possible to pull from FIFO the number of unread samples available in FIFO\_STATUS1 (1Bh) and FIFO\_STATUS2 (1Ch) (DIFF\_FIFO\_[8:0]).

### 6.9.4 Continuous-to-FIFO mode

In continuous-to-FIFO mode (FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0] = 011), FIFO behavior changes according to the trigger event detected in one of the following interrupt events:

- Single tap
- Double tap
- Wake-up
- Free-fall
- D6D

When the selected trigger bit is equal to 1, FIFO operates in FIFO mode.

When the selected trigger bit is equal to 0, FIFO operates in continuous mode.

### 6.9.5 ContinuousWTM-to-full mode

In continuousWTM-to-full mode (FIFO\_CTRL4 (0Ah)(FIFO\_MODE\_[2:0] = 010), FIFO behavior changes according to the trigger event detected in one of the following interrupt events:

- Single tap
- Double tap
- Wake-up
- Free-fall
- D6D

When the selected trigger bit is equal to 0, FIFO operates in continuous mode with the FIFO size limited to the FIFO watermark level (defined by the WTM[7:0] bits in the FIFO\_CTRL1 (07h) register).

When the selected trigger bit is equal to 1, FIFO continues to store data until it is full.



### 6.9.6 Bypass-to-continuous mode

In bypass-to-continuous mode (`FIFO_CTRL4 (0Ah)`(`FIFO_MODE_[2:0] = 100`), data measurement storage inside FIFO operates in Continuous mode when selected triggers are equal to 1, otherwise FIFO content is reset (bypass mode).

FIFO behavior changes according to the trigger event detected in one of the following interrupt events:

- Single tap
- Double tap
- Wake-up
- Free-fall
- D6D

### 6.9.7 Bypass-to-FIFO mode

In bypass-to-FIFO mode (`FIFO_CTRL4 (0Ah)`(`FIFO_MODE_[2:0] = 111`), data measurement storage inside FIFO operates in FIFO mode when selected triggers are equal to 1, otherwise FIFO content is reset (bypass mode).

FIFO behavior changes according to the trigger event detected in one of the following interrupt events:

- Single tap
- Double tap
- Wake-up
- Free-fall
- D6D

### 6.9.8 FIFO reading procedure

The data stored in FIFO are accessible from dedicated registers and each FIFO word is composed of 7 bytes: one tag byte (`FIFO_DATA_OUT_TAG (78h)`), in order to identify the sensor, and 6 bytes of fixed data (`FIFO_DATA_OUT` registers from `(79h)` to `(7Eh)`).

The `DIFF_FIFO_[8:0]` field in the `FIFO_STATUS1 (1Bh)` and `FIFO_STATUS2 (1Ch)` registers contains the number of words (1 byte TAG + 6 bytes DATA) collected in FIFO.

In addition, it is possible to configure a counter of the batch events of accelerometer or gyroscope sensors. The flag `COUNTER_BDR_IA` in `FIFO_STATUS2 (1Ch)` alerts that the counter reaches a selectable threshold (`CNT_BDR_TH_[9:0]` field in `COUNTER_BDR_REG1 (0Bh)` and `COUNTER_BDR_REG2 (0Ch)`). This allows triggering the reading of FIFO with the desired latency of one single sensor. The sensor is selectable using the `TRIG_COUNTER_BDR_[1:0]` bits in `COUNTER_BDR_REG1 (0Bh)`. As for the other FIFO status events, the flag `COUNTER_BDR_IA` can be routed on the INT1 or INT2 pins by asserting the corresponding bits (`INT1_CNT_BDR` of `INT1_CTRL (0Dh)` and `INT2_CNT_BDR` of `INT2_CTRL (0Eh)`).

In order to maximize the amount of accelerometer and gyroscope data in FIFO, the user can enable the compression algorithm by setting to 1 both the `FIFO_COMPR_EN` bit in `EMB_FUNC_EN_B (05h)` (embedded functions registers bank) and the `FIFO_COMPR_RT_EN` bit in `FIFO_CTRL2 (08h)`. When compression is enabled, it is also possible to force writing noncompressed data at a selectable rate using the `UNCOMPR_RATE_[1:0]` field in `FIFO_CTRL2 (08h)`.

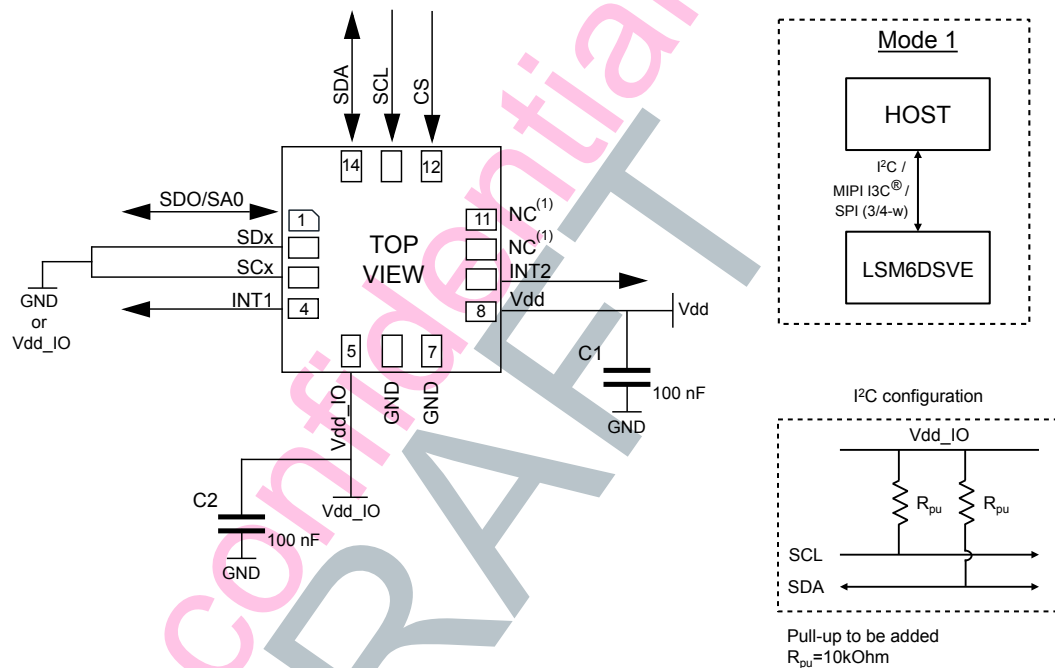
Meta information about accelerometer and gyroscope sensor configuration changes can be managed by enabling the `ODR_CHG_EN` bit in `FIFO_CTRL2 (08h)`.



## 7 Application hints

### 7.1 LSM6DSVE electrical connections in mode 1

Figure 22. LSM6DSVE electrical connections in mode 1



1. Leave pin electrically unconnected and soldered to PCB.

The device core is supplied through the Vdd line. Power supply decoupling capacitors (C1, C2 = 100 nF ceramic) should be placed as near as possible to the supply pin of the device (common design practice).

The functionality of the device and the measured acceleration/angular rate data is selectable and accessible through the SPI/I<sup>2</sup>C/MIPI I3C<sup>®</sup> interface.

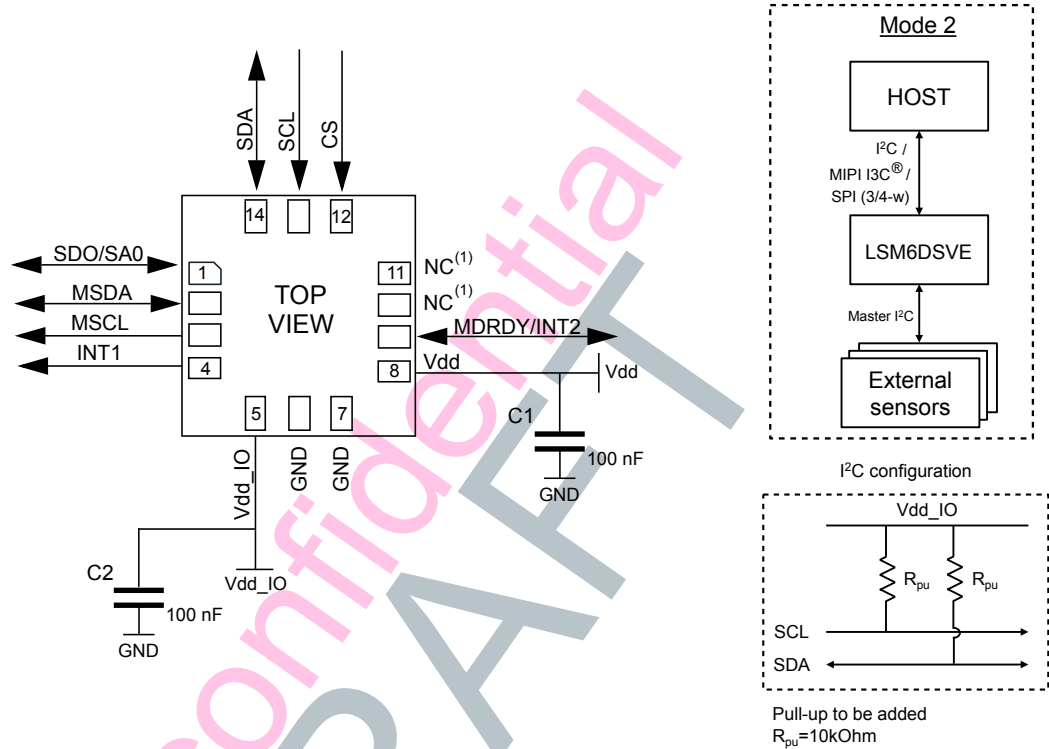
The functions, the threshold, and the timing of the two interrupt pins for each sensor can be completely programmed by the user through the SPI/I<sup>2</sup>C/MIPI I3C<sup>®</sup> interface.





## 7.2 LSM6DSVE electrical connections in mode 2

Figure 23. LSM6DSVE electrical connections in mode 2



1. Leave pin electrically unconnected and soldered to PCB.

The device core is supplied through the Vdd line. Power supply decoupling capacitors (C1, C2 = 100 nF ceramic) should be placed as near as possible to the supply pin of the device (common design practice).

The functionality of the device and the measured acceleration/angular rate data is selectable and accessible through the SPI/I<sup>2</sup>C/MIPI I3C® primary interface.

The functions, the threshold, and the timing of the two interrupt pins for each sensor can be completely programmed by the user through the SPI/I<sup>2</sup>C/MIPI I3C® primary interface.

Table 21. Internal pin status

| pin# | Name   | Mode 1 function                                                                                                                                              | Mode 2 function                                                                                                                                              | Pin status mode 1                                                                                       | Pin status mode 2                                                                                       |
|------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | SDO    | SPI 4-wire interface serial data output (SDO)                                                                                                                | SPI 4-wire interface serial data output (SDO)                                                                                                                | Default: input without pull-up<br>Pull-up is enabled if bit SDO_PU_EN = 1 in register PIN_CTRL (02h).   | Default: input without pull-up<br>Pull-up is enabled if bit SDO_PU_EN = 1 in register PIN_CTRL (02h).   |
|      | SA0    | I <sup>2</sup> C least significant bit of the device address (SA0)<br>MIPI I3C <sup>®</sup> least significant bit of the static address (SA0)                | I <sup>2</sup> C least significant bit of the device address (SA0)<br>MIPI I3C <sup>®</sup> least significant bit of the static address (SA0)                |                                                                                                         |                                                                                                         |
| 2    | SDx    | Connect to Vdd_IO or GND                                                                                                                                     | I <sup>2</sup> C serial data master (MSDA)                                                                                                                   | Default: input without pull-up<br>Pull-up is enabled if bit SHUB_PU_EN = 1 in register IF_CFG (03h).    | Default: input without pull-up<br>Pull-up is enabled if bit SHUB_PU_EN = 1 in register IF_CFG (03h).    |
| 3    | SCx    | Connect to Vdd_IO or GND                                                                                                                                     | I <sup>2</sup> C serial clock master (MSCL)                                                                                                                  | Default: input without pull-up<br>Pull-up is enabled if bit SHUB_PU_EN = 1 in register IF_CFG (03h).    | Default: input without pull-up<br>Pull-up is enabled if bit SHUB_PU_EN = 1 in register IF_CFG (03h).    |
| 4    | INT1   | Programmable interrupt 1                                                                                                                                     | Programmable interrupt 1                                                                                                                                     | Default: output forced to ground                                                                        | Default: output forced to ground                                                                        |
| 5    | Vdd_IO | Power supply for I/O pins                                                                                                                                    | Power supply for I/O pins                                                                                                                                    |                                                                                                         |                                                                                                         |
| 6    | GND    | 0 V supply                                                                                                                                                   | 0 V supply                                                                                                                                                   |                                                                                                         |                                                                                                         |
| 7    | GND    | 0 V supply                                                                                                                                                   | 0 V supply                                                                                                                                                   |                                                                                                         |                                                                                                         |
| 8    | Vdd    | Power supply                                                                                                                                                 | Power supply                                                                                                                                                 |                                                                                                         |                                                                                                         |
| 9    | INT2   | Programmable interrupt 2 (INT2) / Data enabled (DEN)                                                                                                         | Programmable interrupt 2 (INT2) / Data enabled (DEN) / I <sup>2</sup> C master external synchronization signal (MDRDY)                                       | Default: output forced to ground                                                                        | Default: output forced to ground                                                                        |
| 10   | NC     | Connect to Vdd_IO or leave unconnected                                                                                                                       | Connect to Vdd_IO or leave unconnected                                                                                                                       |                                                                                                         |                                                                                                         |
| 11   | NC     | Connect to Vdd_IO or leave unconnected                                                                                                                       | Connect to Vdd_IO or leave unconnected                                                                                                                       |                                                                                                         |                                                                                                         |
| 12   | CS     | I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) | I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) | Default: input with pull-up<br>Pull-up is disabled if bit I2C_I3C_disable = 1 in register IF_CFG (03h). | Default: input with pull-up<br>Pull-up is disabled if bit I2C_I3C_disable = 1 in register IF_CFG (03h). |
| 13   | SCL    | I <sup>2</sup> C/MIPI I3C <sup>®</sup> serial clock (SCL) / SPI serial port clock (SPC)                                                                      | I <sup>2</sup> C/MIPI I3C <sup>®</sup> serial clock (SCL) / SPI serial port clock (SPC)                                                                      | Default: input without pull-up                                                                          | Default: input without pull-up                                                                          |
| 14   | SDA    | I <sup>2</sup> C/MIPI I3C <sup>®</sup> serial data (SDA) / SPI serial data input (SDI) / 3-wire interface serial data output (SDO)                           | I <sup>2</sup> C/MIPI I3C <sup>®</sup> serial data (SDA) / SPI serial data input (SDI) / 3-wire interface serial data output (SDO)                           | Default: input without pull-up<br>Pull-up is enabled if bit SDA_PU_EN = 1 in register IF_CFG (03h).     | Default: input without pull-up<br>Pull-up is enabled if bit SDA_PU_EN = 1 in register IF_CFG (03h).     |

Internal pull-up value is from 30 kΩ to 50 kΩ, depending on Vdd\_IO.





## 8 Register mapping

The table given below provides a list of the 8/16-bit registers embedded in the device and the corresponding addresses.

All these registers are accessible from the primary SPI/I<sup>2</sup>C/MIPI I3C<sup>®</sup> interface only.

**Table 22. Registers address map**

| Name             | Type | Register address |           | Default  | Comment |
|------------------|------|------------------|-----------|----------|---------|
|                  |      | Hex              | Binary    |          |         |
| FUNC_CFG_ACCESS  | R/W  | 01               | 00000001  | 00000000 |         |
| PIN_CTRL         | R/W  | 02               | 00000010  | 00100011 |         |
| IF_CFG           | R/W  | 03               | 00000011  | 00000000 |         |
| RESERVED         | -    | 04-05            |           |          |         |
| ODR_TRIG_CFG     | R/W  | 06               | 00000110  | 00000000 |         |
| FIFO_CTRL1       | R/W  | 07               | 00000111  | 00000000 |         |
| FIFO_CTRL2       | R/W  | 08               | 00001000  | 00000000 |         |
| FIFO_CTRL3       | R/W  | 09               | 00001001  | 00000000 |         |
| FIFO_CTRL4       | R/W  | 0A               | 00001010  | 00000000 |         |
| COUNTER_BDR_REG1 | R/W  | 0B               | 00001011  | 00000000 |         |
| COUNTER_BDR_REG2 | R/W  | 0C               | 00001100  | 00000000 |         |
| INT1_CTRL        | R/W  | 0D               | 00001101  | 00000000 |         |
| INT2_CTRL        | R/W  | 0E               | 00001110  | 00000000 |         |
| WHO_AM_I         | R    | 0F               | 00001111  | 01110000 |         |
| CTRL1            | R/W  | 10               | 00010000  | 00000000 |         |
| CTRL2            | R/W  | 11               | 00010001  | 00000000 |         |
| CTRL3            | R/W  | 12               | 00010010  | 01000100 |         |
| CTRL4            | R/W  | 13               | 00010011  | 00000000 |         |
| CTRL5            | R/W  | 14               | 00010100  | 00000000 |         |
| CTRL6            | R/W  | 15               | 00010101  | 00000000 |         |
| CTRL7            | R/W  | 16               | 00010110  | 00000000 |         |
| CTRL8            | R/W  | 17               | 0001 0111 | 00000000 |         |
| CTRL9            | R/W  | 18               | 00011000  | 00000000 |         |
| CTRL10           | R/W  | 19               | 00011001  | 00000000 |         |
| CTRL_STATUS      | R    | 1A               | 00011010  | output   |         |
| FIFO_STATUS1     | R    | 1B               | 00011011  | output   |         |
| FIFO_STATUS2     | R    | 1C               | 00011100  | output   |         |
| ALL_INT_SRC      | R    | 1D               | 00011101  | output   |         |
| STATUS_REG       | R    | 1E               | 00011110  | output   |         |
| RESERVED         | -    | 1F               |           |          |         |
| OUT_TEMP_L       | R    | 20               | 00100000  | output   |         |
| OUT_TEMP_H       | R    | 21               | 00100001  | output   |         |
| OUTX_L_G         | R    | 22               | 00100010  | output   |         |
| OUTX_H_G         | R    | 23               | 00100011  | output   |         |



| Name                     | Type | Register address |          | Default  | Comment |
|--------------------------|------|------------------|----------|----------|---------|
|                          |      | Hex              | Binary   |          |         |
| OUTY_L_G                 | R    | 24               | 00100100 | output   |         |
| OUTY_H_G                 | R    | 25               | 00100101 | output   |         |
| OUTZ_L_G                 | R    | 26               | 00100110 | output   |         |
| OUTZ_H_G                 | R    | 27               | 00100111 | output   |         |
| OUTX_L_A                 | R    | 28               | 00101000 | output   |         |
| OUTX_H_A                 | R    | 29               | 00101001 | output   |         |
| OUTY_L_A                 | R    | 2A               | 00101010 | output   |         |
| OUTY_H_A                 | R    | 2B               | 00101011 | output   |         |
| OUTZ_L_A                 | R    | 2C               | 00101100 | output   |         |
| OUTZ_H_A                 | R    | 2D               | 00101101 | output   |         |
| UI_OUTX_L_G_EIS          | R    | 2E               | 00101110 | output   |         |
| UI_OUTX_H_G_EIS          | R    | 2F               | 00101111 | output   |         |
| UI_OUTY_L_G_EIS          | R    | 30               | 00110000 | output   |         |
| UI_OUTY_H_G_EIS          | R    | 31               | 00110001 | output   |         |
| UI_OUTZ_L_G_EIS          | R    | 32               | 00110010 | output   |         |
| UI_OUTZ_H_G_EIS          | R    | 33               | 00110011 | output   |         |
| UI_OUTX_L_A_DualC        | R    | 34               | 00110100 | output   |         |
| UI_OUTX_H_A_DualC        | R    | 35               | 00110101 | output   |         |
| UI_OUTY_L_A_DualC        | R    | 36               | 00110110 | output   |         |
| UI_OUTY_H_A_DualC        | R    | 37               | 00110111 | output   |         |
| UI_OUTZ_L_A_DualC        | R    | 38               | 00111000 | output   |         |
| UI_OUTZ_H_A_DualC        | R    | 39               | 00111001 | output   |         |
| RESERVED                 | -    | 3A-3F            |          |          |         |
| TIMESTAMP0               | R    | 40               | 01000000 | output   |         |
| TIMESTAMP1               | R    | 41               | 01000001 | output   |         |
| TIMESTAMP2               | R    | 42               | 01000010 | output   |         |
| TIMESTAMP3               | R    | 43               | 01000011 | output   |         |
| UI_STATUS_REG            | R    | 44               | 01000100 | output   |         |
| WAKE_UP_SRC              | R    | 45               | 01000101 | output   |         |
| TAP_SRC                  | R    | 46               | 01000110 | output   |         |
| D6D_SRC                  | R    | 47               | 01000111 | output   |         |
| STATUS_MASTER_MAINPAGE   | R    | 48               | 01001000 | output   |         |
| EMB_FUNC_STATUS_MAINPAGE | R    | 49               | 01001001 | output   |         |
| FSM_STATUS_MAINPAGE      | R    | 4A               | 01001010 | output   |         |
| RESERVED                 | -    | 4B-4E            |          |          |         |
| INTERNAL_FREQ_FINE       | R    | 4F               | 01001111 | output   |         |
| FUNCTIONS_ENABLE         | RW   | 50               | 01010000 | 00000000 |         |
| DEN                      | R/W  | 51               | 01010001 | 00001110 |         |
| INACTIVITY_DUR           | R/W  | 54               | 01010100 | 00000100 |         |



| Name              | Type | Register address |          | Default  | Comment |
|-------------------|------|------------------|----------|----------|---------|
|                   |      | Hex              | Binary   |          |         |
| INACTIVITY_THS    | R/W  | 55               | 01010101 | 00000000 |         |
| TAP_CFG0          | R/W  | 56               | 01010110 | 00000000 |         |
| TAP_CFG1          | R/W  | 57               | 01010111 | 00000000 |         |
| TAP_CFG2          | R/W  | 58               | 01011000 | 00000000 |         |
| TAP_THS_6D        | R/W  | 59               | 01011001 | 00000000 |         |
| TAP_DUR           | R/W  | 5A               | 01011010 | 00000000 |         |
| WAKE_UP_THS       | R/W  | 5B               | 01011011 | 00000000 |         |
| WAKE_UP_DUR       | R/W  | 5C               | 01011100 | 00000000 |         |
| FREE_FALL         | R/W  | 5D               | 01011101 | 00000000 |         |
| MD1_CFG           | R/W  | 5E               | 01011110 | 00000000 |         |
| MD2_CFG           | R/W  | 5F               | 01011111 | 00000000 |         |
| RESERVED          | -    | 60-61            |          |          |         |
| HAODR_CFG         | R/W  | 62               | 01100010 | 00000000 |         |
| EMB_FUNC_CFG      | R/W  | 63               | 01100011 | 00000000 |         |
| RESERVED          | -    | 64 - 6A          |          |          |         |
| CTRL_EIS          | R/W  | 6B               | 01101011 | 00000000 |         |
| RESERVED          | -    | 6C - 72          |          |          |         |
| X_OFS_USR         | R/W  | 73               | 01110011 | 00000000 |         |
| Y_OFS_USR         | R/W  | 74               | 01110100 | 00000000 |         |
| Z_OFS_USR         | R/W  | 75               | 01110101 | 00000000 |         |
| RESERVED          | -    | 76-77            |          |          |         |
| FIFO_DATA_OUT_TAG | R    | 78               | 01111000 | output   |         |
| FIFO_DATA_OUT_X_L | R    | 79               | 01111001 | output   |         |
| FIFO_DATA_OUT_X_H | R    | 7A               | 01111010 | output   |         |
| FIFO_DATA_OUT_Y_L | R    | 7B               | 01111011 | output   |         |
| FIFO_DATA_OUT_Y_H | R    | 7C               | 01111100 | output   |         |
| FIFO_DATA_OUT_Z_L | R    | 7D               | 01111101 | output   |         |
| FIFO_DATA_OUT_Z_H | R    | 7E               | 01111110 | output   |         |

Reserved registers must not be changed. Writing to those registers may cause permanent damage to the device. The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.



## 9 Register description

The device contains a set of registers which are used to control its behavior and to retrieve linear acceleration, angular rate, and temperature data. The register addresses, made up of 7 bits, are used to identify them and to write the data through the serial interface.

### 9.1 FUNC\_CFG\_ACCESS (01h)

Enable embedded functions register (R/W)

**Table 23. FUNC\_CFG\_ACCESS register**

|                     |                 |                  |                  |                |        |                  |                  |
|---------------------|-----------------|------------------|------------------|----------------|--------|------------------|------------------|
| EMB_FUNC_REG_ACCESS | SHUB_REG_ACCESS | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | FSM_WR_CTRL_EN | SW_POR | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|---------------------|-----------------|------------------|------------------|----------------|--------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 24. FUNC\_CFG\_ACCESS register description**

|                     |                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMB_FUNC_REG_ACCESS | Enables access to the embedded functions configuration registers. <sup>(1)</sup> Default value: 0                                                               |
| SHUB_REG_ACCESS     | Enables access to the sensor hub (I <sup>2</sup> C master) configuration registers. <sup>(2)</sup> Default value: 0                                             |
| FSM_WR_CTRL_EN      | Enables the control of the CTRL registers to FSM (FSM can change some configurations of the device autonomously). Default value: 0<br>(0: disabled; 1: enabled) |
| SW_POR              | Global reset of the device. Default value: 0                                                                                                                    |

1. Details concerning the embedded functions configuration registers are available in [Section 10: Embedded functions register mapping](#) and [Section 11: Embedded functions register description](#).
2. Details concerning the sensor hub registers are available in [Section 14: Sensor hub register mapping](#) and [Section 15: Sensor hub register description](#).



## 9.2

### PIN\_CTRL (02h)

SDO pin pull-up register (R/W). This register is not reset during the software reset procedure (see bit 0 of the CTRL3 (12h) register).

**Table 25. PIN\_CTRL register**

|                  |           |             |                  |                  |                  |                   |                   |
|------------------|-----------|-------------|------------------|------------------|------------------|-------------------|-------------------|
| 0 <sup>(1)</sup> | SDO_PU_EN | IBHR_POR_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | IO_PAD_STRENGTH_1 | IO_PAD_STRENGTH_0 |
|------------------|-----------|-------------|------------------|------------------|------------------|-------------------|-------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 26. PIN\_CTRL register description**

|                      |                                                                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDO_PU_EN            | Enables pull-up on SDO pin. For details, refer to Table 21. Default value: 0<br>(0: SDO pin pull-up disconnected; 1: SDO pin with pull-up)                                                                                                                                                           |
| IBHR_POR_EN          | Selects the action that the device performs after "reset whole chip" I3C pattern. Default value: 1<br>(0: configuration reset (software reset + dynamic address reset);<br>1: global reset (POR reset))                                                                                              |
| IO_PAD_STRENGTH[1:0] | Defines the drive strength for the interrupt pads. Default value: 11<br>(00: lowest strength (recommended for Vdd_IO ≥ 3.0 V);<br>01: intermediate strength (recommended for Vdd_IO ≥ 2.0 V and Vdd_IO < 3 V);<br>10: reserved (do not use);<br>11: highest strength (recommended for Vdd_IO < 2 V)) |



### 9.3 IF\_CFG (03h)

Interface configuration register (R/W). This register is not reset during the software reset procedure (see bit 0 of the CTRL3 (12h) register).

**Table 27. IF\_CFG register**

|           |            |          |           |       |     |                  |                 |
|-----------|------------|----------|-----------|-------|-----|------------------|-----------------|
| SDA_PU_EN | SHUB_PU_EN | ASF_CTRL | H_LACTIVE | PP_OD | SIM | 0 <sup>(1)</sup> | I2C_I3C_disable |
|-----------|------------|----------|-----------|-------|-----|------------------|-----------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 28. IF\_CFG register description**

|                 |                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDA_PU_EN       | Enables pull-up on SDA pin. Default value: 0<br>(0: SDA pin pull-up disconnected;<br>1: SDA pin with pull-up)                                                                                                                     |
| SHUB_PU_EN      | Enables master I <sup>2</sup> C pull-up. Default value: 0<br>(0: internal pull-up on auxiliary I <sup>2</sup> C line disabled;<br>1: internal pull-up on auxiliary I <sup>2</sup> C line enabled)                                 |
| ASF_CTRL        | Enables antispikes filters. Default value: 0<br>(0: antispikes filters are managed by the protocol and turned off after the broadcast address;<br>1: antispikes filters on SCL and SDA lines are always enabled)                  |
| H_LACTIVE       | Interrupt activation level. Default value: 0<br>(0: interrupt output pins active high;<br>1: interrupt output pins active low)                                                                                                    |
| PP_OD           | Push-pull/open-drain selection on INT1 and INT2 pins. Default value: 0<br>(0: push-pull mode;<br>1: open-drain mode)                                                                                                              |
| SIM             | SPI serial interface mode selection. Default value: 0<br>(0: 4-wire interface;<br>1: 3-wire interface)                                                                                                                            |
| I2C_I3C_disable | Disables I <sup>2</sup> C and MIPI I3C <sup>®</sup> interfaces. Default value: 0<br>(0: SPI, I <sup>2</sup> C and MIPI I3C <sup>®</sup> interfaces enabled;<br>1: I <sup>2</sup> C and MIPI I3C <sup>®</sup> interfaces disabled) |

### 9.4 ODR\_TRIG\_CFG (06h)

ODR-triggered mode configuration register (R/W)

**Table 29. ODR\_TRIG\_CFG register**

|                 |                 |                 |                 |                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| ODR_TRIG_NODR_7 | ODR_TRIG_NODR_6 | ODR_TRIG_NODR_5 | ODR_TRIG_NODR_4 | ODR_TRIG_NODR_3 | ODR_TRIG_NODR_2 | ODR_TRIG_NODR_1 | ODR_TRIG_NODR_0 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|

**Table 30. ODR\_TRIG\_CFG register description**

|                     |                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ODR_TRIG_NODR_[7:0] | When ODR-triggered mode is set, these bits are used to define the number of data generated in the reference period.<br>Allowed values for ODR_TRIG_NODR_[7:0] are 0 (default) and values in the range from 4 to 255. |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





## 9.5 FIFO\_CTRL1 (07h)

FIFO control register 1 (R/W)

**Table 31. FIFO\_CTRL1 register**

|       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
| WTM_7 | WTM_6 | WTM_5 | WTM_4 | WTM_3 | WTM_2 | WTM_1 | WTM_0 |
|-------|-------|-------|-------|-------|-------|-------|-------|

**Table 32. FIFO\_CTRL1 register description**

|           |                                                                                                                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WTM_[7:0] | FIFO watermark threshold: 1 LSB = TAG (1 byte) + 1 sensor (6 bytes) written in FIFO.<br>Watermark flag rises when the number of bytes written in the FIFO is greater than or equal to the threshold level. |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.6 FIFO\_CTRL2 (08h)

FIFO control register 2 (R/W)

**Table 33. FIFO\_CTRL2 register**

|             |                  |                  |            |                  |                |                |                         |
|-------------|------------------|------------------|------------|------------------|----------------|----------------|-------------------------|
| STOP_ON_WTM | FIFO_COMPR_RT_EN | 0 <sup>(1)</sup> | ODR_CHG_EN | 0 <sup>(1)</sup> | UNCOMPR_RATE_1 | UNCOMPR_RATE_0 | XL_DualC_BATCH_FROM_FSM |
|-------------|------------------|------------------|------------|------------------|----------------|----------------|-------------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 34. FIFO\_CTRL2 register description**

|                                 |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STOP_ON_WTM                     | Sensing chain FIFO stop values memorization at threshold level. Default value: 0<br>(0: FIFO depth is not limited;<br>1: FIFO depth is limited to threshold level, defined in <a href="#">FIFO_CTRL1 (07h)</a> )                                                                                                                                         |
| FIFO_COMPR_RT_EN <sup>(1)</sup> | Enables/disables compression algorithm runtime. Default value: 0<br>(0: FIFO compression algorithm disabled;<br>1: FIFO compression algorithm enabled)                                                                                                                                                                                                   |
| ODR_CHG_EN                      | Enables ODR CHANGE virtual sensor to be batched in FIFO. Default value: 0<br>(0: ODR CHANGE virtual sensor not batched in FIFO;<br>1: ODR CHANGE virtual sensor batched in FIFO)<br><i>Note: Refer to the product application note for the details regarding operating/power mode configurations, settings, turn-on/off time and on-the-fly changes.</i> |
| UNCOMPR_RATE_[1:0]              | This field configures the compression algorithm to write uncompressed data at each rate.<br>(0: uncompressed data writing is not forced (default);<br>1: uncompressed data every 8 batch data rate;<br>2: uncompressed data every 16 batch data rate;<br>3: uncompressed data every 32 batch data rate)                                                  |
| XL_DualC_BATCH_FROM_FSM         | When dual-channel mode is enabled, this bit enables FSM-triggered batching in FIFO of accelerometer channel 2. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                                             |

1. This bit is activated if the FIFO\_COMPR\_EN bit of [EMB\\_FUNC\\_EN\\_B \(05h\)](#) is set to 1.



## 9.7

### FIFO\_CTRL3 (09h)

FIFO control register 3 (R/W)

**Table 35. FIFO\_CTRL3 register**

| BDR_GY_3 | BDR_GY_2 | BDR_GY_1 | BDR_GY_0 | BDR_XL_3 | BDR_XL_2 | BDR_XL_1 | BDR_XL_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 36. FIFO\_CTRL3 register description**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BDR_GY_[3:0] | <p>Selects batch data rate (write frequency in FIFO) for gyroscope data.</p> <p>(0000: gyroscope not batched in FIFO (default);</p> <p>0001: 1.875 Hz;</p> <p>0010: 7.5 Hz;</p> <p>0011: 15 Hz;</p> <p>0100: 30 Hz;</p> <p>0101: 60 Hz;</p> <p>0110: 120 Hz;</p> <p>0111: 240 Hz;</p> <p>1000: 480 Hz;</p> <p>1001: 960 Hz;</p> <p>1010: 1.92 kHz;</p> <p>1011: 3.84 kHz;</p> <p>1100: 7.68 kHz</p> <p>1101-1111: reserved)</p>         |
| BDR_XL_[3:0] | <p>Selects batch data rate (write frequency in FIFO) for accelerometer data.</p> <p>(0000: accelerometer not batched in FIFO (default);</p> <p>0001: 1.875 Hz;</p> <p>0010: 7.5 Hz;</p> <p>0011: 15 Hz;</p> <p>0100: 30 Hz;</p> <p>0101: 60 Hz;</p> <p>0110: 120 Hz;</p> <p>0111: 240 Hz;</p> <p>1000: 480 Hz;</p> <p>1001: 960 Hz;</p> <p>1010: 1.92 kHz;</p> <p>1011: 3.84 kHz;</p> <p>1100: 7.68 kHz</p> <p>1101-1111: reserved)</p> |



## 9.8

### FIFO\_CTRL4 (0Ah)

FIFO control register 4 (R/W)

**Table 37. FIFO\_CTRL4 register**

|                |                |               |               |               |             |             |             |
|----------------|----------------|---------------|---------------|---------------|-------------|-------------|-------------|
| DEC_TS_BATCH_1 | DEC_TS_BATCH_0 | ODR_T_BATCH_1 | ODR_T_BATCH_0 | G_EIS_FIFO_EN | FIFO_MODE_2 | FIFO_MODE_1 | FIFO_MODE_0 |
|----------------|----------------|---------------|---------------|---------------|-------------|-------------|-------------|

**Table 38. FIFO\_CTRL4 register description**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC_TS_BATCH_[1:0] | <p>Selects decimation for timestamp batching in FIFO. Write rate is the maximum rate between the accelerometer and gyroscope BDR divided by decimation decoder.</p> <p>(00: timestamp not batched in FIFO (default);</p> <p>01: decimation 1: <math>\max(\text{BDR\_XL}[\text{Hz}], \text{BDR\_GY}[\text{Hz}])</math> [Hz];</p> <p>10: decimation 8: <math>\max(\text{BDR\_XL}[\text{Hz}], \text{BDR\_GY}[\text{Hz}]) / 8</math> [Hz];</p> <p>11: decimation 32: <math>\max(\text{BDR\_XL}[\text{Hz}], \text{BDR\_GY}[\text{Hz}]) / 32</math> [Hz])</p>                                                                                                                                                                                |
| ODR_T_BATCH_[1:0]  | <p>Selects batch data rate (write frequency in FIFO) for temperature data</p> <p>(00: temperature not batched in FIFO (default);</p> <p>01: 1.875 Hz;</p> <p>10: 15 Hz;</p> <p>11: 60 Hz)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| G_EIS_FIFO_EN      | <p>Enables FIFO batching of enhanced EIS gyroscope output values. Default value: 0</p> <p>(0: disabled; 1: enabled)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| FIFO_MODE_[2:0]    | <p>FIFO mode selection</p> <p>(000: bypass mode: FIFO disabled (default);</p> <p>001: FIFO mode: stops collecting data when FIFO is full;</p> <p>010: continuousWTM-to-full mode: continuous mode with FIFO watermark size until trigger is deasserted, then data are stored in FIFO until the buffer is full;</p> <p>011: continuous-to-FIFO mode: continuous mode until trigger is deasserted, then FIFO mode;</p> <p>100: bypass-to-continuous mode: bypass mode until trigger is deasserted, then continuous mode;</p> <p>101: reserved;</p> <p>110: continuous mode: if the FIFO is full, the new sample overwrites the older one;</p> <p>111: bypass-to-FIFO mode: bypass mode until trigger is deasserted, then FIFO mode.)</p> |



## 9.9 COUNTER\_BDR\_REG1 (0Bh)

Counter batch data rate register 1 (R/W)

**Table 39. COUNTER\_BDR\_REG1 register**

|                  |                    |                    |                  |                  |                  |              |              |
|------------------|--------------------|--------------------|------------------|------------------|------------------|--------------|--------------|
| 0 <sup>(1)</sup> | TRIG_COUNTER_BDR_1 | TRIG_COUNTER_BDR_0 | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | CNT_BDR_TH_9 | CNT_BDR_TH_8 |
|------------------|--------------------|--------------------|------------------|------------------|------------------|--------------|--------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 40. COUNTER\_BDR\_REG1 register description**

|                        |                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRIG_COUNTER_BDR_[1:0] | Selects the trigger for the internal counter of batch events between the accelerometer, gyroscope and EIS gyroscope.<br>(00: accelerometer batch event;<br>01: gyroscope batch event;<br>10 – 11: gyroscope EIS batch event)                                                               |
| CNT_BDR_TH_[9:8]       | In conjunction with CNT_BDR_TH_[7:0] in <a href="#">COUNTER_BDR_REG2 (0Ch)</a> , sets the threshold for the internal counter of batch events. When this counter reaches the threshold, the counter is reset and the COUNTER_BDR_IA flag in <a href="#">FIFO_STATUS2 (1Ch)</a> is set to 1. |

## 9.10 COUNTER\_BDR\_REG2 (0Ch)

Counter batch data rate register 2 (R/W)

**Table 41. COUNTER\_BDR\_REG2 register**

|              |              |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CNT_BDR_TH_7 | CNT_BDR_TH_6 | CNT_BDR_TH_5 | CNT_BDR_TH_4 | CNT_BDR_TH_3 | CNT_BDR_TH_2 | CNT_BDR_TH_1 | CNT_BDR_TH_0 |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|

**Table 42. COUNTER\_BDR\_REG2 register description**

|                  |                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CNT_BDR_TH_[7:0] | In conjunction with CNT_BDR_TH_[9:8] in <a href="#">COUNTER_BDR_REG1 (0Bh)</a> , sets the threshold for the internal counter of batch events. When this counter reaches the threshold, the counter is reset and the COUNTER_BDR_IA flag in <a href="#">FIFO_STATUS2 (1Ch)</a> is set to 1. |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 9.11 INT1\_CTRL (0Dh)

INT1 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT1 when the MIPI I3C<sup>®</sup> dynamic address is not assigned (I<sup>2</sup>C or SPI is used). Some bits can be also used to trigger an IBI (in-band interrupt) when the MIPI I3C<sup>®</sup> interface is used. The output of the pin is the OR combination of the signals selected here and in MD1\_CFG (5Eh).

**Table 43. INT1\_CTRL register**

|                  |              |                |               |              |                  |             |              |
|------------------|--------------|----------------|---------------|--------------|------------------|-------------|--------------|
| 0 <sup>(1)</sup> | INT1_CNT_BDR | INT1_FIFO_FULL | INT1_FIFO_OVR | INT1_FIFO_TH | 0 <sup>(1)</sup> | INT1_DRDY_G | INT1_DRDY_XL |
|------------------|--------------|----------------|---------------|--------------|------------------|-------------|--------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 44. INT1\_CTRL register description**

|                |                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT1_CNT_BDR   | Enables COUNTER_BDR_IA interrupt on INT1 pin. Default value: 0                                                                                                   |
| INT1_FIFO_FULL | Enables FIFO full flag interrupt on INT1 pin. It can be also used to trigger an IBI when the MIPI I3C <sup>®</sup> interface is used. Default value: 0           |
| INT1_FIFO_OVR  | Enables FIFO overrun interrupt on INT1 pin. It can be also used to trigger an IBI when the MIPI I3C <sup>®</sup> interface is used. Default value: 0             |
| INT1_FIFO_TH   | Enables FIFO threshold interrupt on INT1 pin. It can be also used to trigger an IBI when the MIPI I3C <sup>®</sup> interface is used. Default value: 0           |
| INT1_DRDY_G    | Enables gyroscope data-ready interrupt on INT1 pin. It can be also used to trigger an IBI when the MIPI I3C <sup>®</sup> interface is used. Default value: 0     |
| INT1_DRDY_XL   | Enables accelerometer data-ready interrupt on INT1 pin. It can be also used to trigger an IBI when the MIPI I3C <sup>®</sup> interface is used. Default value: 0 |



## 9.12 INT2\_CTRL (0Eh)

INT2 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT2 when the MIPI I3C<sup>®</sup> dynamic address is not assigned (I<sup>2</sup>C or SPI is used). Some bits can be also used to trigger an IBI when the MIPI I3C<sup>®</sup> interface is used. The output of the pin is the OR combination of the signals selected here and in MD2\_CFG (5Fh).

**Table 45. INT2\_CTRL register**

| INT2_EMB_FUNC_ENDOP | INT2_CNT_BDR | INT2_FIFO_FULL | INT2_FIFO_OVR | INT2_FIFO_TH | INT2_DRDY_G_EIS | INT2_DRDY_G | INT2_DRDY_XL |
|---------------------|--------------|----------------|---------------|--------------|-----------------|-------------|--------------|
|---------------------|--------------|----------------|---------------|--------------|-----------------|-------------|--------------|

**Table 46. INT2\_CTRL register description**

|                     |                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| INT2_EMB_FUNC_ENDOP | Enables routing the embedded functions end of operations signal to the INT2 pin. Default value: 0 |
| INT2_CNT_BDR        | Enables COUNTER_BDR_IA interrupt on INT2. Default value: 0                                        |
| INT2_FIFO_FULL      | Enables FIFO full flag interrupt on INT2 pin. Default value: 0                                    |
| INT2_FIFO_OVR       | Enables FIFO overrun interrupt on INT2 pin. Default value: 0                                      |
| INT2_FIFO_TH        | Enables FIFO threshold interrupt on INT2 pin. Default value: 0                                    |
| INT2_DRDY_G_EIS     | Enables gyroscope EIS data-ready interrupt on INT2 pin. Default value: 0                          |
| INT2_DRDY_G         | Gyroscope data-ready interrupt on INT2 pin. Default value: 0                                      |
| INT2_DRDY_XL        | Accelerometer data-ready interrupt on INT2 pin. Default value: 0                                  |

## 9.13 WHO\_AM\_I (0Fh)

WHO\_AM\_I register (R). This is a read-only register. Its value is fixed at 70h.

**Table 47. WhoAml register**

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|


**9.14**
**CTRL1 (10h)**

Accelerometer control register 1 (R/W)

**Table 48. CTRL1 register**

| 0 <sup>(1)</sup> | OP_MODE_XL_2 | OP_MODE_XL_1 | OP_MODE_XL_0 | ODR_XL_3 | ODR_XL_2 | ODR_XL_1 | ODR_XL_0 |
|------------------|--------------|--------------|--------------|----------|----------|----------|----------|
|------------------|--------------|--------------|--------------|----------|----------|----------|----------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 49. CTRL1 register description**

|                 |                                                                                                                                                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OP_MODE_XL[2:0] | Accelerometer operating mode selection.<br>(000: high-performance mode (default);<br>001: high-accuracy ODR mode;<br>010: reserved;<br>011: ODR-triggered mode;<br>100: low-power mode 1 (2 mean);<br>101: low-power mode 2 (4 mean);<br>110: low-power mode 3 (8 mean);<br>111: normal mode) |
| ODR_XL[3:0]     | Accelerometer ODR selection (see Table 50)                                                                                                                                                                                                                                                    |

**Table 50. Accelerometer ODR selection**

| ODR_XL_3 | ODR_XL_2 | ODR_XL_1 | ODR_XL_0 | ODR selection [Hz]                                |
|----------|----------|----------|----------|---------------------------------------------------|
| 0        | 0        | 0        | 0        | Power-down (default)                              |
| 0        | 0        | 0        | 1        | 1.875 Hz (low-power mode)                         |
| 0        | 0        | 1        | 0        | 7.5 Hz (high-performance, normal mode)            |
| 0        | 0        | 1        | 1        | 15 Hz (low-power, high-performance, normal mode)  |
| 0        | 1        | 0        | 0        | 30 Hz (low-power, high-performance, normal mode)  |
| 0        | 1        | 0        | 1        | 60 Hz (low-power, high-performance, normal mode)  |
| 0        | 1        | 1        | 0        | 120 Hz (low-power, high-performance, normal mode) |
| 0        | 1        | 1        | 1        | 240 Hz (low-power, high-performance, normal mode) |
| 1        | 0        | 0        | 0        | 480 Hz (high-performance, normal mode)            |
| 1        | 0        | 0        | 1        | 960 Hz (high-performance, normal mode)            |
| 1        | 0        | 1        | 0        | 1.92 kHz (high-performance, normal mode)          |
| 1        | 0        | 1        | 1        | 3.84 kHz (high-performance mode)                  |
| 1        | 1        | 0        | 0        | 7.68 kHz (high-performance mode)                  |
| Others   |          |          |          | Reserved                                          |


**9.15**
**CTRL2 (11h)**

Gyroscope control register 2 (R/W)

**Table 51. CTRL2 register**

| 0 <sup>(1)</sup> | OP_MODE_<br>G_2 | OP_MODE_<br>G_1 | OP_MODE_<br>G_0 | ODR_G_3 | ODR_G_2 | ODR_G_1 | ODR_G_0 |
|------------------|-----------------|-----------------|-----------------|---------|---------|---------|---------|
|------------------|-----------------|-----------------|-----------------|---------|---------|---------|---------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 52. CTRL2 register description**

|                 |                                                                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OP_MODE_G_[2:0] | Gyroscope operating mode selection.<br>(000: high-performance mode (default);<br>001: high-accuracy ODR mode;<br>010: reserved;<br>011: ODR-triggered mode;<br>100: sleep mode;<br>101: low-power mode;<br>110-111: reserved) |
| ODR_G_[3:0]     | Gyroscope output data rate selection.<br>(See Table 53)                                                                                                                                                                       |

**Table 53. Gyroscope ODR selection**

| ODR_G_3 | ODR_G_2 | ODR_G_1 | ODR_G_0 | ODR [Hz]                                  |
|---------|---------|---------|---------|-------------------------------------------|
| 0       | 0       | 0       | 0       | Power-down (default)                      |
| 0       | 0       | 1       | 0       | 7.5 Hz (low-power, high-performance mode) |
| 0       | 0       | 1       | 1       | 15 Hz (low-power, high-performance mode)  |
| 0       | 1       | 0       | 0       | 30 Hz (low-power, high-performance mode)  |
| 0       | 1       | 0       | 1       | 60 Hz (low-power, high-performance mode)  |
| 0       | 1       | 1       | 0       | 120 Hz (low-power, high-performance mode) |
| 0       | 1       | 1       | 1       | 240 Hz (low-power, high-performance mode) |
| 1       | 0       | 0       | 0       | 480 Hz (high-performance mode)            |
| 1       | 0       | 0       | 1       | 960 Hz (high-performance mode)            |
| 1       | 0       | 1       | 0       | 1.92 kHz (high-performance mode)          |
| 1       | 0       | 1       | 1       | 3.84 kHz (high-performance mode)          |
| 1       | 1       | 0       | 0       | 7.68 kHz (high-performance mode)          |
| Others  |         |         |         | Reserved                                  |





## 9.16

### CTRL3 (12h)

Control register 3 (R/W)

**Table 54. CTRL3 register**

|      |     |                  |                  |                  |        |                  |          |
|------|-----|------------------|------------------|------------------|--------|------------------|----------|
| BOOT | BDU | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | IF_INC | 0 <sup>(1)</sup> | SW_RESET |
|------|-----|------------------|------------------|------------------|--------|------------------|----------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 55. CTRL3 register description**

|          |                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOOT     | Reboots memory content. This bit is automatically cleared. Default value: 0<br>(0: normal mode; 1: reboot memory content)                                                            |
| BDU      | Block data update. Default value: 1<br>(0: continuous update;<br>1: output registers are not updated until LSB and MSB have been read)                                               |
| IF_INC   | Register address automatically incremented during a multiple byte access with a serial interface (I <sup>2</sup> C, MIPI I3C, or SPI). Default value: 1<br>(0: disabled; 1: enabled) |
| SW_RESET | Software reset, resets all control registers to their default value. This bit is automatically cleared. Default value: 0<br>(0: normal mode; 1: reset device)                        |


**9.17**
**CTRL4 (13h)**

Control register 4 (R/W)

**Table 56. CTRL4 register**

|                  |                  |                  |              |           |                |             |            |
|------------------|------------------|------------------|--------------|-----------|----------------|-------------|------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | INT2_on_INT1 | DRDY_MASK | INT2_DRDY_TEMP | DRDY_PULSED | INT2_IN_LH |
|------------------|------------------|------------------|--------------|-----------|----------------|-------------|------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 57. CTRL4 register description**

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT2_on_INT1   | <p>Enables routing the embedded functions interrupt signals to the INT1 pin. Default value: 0</p> <ul style="list-style-type: none"> <li>The corresponding bits in the INT2 control registers need to be enabled.</li> <li>These interrupts are in OR with those enabled on the INT1 pin.</li> <li>They are not fed to the INT2 pin.</li> <li>The movable interrupts are:               <ul style="list-style-type: none"> <li>INT2_DRDY_G_EIS and INT2_EMB_FUNC_ENDOP enabled through INT2_CTRL (0Eh)</li> <li>INT2_TIMESTAMP enabled through MD2_CFG (5Fh)</li> <li>INT2_DRDY_TEMP enabled through CTRL4 (13h)</li> </ul> </li> </ul> |
| DRDY_MASK      | <p>Enables / masks data-ready signal. Default value: 0</p> <p>(0: disabled;<br/>1: masks DRDY signals (both accelerometer and gyroscope) until filter settling ends (accelerometer and gyroscope independently masked))</p> <p><i>Note: Refer to the product application note for the details regarding operating/power mode configurations, settings, turn-on/off time and on-the-fly changes.</i></p>                                                                                                                                                                                                                                 |
| INT2_DRDY_TEMP | <p>Enables temperature sensor data-ready interrupt on the INT2 pin. It can be also used to trigger an IBI when the MIPI I3C® interface is used and INT2_ON_INT1 = 1 in CTRL4_C (13h). Default value: 0</p> <p>(0: disabled; 1: enabled)</p>                                                                                                                                                                                                                                                                                                                                                                                             |
| DRDY_PULSED    | <p>Enables pulsed data-ready mode. Default value: 0</p> <p>(0: data-ready latched mode (returns to 0 only after the higher part of the associated output register has been read);<br/>1: data-ready pulsed mode (the data-ready pulses are 65 µs long))</p>                                                                                                                                                                                                                                                                                                                                                                             |
| INT2_IN_LH     | <p>Set to 1 in order to change the polarity of the INT2 pin input trigger for DEN or embedded functions. Default value: 0</p> <p>(0: trigger for DEN and embedded functions pin is active low;<br/>1: trigger for DEN and embedded functions pin is active high)</p>                                                                                                                                                                                                                                                                                                                                                                    |



## 9.18

### CTRL5 (14h)

Control register 5 (R/W)

**Table 58. CTRL5 register**

|                  |                  |                  |                  |                  |               |               |            |
|------------------|------------------|------------------|------------------|------------------|---------------|---------------|------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | BUS_ACT_SEL_1 | BUS_ACT_SEL_0 | INT_EN_I3C |
|------------------|------------------|------------------|------------------|------------------|---------------|---------------|------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 59. CTRL5 register description**

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| BUS_ACT_SEL_[1:0] | Bus available time selection for IBI (in-band interrupt):<br>00: 50 µs (default);<br>01: 2 µs;<br>10: 1 ms;<br>11: 50 ms) |
| INT_EN_I3C        | Enables the INT pin when I3C is enabled. Default value: 0<br>(0: disabled; 1: enabled)                                    |

## 9.19

### CTRL6 (15h)

Control register 6 (R/W)

**Table 60. CTRL6 register**

|                  |             |             |             |        |        |        |        |
|------------------|-------------|-------------|-------------|--------|--------|--------|--------|
| 0 <sup>(1)</sup> | LPF1_G_BW_2 | LPF1_G_BW_1 | LPF1_G_BW_0 | FS_G_3 | FS_G_2 | FS_G_1 | FS_G_0 |
|------------------|-------------|-------------|-------------|--------|--------|--------|--------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 61. CTRL6 register description**

|                 |                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LPF1_G_BW_[2:0] | Gyroscope low-pass filter (LPF1) bandwidth selection<br>Table 62 shows the selectable bandwidth values (available if EIS is disabled).                                                       |
| FS_G_[3:0]      | Gyroscope UI chain full-scale selection:<br>(0000: ±125 dps (default);<br>0001: ±250 dps;<br>0010: ±500 dps;<br>0011: ±1000 dps;<br>0100: ±2000 dps;<br>1100: ±4000 dps<br>Others: reserved) |


**Table 62. Gyroscope LPF1 + LPF2 bandwidth selection**

| LPF1_G_BW_[2:0] | 60 Hz | 120 Hz | 240 Hz | 480 Hz | 960 Hz | 1.92 kHz | 3.84 kHz | 7.68 kHz |
|-----------------|-------|--------|--------|--------|--------|----------|----------|----------|
| 000             | 24.6  | 49.4   | 96     | 175    | 241    | 273      | 280      | 281      |
| 001             | 24.6  | 49.4   | 96     | 157    | 195    | 210      | 213      | 213      |
| 010             | 24.6  | 49.4   | 96     | 131    | 149    | 155      | 156      | 156      |
| 011             | 24.6  | 49.4   | 96     | 188    | 310    | 387      | 403      | 407      |
| 100             | 24.6  | 49.4   | 78.4   | 94     | 100    | 101      | 102      | 102      |
| 101             | 24.6  | 42.6   | 53     | 56.7   | 57.9   | 58.2     | 58.3     | 58       |
| 110             | 18.0  | 24.2   | 27.3   | 28.4   | 28.7   | 28.8     | 28.8     | 28.8     |
| 111             | 12.1  | 13.6   | 14.2   | 14.3   | 14.4   | 14.4     | 14.4     | 14.4     |

## 9.20

### CTRL7 (16h)

Control register 7 (R/W)

**Table 63. CTRL7 register**

|                  |                  |                  |                  |                  |                  |                  |           |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|-----------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | LPF1_G_EN |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|-----------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 64. CTRL7 register description**

|           |                                           |
|-----------|-------------------------------------------|
| LPF1_G_EN | Enables the gyroscope digital LPF1 filter |
|-----------|-------------------------------------------|



## 9.21

### CTRL8 (17h)

Control register 8 (R/W)

**Table 65. CTRL8 register**

|                 |                 |                 |                  |             |                  |         |         |
|-----------------|-----------------|-----------------|------------------|-------------|------------------|---------|---------|
| HP_LPF2_XL_BW_2 | HP_LPF2_XL_BW_1 | HP_LPF2_XL_BW_0 | 0 <sup>(1)</sup> | XL_DualC_EN | 0 <sup>(1)</sup> | FS_XL_1 | FS_XL_0 |
|-----------------|-----------------|-----------------|------------------|-------------|------------------|---------|---------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 66. CTRL8 register description**

|                     |                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HP_LPF2_XL_BW_[2:0] | Accelerometer LPF2 and HP filter configuration and cutoff setting. Refer to Table 67.                                                                                                                                           |
| XL_DualC_EN         | Enables dual-channel mode. When this bit is set to 1, data with the maximum full scale are sent to the output registers at addresses 34h to 39h. The UI processing chain is used. Default value: 0<br>(0: disabled; 1: enabled) |
| FS_XL_[1:0]         | Accelerometer full-scale selection:<br>(00: $\pm 2$ g;<br>01: $\pm 4$ g;<br>10: $\pm 8$ g;<br>11: $\pm 16$ g)                                                                                                                   |

**Table 67. Accelerometer bandwidth configurations**

| Filter type | HP_SLOPE_XL_EN | LPF2_XL_EN | HP_LPF2_XL_BW_[2:0] | Bandwidth            |
|-------------|----------------|------------|---------------------|----------------------|
| Low pass    | 0              | 0          | -                   | ODR/2 <sup>(1)</sup> |
|             |                |            | 000                 | ODR/4                |
|             |                |            | 001                 | ODR/10               |
|             |                |            | 010                 | ODR/20               |
|             |                |            | 011                 | ODR/45               |
|             |                |            | 100                 | ODR/100              |
|             |                |            | 101                 | ODR/200              |
|             |                |            | 110                 | ODR/400              |
| High pass   | 1              | 1          | 111                 | ODR/800              |
|             |                |            | 000                 | SLOPE (ODR/4)        |
|             |                |            | 001                 | ODR/10               |
|             |                |            | 010                 | ODR/20               |
|             |                |            | 011                 | ODR/45               |
|             |                |            | 100                 | ODR/100              |
|             |                |            | 101                 | ODR/200              |
|             |                |            | 110                 | ODR/400              |
|             |                |            | 111                 | ODR/800              |

1. This value is ODR/2 when the accelerometer is in high-performance mode, high-accuracy ODR mode and normal mode. It is equal to 2300 Hz when the accelerometer is in low-power mode 1 (2 mean), 912 Hz in low-power mode 2 (4 mean) and 431 Hz in low-power mode 3 (8 mean).



## 9.22

### CTRL9 (18h)

Control register 9 (R/W)

**Table 68. CTRL9 register**

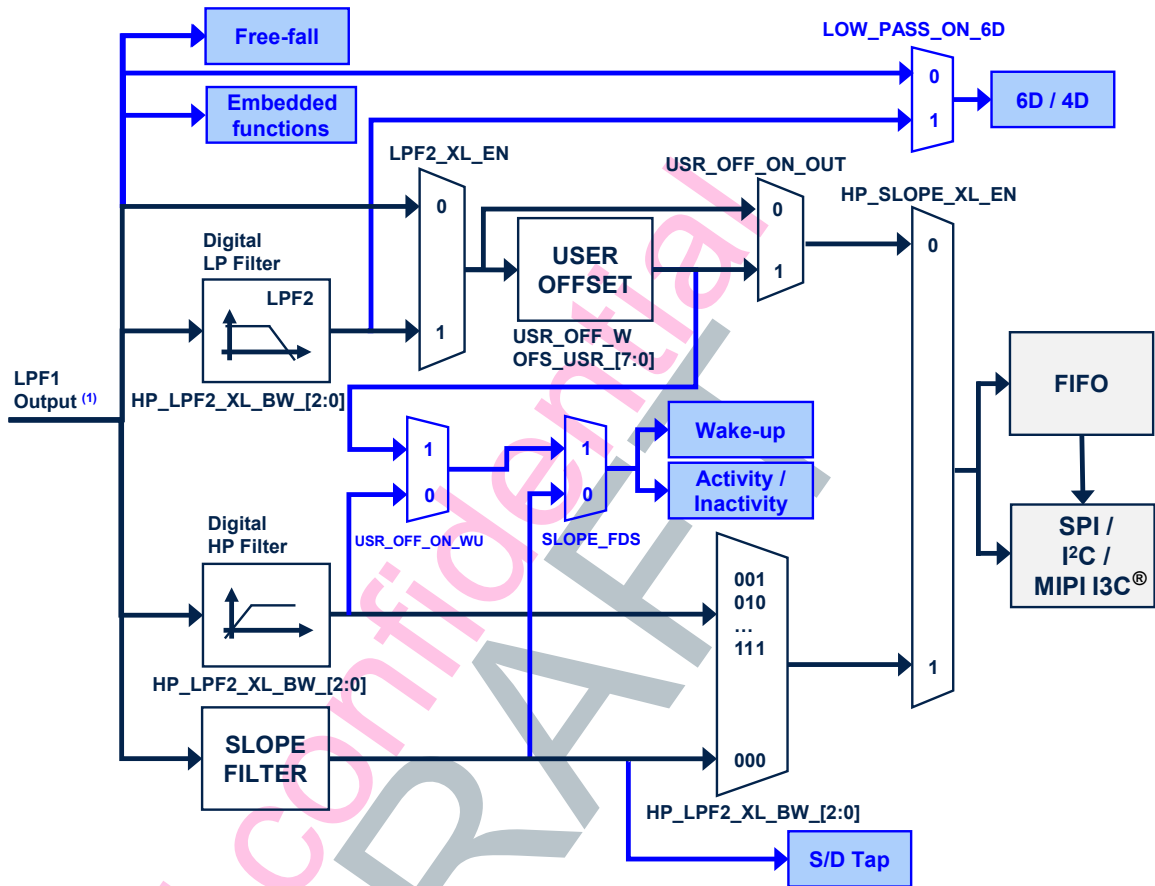
|                  |                |                   |                |            |                  |           |                |
|------------------|----------------|-------------------|----------------|------------|------------------|-----------|----------------|
| 0 <sup>(1)</sup> | HP_REF_MODE_XL | XL_FASTSETTL_MODE | HP_SLOPE_XL_EN | LPF2_XL_EN | 0 <sup>(1)</sup> | USR_OFF_W | USR_OFF_ON_OUT |
|------------------|----------------|-------------------|----------------|------------|------------------|-----------|----------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 69. CTRL9 register description**

|                   |                                                                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HP_REF_MODE_XL    | Enables accelerometer high-pass filter reference mode (valid for high-pass path - HP_SLOPE_XL_EN bit must be 1). Default value: 0<br>(0: disabled, 1: enabled) <sup>(1)</sup>                                                                     |
| XL_FASTSETTL_MODE | Enables accelerometer LPF2 and HPF fast-settling mode. The filter sets the first sample after writing this bit. Active only during device exit from power-down mode. Default value: 0<br>(0: disabled, 1: enabled)                                |
| HP_SLOPE_XL_EN    | Accelerometer slope filter / high-pass filter selection. Refer to Figure 24. Default value: 0<br>(0: low-pass filter path selected;<br>1: high-pass filter path selected)                                                                         |
| LPF2_XL_EN        | Accelerometer high-resolution selection. Refer to Figure 24. Default value: 0<br>(0: output from first stage digital filtering selected;<br>1: output from LPF2 second filtering stage selected)                                                  |
| USR_OFF_W         | Weight of XL user offset bits of registers X_OFS_USR (73h), Y_OFS_USR (74h), Z_OFS_USR (75h). Default value: 0<br>(0: 2 <sup>-10</sup> g/LSB;<br>1: 2 <sup>-6</sup> g/LSB)                                                                        |
| USR_OFF_ON_OUT    | Enables accelerometer user offset correction block; it is valid for the low-pass path. Refer to Figure 24. Default value: 0<br>(0: accelerometer user offset correction block bypassed;<br>1: accelerometer user offset correction block enabled) |

1. When enabled, the first output data has to be discarded.


**Figure 24. Accelerometer block diagram**


1. The cutoff value of the LPF1 output is  $ODR/2$  when the accelerometer is in high-performance mode, high-accuracy ODR mode or normal mode. This value is equal to 2300 Hz when the accelerometer is in low-power mode 1 (2 mean), 912 Hz in low-power mode 2 (4 mean) or 431 Hz in low-power mode 3 (8 mean).



## 9.23

### CTRL10 (19h)

Control register 10 (R/W)

**Table 70. CTRL10 register**

|                  |                |                  |                  |        |        |         |         |
|------------------|----------------|------------------|------------------|--------|--------|---------|---------|
| 0 <sup>(1)</sup> | EMB_FUNC_DEBUG | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | ST_G_1 | ST_G_0 | ST_XL_1 | ST_XL_0 |
|------------------|----------------|------------------|------------------|--------|--------|---------|---------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 71. CTRL10 register description**

|                |                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| EMB_FUNC_DEBUG | Enables debug mode for the embedded functions.<br>(0: disabled; 1: enabled)                                                                       |
| ST_G_[1:0]     | Gyroscope self-test selection<br>(00: normal mode (default);<br>01: positive sign self-test;<br>10: negative sign self-test;<br>11: reserved)     |
| ST_XL_[1:0]    | Accelerometer self-test selection<br>(00: normal mode (default);<br>01: positive sign self-test;<br>10: negative sign self-test;<br>11: reserved) |

## 9.24

### CTRL\_STATUS (1Ah)

(R)

**Table 72. CTRL\_STATUS register**

|   |   |   |   |   |                    |   |   |
|---|---|---|---|---|--------------------|---|---|
| 0 | 0 | 0 | 0 | 0 | FSM_WR_CTRL_STATUS | - | 0 |
|---|---|---|---|---|--------------------|---|---|

**Table 73. CTRL\_STATUS register description**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSM_WR_CTRL_STATUS | This flag indicates the current controller of the device configuration registers. This flag must be used as an acknowledge flag when the value of the FSM_WR_CTRL_EN bit in the <a href="#">FUNC_CFG_ACCESS (01h)</a> register is changed. Default value: 0<br>(0: all registers and configurations are writable from the standard interface;<br>1: some registers and configurations are under FSM control and are in read-only mode from the standard interface). |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





## 9.25 FIFO\_STATUS1 (1Bh)

FIFO status register 1 (R)

**Table 74. FIFO\_STATUS1 register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| DIFF_FIFO_7 | DIFF_FIFO_6 | DIFF_FIFO_5 | DIFF_FIFO_4 | DIFF_FIFO_3 | DIFF_FIFO_2 | DIFF_FIFO_1 | DIFF_FIFO_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 75. FIFO\_STATUS1 register description**

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| DIFF_FIFO_[7:0] | Number of unread sensor data (TAG + 6 bytes) stored in FIFO<br>In conjunction with DIFF_FIFO_8 in FIFO_STATUS2 (1Ch). |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|

## 9.26 FIFO\_STATUS2 (1Ch)

FIFO status register 2 (R)

**Table 76. FIFO\_STATUS2 register**

|             |             |              |                |                  |   |   |             |
|-------------|-------------|--------------|----------------|------------------|---|---|-------------|
| FIFO_WTM_IA | FIFO_OVR_IA | FIFO_FULL_IA | COUNTER_BDR_IA | FIFO_OVR_LATCHED | 0 | 0 | DIFF_FIFO_8 |
|-------------|-------------|--------------|----------------|------------------|---|---|-------------|

**Table 77. FIFO\_STATUS2 register description**

|                  |                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIFO_WTM_IA      | FIFO watermark status. Default value: 0<br>(0: FIFO filling is lower than WTM;<br>1: FIFO filling is equal to or greater than WTM)<br>Watermark is set through bits WTM[7:0] in FIFO_CTRL2 (08h) and FIFO_CTRL1 (07h). |
| FIFO_OVR_IA      | FIFO overrun status. Default value: 0<br>(0: FIFO is not completely filled; 1: FIFO is completely filled)                                                                                                              |
| FIFO_FULL_IA     | Smart FIFO full status. Default value: 0<br>(0: FIFO is not full; 1: FIFO will be full at the next ODR)                                                                                                                |
| COUNTER_BDR_IA   | Counter BDR reaches the CNT_BDR_TH_[10:0] threshold set in COUNTER_BDR_REG1 (0Bh) and COUNTER_BDR_REG2 (0Ch). Default value: 0<br>This bit is reset when these registers are read.                                     |
| FIFO_OVR_LATCHED | Latched FIFO overrun status. Default value: 0<br>This bit is reset when this register is read.                                                                                                                         |
| DIFF_FIFO_8      | Number of unread sensor data (TAG + 6 bytes) stored in FIFO. Default value: 00<br>In conjunction with DIFF_FIFO[7:0] in FIFO_STATUS1 (1Bh)                                                                             |



## 9.27

### ALL\_INT\_SRC (1Dh)

Source register for all interrupts (R)

**Table 78. ALL\_INT\_SRC register**

|             |         |                 |        |   |        |       |       |
|-------------|---------|-----------------|--------|---|--------|-------|-------|
| EMB_FUNC_IA | SHUB_IA | SLEEP_CHANGE_IA | D6D_IA | 0 | TAP_IA | WU_IA | FF_IA |
|-------------|---------|-----------------|--------|---|--------|-------|-------|

**Table 79. ALL\_INT\_SRC register description**

|                 |                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMB_FUNC_IA     | Embedded functions interrupt status. Default value: 0<br>(0: embedded functions event not detected;<br>1: embedded functions event detected)                                                                  |
| SHUB_IA         | Sensor hub (I <sup>2</sup> C master) interrupt status. Default value: 0<br>(0: sensor hub interrupt not generated;<br>1: sensor hub interrupt generated)                                                      |
| SLEEP_CHANGE_IA | Detects change event in activity/inactivity status. Default value: 0<br>(0: change status not detected; 1: change status detected)                                                                            |
| D6D_IA          | Interrupt active for change in position of portrait, landscape, face-up, face-down. Default value: 0<br>(0: change in position not detected; 1: change in position detected)                                  |
| TAP_IA          | Single or double-tap event detection status depending on SINGLE_DOUBLE_TAP_bit value (see <a href="#">WAKE_UP_THS (5Bh)</a> register). Default value: 0<br>(0: tap event not detected; 1: tap event detected) |
| WU_IA           | Wake-up event status. Default value: 0<br>(0: event not detected, 1: event detected)                                                                                                                          |
| FF_IA           | Free-fall event status. Default value: 0<br>(0: event not detected, 1: event detected)                                                                                                                        |



## 9.28 STATUS\_REG (1Eh)

The STATUS\_REG register is read by the primary interface SPI/I<sup>2</sup>C & MIPI I3C® (R).

**Table 80. STATUS\_REG register**

|                    |   |   |         |   |     |     |      |
|--------------------|---|---|---------|---|-----|-----|------|
| TIMESTAMP_ENDCOUNT | 0 | 0 | GDA_EIS | 0 | TDA | GDA | XLDA |
|--------------------|---|---|---------|---|-----|-----|------|

**Table 81. STATUS\_REG register description**

|                    |                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TIMESTAMP_ENDCOUNT | Alerts timestamp overflow within 5.6 ms                                                                                                                                        |
| GDA_EIS            | Enhanced EIS gyroscope new data available. Default value: 0<br>(0: no set of data available at gyroscope output;<br>1: a new set of data is available at gyroscope output)     |
| TDA                | Temperature new data available. Default: 0<br>(0: no set of data is available at temperature sensor output;<br>1: a new set of data is available at temperature sensor output) |
| GDA                | Gyroscope new data available. Default value: 0<br>(0: no set of data available at gyroscope output;<br>1: a new set of data is available at gyroscope output)                  |
| XLDA               | Accelerometer new data available. Default value: 0<br>(0: no set of data available at accelerometer output;<br>1: a new set of data is available at accelerometer output)      |



## 9.29 OUT\_TEMP\_L (20h), OUT\_TEMP\_H (21h)

Temperature data output register (R). L and H registers together express a 16-bit word in two's complement.

**Table 82. OUT\_TEMP\_L register**

|       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
| Temp7 | Temp6 | Temp5 | Temp4 | Temp3 | Temp2 | Temp1 | Temp0 |
|-------|-------|-------|-------|-------|-------|-------|-------|

**Table 83. OUT\_TEMP\_H register**

|        |        |        |        |        |        |       |       |
|--------|--------|--------|--------|--------|--------|-------|-------|
| Temp15 | Temp14 | Temp13 | Temp12 | Temp11 | Temp10 | Temp9 | Temp8 |
|--------|--------|--------|--------|--------|--------|-------|-------|

**Table 84. OUT\_TEMP register description**

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| Temp[15:0] | Temperature sensor output data<br>The value is expressed in two's complement. |
|------------|-------------------------------------------------------------------------------|

## 9.30 OUTX\_L\_G (22h) and OUTX\_H\_G (23h)

Angular rate sensor pitch axis (X) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL6 (15h)) and ODR settings (CTRL2 (11h)) of the gyroscope user interface.

**Table 85. OUTX\_L\_G register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 86. OUTX\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 87. OUTX\_G register description**

|         |                                                             |
|---------|-------------------------------------------------------------|
| D[15:0] | Gyroscope UI chain pitch axis (X) angular rate output value |
|---------|-------------------------------------------------------------|



### 9.31 OUTY\_L\_G (24h) and OUTY\_H\_G (25h)

Angular rate sensor roll axis (Y) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL6 (15h)) and ODR settings (CTRL2 (11h)) of the gyroscope user interface.

**Table 88. OUTY\_L\_G register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 89. OUTY\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 90. OUTY\_G register description**

|         |                                                            |
|---------|------------------------------------------------------------|
| D[15:0] | Gyroscope UI chain roll axis (Y) angular rate output value |
|---------|------------------------------------------------------------|

### 9.32 OUTZ\_L\_G (26h) and OUTZ\_H\_G (27h)

Angular rate sensor yaw axis (Z) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL6 (15h)) and ODR settings (CTRL2 (11h)) of the gyroscope user interface.

**Table 91. OUTZ\_L\_G register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 92. OUTZ\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 93. OUTZ\_H\_G register description**

|         |                                                           |
|---------|-----------------------------------------------------------|
| D[15:0] | Gyroscope UI chain yaw axis (Z) angular rate output value |
|---------|-----------------------------------------------------------|



### 9.33 OUTX\_L\_A (28h) and OUTX\_H\_A (29h)

Linear acceleration sensor X-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL8 (17h)) and ODR settings (CTRL1 (10h)) of the accelerometer user interface.

**Table 94. OUTX\_L\_A register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 95. OUTX\_H\_A register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 96. OUTX\_A register description**

|         |                                                                |
|---------|----------------------------------------------------------------|
| D[15:0] | Accelerometer UI chain X-axis linear acceleration output value |
|---------|----------------------------------------------------------------|

### 9.34 OUTY\_L\_A (2Ah) and OUTY\_H\_A (2Bh)

Linear acceleration sensor Y-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL8 (17h)) and ODR settings (CTRL1 (10h)) of the accelerometer user interface.

**Table 97. OUTY\_L\_A register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 98. OUTY\_H\_A register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 99. OUTY\_A register description**

|         |                                                                |
|---------|----------------------------------------------------------------|
| D[15:0] | Accelerometer UI chain Y-axis linear acceleration output value |
|---------|----------------------------------------------------------------|



### 9.35 OUTZ\_L\_A (2Ch) and OUTZ\_H\_A (2Dh)

Linear acceleration sensor Z-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale (CTRL8 (17h)) and ODR settings (CTRL1 (10h)) of the accelerometer user interface.

**Table 100. OUTZ\_L\_A register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 101. OUTZ\_H\_A register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 102. OUTZ\_A register description**

|         |                                                                |
|---------|----------------------------------------------------------------|
| D[15:0] | Accelerometer UI chain Z-axis linear acceleration output value |
|---------|----------------------------------------------------------------|

### 9.36 UI\_OUTX\_L\_G\_EIS (2Eh) and UI\_OUTX\_H\_G\_EIS (2Fh)

Angular rate sensor pitch axis (X) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the gyroscope full-scale and ODR settings of the EIS gyroscope channel.

**Table 103. UI\_OUTX\_L\_G\_EIS register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 104. UI\_OUTX\_H\_G\_EIS register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 105. UI\_OUTX\_G\_EIS register description**

|         |                                                               |
|---------|---------------------------------------------------------------|
| D[15:0] | Gyroscope pitch axis EIS output expressed in two's complement |
|---------|---------------------------------------------------------------|



### 9.37 UI\_OUTY\_L\_G\_EIS (30h) and UI\_OUTY\_H\_G\_EIS (31h)

Angular rate sensor roll axis (Y) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the gyroscope full-scale and ODR settings of the EIS gyroscope channel.

**Table 106. UI\_OUTY\_L\_G\_EIS register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 107. UI\_OUTY\_H\_G\_EIS register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 108. UI\_OUTY\_G\_EIS register description**

|         |                                                              |
|---------|--------------------------------------------------------------|
| D[15:0] | Gyroscope roll axis EIS output expressed in two's complement |
|---------|--------------------------------------------------------------|

### 9.38 UI\_OUTZ\_L\_G\_EIS (32h) and UI\_OUTZ\_H\_G\_EIS (33h)

Angular rate sensor yaw axis (Z) angular rate output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the gyroscope full-scale and ODR settings of the EIS gyroscope channel.

**Table 109. UI\_OUTZ\_L\_G\_EIS register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 110. UI\_OUTZ\_H\_G\_EIS register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 111. UI\_OUTZ\_G\_EIS register description**

|         |                                                             |
|---------|-------------------------------------------------------------|
| D[15:0] | Gyroscope yaw axis EIS output expressed in two's complement |
|---------|-------------------------------------------------------------|





### 9.39 UI\_OUTX\_L\_A\_DualC (34h) and UI\_OUTX\_H\_A\_DualC (35h)

Linear acceleration sensor X-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the accelerometer dual-channel mode configuration.

**Table 112. UI\_OUTX\_L\_A\_DualC register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 113. UI\_OUTX\_H\_A\_DualC register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 114. UI\_OUTX\_A\_DualC register description**

|         |                                                                 |
|---------|-----------------------------------------------------------------|
| D[15:0] | Accelerometer X-axis DualC output expressed in two's complement |
|---------|-----------------------------------------------------------------|

### 9.40 UI\_OUTY\_L\_A\_DualC (36h) and UI\_OUTY\_H\_A\_DualC (37h)

Linear acceleration sensor Y-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the accelerometer dual-channel mode configuration.

**Table 115. UI\_OUTY\_L\_A\_DualC register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 116. UI\_OUTY\_H\_A\_DualC register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 117. UI\_OUTY\_A\_DualC register description**

|         |                                                                 |
|---------|-----------------------------------------------------------------|
| D[15:0] | Accelerometer Y-axis DualC output expressed in two's complement |
|---------|-----------------------------------------------------------------|



#### 9.41 UI\_OUTZ\_L\_A\_DualC (38h) and UI\_OUTZ\_H\_A\_DualC (39h)

Linear acceleration sensor Z-axis output register (R). The value is expressed as a 16-bit word in two's complement.

Data are according to the accelerometer dual-channel mode configuration.

**Table 118. UI\_OUTZ\_L\_A\_DualC register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 119. UI\_OUTZ\_H\_A\_DualC register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 120. UI\_OUTZ\_A\_DualC register description**

|         |                                                                 |
|---------|-----------------------------------------------------------------|
| D[15:0] | Accelerometer Z-axis DualC output expressed in two's complement |
|---------|-----------------------------------------------------------------|

#### 9.42 TIMESTAMP0 (40h), TIMESTAMP1 (41h), TIMESTAMP2 (42h), and TIMESTAMP3 (43h)

Timestamp first data output register (R). The value is expressed as a 32-bit word and the bit resolution is 21.7  $\mu$ s (typical).

**Table 121. TIMESTAMP output registers**

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| D31 | D30 | D29 | D28 | D27 | D26 | D25 | D24 |
| D23 | D22 | D21 | D20 | D19 | D18 | D17 | D16 |
| D15 | D14 | D13 | D12 | D11 | D10 | D9  | D8  |
| D7  | D6  | D5  | D4  | D3  | D2  | D1  | D0  |

**Table 122. TIMESTAMP output register description**

|         |                                                           |
|---------|-----------------------------------------------------------|
| D[31:0] | Timestamp output registers: 1LSB = 21.7 $\mu$ s (typical) |
|---------|-----------------------------------------------------------|



## 9.43 UI\_STATUS\_REG (44h)

**Table 123. UI\_STATUS\_REG register**

|   |   |   |   |   |               |   |   |
|---|---|---|---|---|---------------|---|---|
| 0 | 0 | 0 | 0 | 0 | GYRO_SETTLING | 0 | 0 |
|---|---|---|---|---|---------------|---|---|

**Table 124. UI\_STATUS\_REG register description**

|               |                                                         |
|---------------|---------------------------------------------------------|
| GYRO_SETTLING | High when the gyroscope output is in the settling phase |
|---------------|---------------------------------------------------------|

## 9.44 WAKE\_UP\_SRC (45h)

Wake-up interrupt source register (R)

**Table 125. WAKE\_UP\_SRC register**

|   |                 |       |             |       |      |      |      |
|---|-----------------|-------|-------------|-------|------|------|------|
| 0 | SLEEP_CHANGE_IA | FF_IA | SLEEP_STATE | WU_IA | X_WU | Y_WU | Z_WU |
|---|-----------------|-------|-------------|-------|------|------|------|

**Table 126. WAKE\_UP\_SRC register description**

|                 |                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| SLEEP_CHANGE_IA | Detects change event in activity/inactivity status. Default value: 0<br>(0: change status not detected; 1: change status detected)           |
| FF_IA           | Free-fall event detection status. Default: 0<br>(0: free-fall event not detected; 1: free-fall event detected)                               |
| SLEEP_STATE     | Sleep status bit. Default value: 0<br>(0: Activity status; 1: Inactivity status)                                                             |
| WU_IA           | Wake-up event detection status. Default value: 0<br>(0: wake-up event not detected; 1: wake-up event detected.)                              |
| X_WU            | Wake-up event detection status on X-axis. Default value: 0<br>(0: wake-up event on X-axis not detected; 1: wake-up event on X-axis detected) |
| Y_WU            | Wake-up event detection status on Y-axis. Default value: 0<br>(0: wake-up event on Y-axis not detected; 1: wake-up event on Y-axis detected) |
| Z_WU            | Wake-up event detection status on Z-axis. Default value: 0<br>(0: wake-up event on Z-axis not detected; 1: wake-up event on Z-axis detected) |


**9.45**
**TAP\_SRC (46h)**

Tap source register (R)

**Table 127. TAP\_SRC register**

|   |        |            |            |          |       |       |       |
|---|--------|------------|------------|----------|-------|-------|-------|
| 0 | TAP_IA | SINGLE_TAP | DOUBLE_TAP | TAP_SIGN | X_TAP | Y_TAP | Z_TAP |
|---|--------|------------|------------|----------|-------|-------|-------|

**Table 128. TAP\_SRC register description**

|            |                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAP_IA     | Tap event detection status. Default: 0<br>(0: tap event not detected; 1: tap event detected)                                                                                  |
| SINGLE_TAP | Single-tap event status. Default value: 0<br>(0: single tap event not detected; 1: single tap event detected)                                                                 |
| DOUBLE_TAP | Double-tap event detection status. Default value: 0<br>(0: double-tap event not detected; 1: double-tap event detected.)                                                      |
| TAP_SIGN   | Sign of acceleration detected by tap event. Default: 0<br>(0: positive sign of acceleration detected by tap event;<br>1: negative sign of acceleration detected by tap event) |
| X_TAP      | Tap event detection status on X-axis. Default value: 0<br>(0: tap event on X-axis not detected; 1: tap event on X-axis detected)                                              |
| Y_TAP      | Tap event detection status on Y-axis. Default value: 0<br>(0: tap event on Y-axis not detected; 1: tap event on Y-axis detected)                                              |
| Z_TAP      | Tap event detection status on Z-axis. Default value: 0<br>(0: tap event on Z-axis not detected; 1: tap event on Z-axis detected)                                              |


**9.46**
**D6D\_SRC (47h)**

Portrait, landscape, face-up and face-down source register (R)

**Table 129. D6D\_SRC register**

|   |        |    |    |    |    |    |    |
|---|--------|----|----|----|----|----|----|
| 0 | D6D_IA | ZH | ZL | YH | YL | XH | XL |
|---|--------|----|----|----|----|----|----|

**Table 130. D6D\_SRC register description**

|        |                                                                                                                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D6D_IA | Interrupt active for change position portrait, landscape, face-up, face-down. Default value: 0<br>(0: change position not detected; 1: change position detected) |
| ZH     | Z-axis high event (over threshold). Default value: 0<br>(0: event not detected; 1: event (over threshold) detected)                                              |
| ZL     | Z-axis low event (under threshold). Default value: 0<br>(0: event not detected; 1: event (under threshold) detected)                                             |
| YH     | Y-axis high event (over threshold). Default value: 0<br>(0: event not detected; 1: event (over-threshold) detected)                                              |
| YL     | Y-axis low event (under threshold). Default value: 0<br>(0: event not detected; 1: event (under threshold) detected)                                             |
| XH     | X-axis high event (over threshold). Default value: 0<br>(0: event not detected; 1: event (over threshold) detected)                                              |
| XL     | X-axis low event (under threshold). Default value: 0<br>(0: event not detected; 1: event (under threshold) detected)                                             |



## 9.47 EMB\_FUNC\_STATUS\_MAINPAGE (49h)

Embedded function status register (R)

**Table 131. EMB\_FUNC\_STATUS\_MAINPAGE register**

|           |   |           |         |             |   |   |   |
|-----------|---|-----------|---------|-------------|---|---|---|
| IS_FSM_LC | 0 | IS_SIGMOT | IS_TILT | IS_STEP_DET | 0 | 0 | 0 |
|-----------|---|-----------|---------|-------------|---|---|---|

**Table 132. EMB\_FUNC\_STATUS\_MAINPAGE register description**

|             |                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------|
| IS_FSM_LC   | Interrupt status bit for FSM long counter timeout interrupt event.<br>(1: interrupt detected; 0: no interrupt) |
| IS_SIGMOT   | Interrupt status bit for significant motion detection<br>(1: interrupt detected; 0: no interrupt)              |
| IS_TILT     | Interrupt status bit for tilt detection<br>(1: interrupt detected; 0: no interrupt)                            |
| IS_STEP_DET | Interrupt status bit for step detection<br>(1: interrupt detected; 0: no interrupt)                            |

## 9.48 FSM\_STATUS\_MAINPAGE (4Ah)

Finite state machine status register (R)

**Table 133. FSM\_STATUS\_MAINPAGE register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| IS_FSM8 | IS_FSM7 | IS_FSM6 | IS_FSM5 | IS_FSM4 | IS_FSM3 | IS_FSM2 | IS_FSM1 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 134. FSM\_STATUS\_MAINPAGE register description**

|         |                                                                                         |
|---------|-----------------------------------------------------------------------------------------|
| IS_FSM8 | Interrupt status bit for FSM8 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM7 | Interrupt status bit for FSM7 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM6 | Interrupt status bit for FSM6 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM5 | Interrupt status bit for FSM5 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM4 | Interrupt status bit for FSM4 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM3 | Interrupt status bit for FSM3 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM2 | Interrupt status bit for FSM2 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM1 | Interrupt status bit for FSM1 interrupt event. (1: interrupt detected; 0: no interrupt) |



## 9.49 INTERNAL\_FREQ\_FINE (4Fh)

Internal frequency register (R)

**Table 135. INTERNAL\_FREQ\_FINE register**

| FREQ_FINE_7 | FREQ_FINE_6 | FREQ_FINE_5 | FREQ_FINE_4 | FREQ_FINE_3 | FREQ_FINE_2 | FREQ_FINE_1 | FREQ_FINE_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 136. INTERNAL\_FREQ\_FINE register description**

| FREQ_FINE_[7:0] | Difference in percentage of the effective ODR (and timestamp rate) with respect to the typical.<br>Step: 0.13%. 8-bit format, two's complement. |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

The actual timestamp resolution and the actual output data rate can be calculated using the following formulas:

$$t_{actual}[s] = \frac{1}{46080 \cdot (1 + 0.0013 \cdot FREQ\_FINE)}$$

$$ODR_{actual}[Hz] = \frac{7680 \cdot (1 + 0.0013 \cdot FREQ\_FINE)}{ODR_{coeff}}$$

**Table 137. ODR<sub>coeff</sub> values**

| Selected ODR [Hz] | ODRcoeff |
|-------------------|----------|
| 7.5               | 1024     |
| 15                | 512      |
| 30                | 256      |
| 60                | 128      |
| 120               | 64       |
| 240               | 32       |
| 480               | 16       |
| 960               | 8        |
| 1.92 kHz          | 4        |
| 3.84 kHz          | 2        |
| 7.68 kHz          | 1        |



## 9.50 FUNCTIONS\_ENABLE (50h)

Enable interrupt functions register (R/W)

**Table 138. FUNCTIONS\_ENABLE register**

|                   |              |                  |                  |                     |                  |            |            |
|-------------------|--------------|------------------|------------------|---------------------|------------------|------------|------------|
| INTERRUPTS_ENABLE | TIMESTAMP_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | DIS_RST_LIR_ALL_INT | 0 <sup>(1)</sup> | INACT_EN_1 | INACT_EN_0 |
|-------------------|--------------|------------------|------------------|---------------------|------------------|------------|------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 139. FUNCTIONS\_ENABLE register description**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INTERRUPTS_ENABLE   | Enables basic interrupts (6D/4D, free-fall, wake-up, tap, activity/inactivity). Default value: 0<br>(0: interrupt disabled; 1: interrupt enabled)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| TIMESTAMP_EN        | Enables timestamp counter. The counter is readable in <a href="#">TIMESTAMP0 (40h)</a> , <a href="#">TIMESTAMP1 (41h)</a> , <a href="#">TIMESTAMP2 (42h)</a> , and <a href="#">TIMESTAMP3 (43h)</a> . Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DIS_RST_LIR_ALL_INT | When this bit is set to 1, reading the <a href="#">ALL_INT_SRC (1Dh)</a> register does not reset the latched interrupt signals. This can be useful in order to not reset some status flags before reading the corresponding status register. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| INACT_EN_[1:0]      | Enables activity/inactivity (sleep) function. Default value: 00<br>(00: stationary/motion-only interrupts generated, accelerometer and gyroscope configuration do not change;<br>01: sets accelerometer to low-power mode 1 with accelerometer ODR selected through the <a href="#">XL_INACT_ODR [1:0]</a> bits of the <a href="#">INACTIVITY_DUR (54h)</a> register, gyroscope configuration does not change;<br>10: sets accelerometer to low-power mode 1 with accelerometer ODR selected through the <a href="#">XL_INACT_ODR [1:0]</a> bits of the <a href="#">INACTIVITY_DUR (54h)</a> register, gyroscope in sleep mode;<br>11: sets accelerometer to low-power mode 1 with accelerometer ODR selected through the <a href="#">XL_INACT_ODR [1:0]</a> bits of the <a href="#">INACTIVITY_DUR (54h)</a> register, gyroscope in power-down mode) |





## 9.51

### DEN (51h)

DEN configuration register (R/W)

**Table 140. DEN register**

|                  |         |         |           |       |       |       |          |
|------------------|---------|---------|-----------|-------|-------|-------|----------|
| 0 <sup>(1)</sup> | LVL1_EN | LVL2_EN | DEN_XL_EN | DEN_X | DEN_Y | DEN_Z | DEN_XL_G |
|------------------|---------|---------|-----------|-------|-------|-------|----------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 141. DEN register description**

|           |                                                                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LVL1_EN   | Enables DEN data level-sensitive trigger. Refer to Table 142.                                                                                                                                                                      |
| LVL2_EN   | Enables DEN level-sensitive latched. Refer to Table 142.                                                                                                                                                                           |
| DEN_XL_EN | Extends DEN functionality to accelerometer sensor. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                   |
| DEN_X     | DEN value stored in LSB of X-axis. Default value: 1<br>(0: DEN not stored in X-axis LSB; 1: DEN stored in X-axis LSB)                                                                                                              |
| DEN_Y     | DEN value stored in LSB of Y-axis. Default value: 1<br>(0: DEN not stored in Y-axis LSB; 1: DEN stored in Y-axis LSB)                                                                                                              |
| DEN_Z     | DEN value stored in LSB of Z-axis. Default value: 1<br>(0: DEN not stored in Z-axis LSB; 1: DEN stored in Z-axis LSB)                                                                                                              |
| DEN_XL_G  | DEN stamping sensor selection. Default value: 0<br>(0: DEN pin info stamped in the gyroscope axis selected by bits DEN_X, DEN_Y, DEN_Z;<br>1: DEN pin info stamped in the accelerometer axis selected by bits DEN_X, DEN_Y, DEN_Z) |

**Table 142. Trigger mode selection**

| LVL1_EN, LVL2_EN | Trigger mode                             |
|------------------|------------------------------------------|
| 10               | Level-sensitive trigger mode is selected |
| 11               | Level-sensitive latched mode is selected |



## 9.52 INACTIVITY\_DUR (54h)

Activity/inactivity configuration register (R/W)

**Table 143. INACTIVITY\_DUR register**

|                     |                  |                  |                  |                |                |             |             |
|---------------------|------------------|------------------|------------------|----------------|----------------|-------------|-------------|
| SLEEP_STATUS_ON_INT | WU_INACT_THS_W_2 | WU_INACT_THS_W_1 | WU_INACT_THS_W_0 | XL_INACT_ODR_1 | XL_INACT_ODR_0 | INACT_DUR_1 | INACT_DUR_0 |
|---------------------|------------------|------------------|------------------|----------------|----------------|-------------|-------------|

**Table 144. INACTIVITY\_DUR register description**

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLEEP_STATUS_ON_INT  | Activity/inactivity interrupt mode configuration.<br>If the INT1_SLEEP_CHANGE or INT2_SLEEP_CHANGE bit is enabled, drives the sleep status or sleep change on the INT pin. Default value: 0<br>(0: sleep change notification on INT pin;<br>1: sleep status reported on INT pin)                                                                                                                                                            |
| WU_INACT_THS_W_[2:0] | Weight of 1 LSB of wake-up (WU_THS) and activity/inactivity (INACT_THS) threshold.<br>(000: 7.8125 mg/LSB (default);<br>001: 15.625 mg/LSB;<br>010: 31.25 mg/LSB;<br>011: 62.5 mg/LSB;<br>100: 125 mg/LSB;<br>101 - 110 - 111: 250 mg/LSB)                                                                                                                                                                                                  |
| XL_INACT_ODR_[1:0]   | Selects the ODR_XL target during inactivity.<br>(00: 1.875 Hz;<br>01: 15 Hz (default);<br>10: 30 Hz;<br>11: 60 Hz)                                                                                                                                                                                                                                                                                                                          |
| INACT_DUR_[1:0]      | Duration in the transition from stationary to motion (from inactivity to activity).<br>(00: transition to motion (activity) immediately at first overthreshold event (default);<br>01: transition to motion (activity) after two consecutive overthreshold events;<br>10: transition to motion (activity) after three consecutive overthreshold events;<br>11: transition to motion (activity) after four consecutive overthreshold events) |

## 9.53 INACTIVITY\_THS (55h)

Activity/inactivity threshold setting register (R/W)

**Table 145. INACTIVITY\_THS register**

|                  |                  |             |             |             |             |             |             |
|------------------|------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | INACT_THS_5 | INACT_THS_4 | INACT_THS_3 | INACT_THS_2 | INACT_THS_1 | INACT_THS_0 |
|------------------|------------------|-------------|-------------|-------------|-------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 146. INACTIVITY\_THS register description**

|                 |                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INACT_THS_[5:0] | Activity/inactivity threshold. The resolution of the threshold depends on the value of WU_INACT_THS_W_[2:0] in the <a href="#">INACTIVITY_DUR (54h)</a> register. Default value: 000000 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 9.54

### TAP\_CFG0 (56h)

Tap configuration register 0 (R/W)

**Table 147. TAP\_CFG0 register**

| 0 <sup>(1)</sup> | LOW_PASS_ON_6D | HW_FUNC_MASK_XL_SETTL | SLOPE_FDS | TAP_X_EN | TAP_Y_EN | TAP_Z_EN | LIR |
|------------------|----------------|-----------------------|-----------|----------|----------|----------|-----|
|------------------|----------------|-----------------------|-----------|----------|----------|----------|-----|

1. This bit must be set to 0 for the correct operation of the device.

**Table 148. TAP\_CFG0 register description**

|                       |                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LOW_PASS_ON_6D        | LPF2 filter on 6D function selection. Refer to Figure 24. Default value: 0<br>(0: ODR/2 low-pass filtered data sent to 6D interrupt function;<br>1: LPF2 output data sent to 6D interrupt function)                                                                                                                                                                                      |
| HW_FUNC_MASK_XL_SETTL | Enables masking the execution trigger of the basic interrupt functions (6D/4D, free-fall, wake-up, tap, activity/inactivity) when accelerometer data are settling. Default value: 0<br>(0: disabled; 1: enabled)<br><i>Note: Refer to the product application note for the details regarding operating/power mode configurations, settings, turn-on/off time and on-the-fly changes.</i> |
| SLOPE_FDS             | HPF or slope filter selection on wake-up and activity/inactivity functions. Refer to Figure 24. Default value: 0<br>(0: slope filter applied; 1: HPF applied)                                                                                                                                                                                                                            |
| TAP_X_EN              | Enables X direction in tap recognition. Default value: 0<br>(0: X direction disabled; 1: X direction enabled)                                                                                                                                                                                                                                                                            |
| TAP_Y_EN              | Enables Y direction in tap recognition. Default value: 0<br>(0: Y direction disabled; 1: Y direction enabled)                                                                                                                                                                                                                                                                            |
| TAP_Z_EN              | Enables Z direction in tap recognition. Default value: 0<br>(0: Z direction disabled; 1: Z direction enabled)                                                                                                                                                                                                                                                                            |
| LIR                   | Latched interrupt. Default value: 0<br>(0: interrupt request not latched; 1: interrupt request latched)                                                                                                                                                                                                                                                                                  |



## 9.55

### TAP\_CFG1 (57h)

Tap configuration register 1 (R/W)

**Table 149. TAP\_CFG1 register**

|                |                |                |             |             |             |             |             |
|----------------|----------------|----------------|-------------|-------------|-------------|-------------|-------------|
| TAP_PRIORITY_2 | TAP_PRIORITY_1 | TAP_PRIORITY_0 | TAP_THS_X_4 | TAP_THS_X_3 | TAP_THS_X_2 | TAP_THS_X_1 | TAP_THS_X_0 |
|----------------|----------------|----------------|-------------|-------------|-------------|-------------|-------------|

**Table 150. TAP\_CFG1 register description**

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| TAP_PRIORITY_[2:0] | Selection of axis priority for tap detection (see Table 151)                            |
| TAP_THS_X_[4:0]    | X-axis tap recognition threshold. Default value: 0<br>1 LSB = FS_XL / (2 <sup>5</sup> ) |

**Table 151. TAP priority decoding**

| TAP_PRIORITY_[2:0] | Max. priority | Mid. priority | Min. priority |
|--------------------|---------------|---------------|---------------|
| 000                | X             | Y             | Z             |
| 001                | Y             | X             | Z             |
| 010                | X             | Z             | Y             |
| 011                | Z             | Y             | X             |
| 100                | X             | Y             | Z             |
| 101                | Y             | Z             | X             |
| 110                | Z             | X             | Y             |
| 111                | Z             | Y             | X             |

## 9.56

### TAP\_CFG2 (58h)

Tap configuration register 2 (R/W)

**Table 152. TAP\_CFG2 register**

|                  |                  |                  |             |             |             |             |             |
|------------------|------------------|------------------|-------------|-------------|-------------|-------------|-------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | TAP_THS_Y_4 | TAP_THS_Y_3 | TAP_THS_Y_2 | TAP_THS_Y_1 | TAP_THS_Y_0 |
|------------------|------------------|------------------|-------------|-------------|-------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 153. TAP\_CFG2 register description**

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| TAP_THS_Y_[4:0] | Y-axis tap recognition threshold. Default value: 0<br>1 LSB = FS_XL / (2 <sup>5</sup> ) |
|-----------------|-----------------------------------------------------------------------------------------|


**9.57**
**TAP\_THS\_6D (59h)**

Portrait/landscape position and tap function threshold register (R/W)

**Table 154. TAP\_THS\_6D register**

|        |            |            |             |             |             |             |             |
|--------|------------|------------|-------------|-------------|-------------|-------------|-------------|
| D4D_EN | SIXD_THS_1 | SIXD_THS_0 | TAP_THS_Z_4 | TAP_THS_Z_3 | TAP_THS_Z_2 | TAP_THS_Z_1 | TAP_THS_Z_0 |
|--------|------------|------------|-------------|-------------|-------------|-------------|-------------|

**Table 155. TAP\_THS\_6D register description**

|                 |                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------|
| D4D_EN          | Enables 4D orientation detection. Z-axis position detection is disabled. Default value: 0 (0: disabled; 1: enabled) |
| SIXD_THS_[1:0]  | Threshold for 4D/6D function. Default value: 00<br>For details, refer to <a href="#">Table 156</a> .                |
| TAP_THS_Z_[4:0] | Z-axis recognition threshold. Default value: 0<br>1 LSB = $FS_{XL} / (2^5)$                                         |

**Table 156. Threshold for D4D/D6D function**

| SIXD_THS_[1:0] | Threshold value |
|----------------|-----------------|
| 00             | 80 degrees      |
| 01             | 70 degrees      |
| 10             | 60 degrees      |
| 11             | 50 degrees      |



## 9.58 TAP\_DUR (5Ah)

Tap recognition function setting register (R/W)

**Table 157. TAP\_DUR register**

| DUR_3 | DUR_2 | DUR_1 | DUR_0 | QUIET_1 | QUIET_0 | SHOCK_1 | SHOCK_0 |
|-------|-------|-------|-------|---------|---------|---------|---------|
|-------|-------|-------|-------|---------|---------|---------|---------|

**Table 158. TAP\_DUR register description**

|             |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DUR_[3:0]   | Duration of maximum time gap for double-tap recognition. Default: 0000<br>When double-tap recognition is enabled, this register expresses the maximum time between two consecutive detected taps to determine a double-tap event. The default value of these bits is 0000b which corresponds to 16/ODR_XL time. If the DUR_[3:0] bits are set to a different value, 1LSB corresponds to 32/ODR_XL time. |
| QUIET_[1:0] | Expected quiet time after a tap detection. Default value: 00<br>Quiet time is the time after the first detected tap in which there must not be any overthreshold event. The default value of these bits is 00b which corresponds to 2/ODR_XL time. If the QUIET_[1:0] bits are set to a different value, 1LSB corresponds to 4/ODR_XL time.                                                             |
| SHOCK_[1:0] | Maximum duration of overthreshold event. Default value: 00<br>Maximum duration is the maximum time of an overthreshold signal detection to be recognized as a tap event. The default value of these bits is 00b which corresponds to 4/ODR_XL time. If the SHOCK_[1:0] bits are set to a different value, 1LSB corresponds to 8/ODR_XL time.                                                            |

## 9.59 WAKE\_UP\_THS (5Bh)

Single/double-tap selection and wake-up configuration (R/W)

**Table 159. WAKE\_UP\_THS register**

| SINGLE_DOUBLE_TAP | USR_OFF_ON_WU | WK_THS_5 | WK_THS_4 | WK_THS_3 | WK_THS_2 | WK_THS_1 | WK_THS_0 |
|-------------------|---------------|----------|----------|----------|----------|----------|----------|
|-------------------|---------------|----------|----------|----------|----------|----------|----------|

**Table 160. WAKE\_UP\_THS register description**

|                   |                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SINGLE_DOUBLE_TAP | Enables single/double-tap event. Default value: 0<br>(0: only single-tap event enabled;<br>1: both single and double-tap events enabled)                                                                       |
| USR_OFF_ON_WU     | Drives the low-pass filtered data with user offset correction (instead of high-pass filtered data) to the wake-up and the activity/inactivity functions. Refer to <a href="#">Figure 24</a> . Default value: 0 |
| WK_THS_[5:0]      | Wake-up threshold. The resolution of the threshold depends on the value of WU_INACT_THS_W_[2:0] in the <a href="#">INACTIVITY_DUR (54h)</a> register. Default value: 000000                                    |



## 9.60 WAKE\_UP\_DUR (5Ch)

Free-fall, wake-up, and sleep mode functions duration setting register (R/W)

**Table 161. WAKE\_UP\_DUR register**

|          |            |            |                  |             |             |             |             |
|----------|------------|------------|------------------|-------------|-------------|-------------|-------------|
| FF_DUR_5 | WAKE_DUR_1 | WAKE_DUR_0 | 0 <sup>(1)</sup> | SLEEP_DUR_3 | SLEEP_DUR_2 | SLEEP_DUR_1 | SLEEP_DUR_0 |
|----------|------------|------------|------------------|-------------|-------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 162. WAKE\_UP\_DUR register description**

|                |                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FF_DUR_5       | Free-fall duration event. Default: 0<br>For the complete configuration of the free-fall duration, refer to FF_DUR_4:0] in the <a href="#">FREE_FALL (5Dh)</a> configuration.<br>1 LSB = 1/ODR_XL time |
| WAKE_DUR_1:0]  | Wake-up duration event. Default: 00<br>1 LSB = 1/ODR_XL time                                                                                                                                          |
| SLEEP_DUR_3:0] | Duration to go in sleep mode. Default value: 0000 (this corresponds to 18 ODR)<br>1 LSB = 2 + 512/ODR_XL time                                                                                         |

## 9.61 FREE\_FALL (5Dh)

Free-fall function duration setting register (R/W)

**Table 163. FREE\_FALL register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| FF_DUR_4 | FF_DUR_3 | FF_DUR_2 | FF_DUR_1 | FF_DUR_0 | FF_THS_2 | FF_THS_1 | FF_THS_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 164. FREE\_FALL register description**

|             |                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FF_DUR_4:0] | Free-fall duration event. Default: 00000<br>For the complete configuration of the free-fall duration, refer to FF_DUR_5 in the <a href="#">WAKE_UP_DUR (5Ch)</a> configuration. |
| FF_THS_2:0] | Free-fall threshold setting. Default: 000<br>For details refer to <a href="#">Table 165</a> .                                                                                   |

**Table 165. Threshold for free-fall function**

| FF_THS_2:0] | Threshold value |
|-------------|-----------------|
| 000         | 156 mg          |
| 001         | 219 mg          |
| 010         | 250 mg          |
| 011         | 312 mg          |
| 100         | 344 mg          |
| 101         | 406 mg          |
| 110         | 469 mg          |
| 111         | 500 mg          |



## 9.62

### MD1\_CFG (5Eh)

Functions routing to INT1 pin register (R/W). Each bit in this register enables a signal to be carried over the INT1 pin. The output of the pin is the OR combination of the signals selected here and in the **INT1\_CTRL (0Dh)** register.

**Table 166. MD1\_CFG register**

| INT1_SLEEP_CHANGE | INT1_SINGLE_TAP | INT1_WU | INT1_FF | INT1_DOUBLE_TAP | INT1_6D | INT1_EMB_FUNC | INT1_SHUB |
|-------------------|-----------------|---------|---------|-----------------|---------|---------------|-----------|
|-------------------|-----------------|---------|---------|-----------------|---------|---------------|-----------|

**Table 167. MD1\_CFG register description**

|                                  |                                                                                                                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT1_SLEEP_CHANGE <sup>(1)</sup> | Routing activity/inactivity recognition event to INT1. Default: 0<br>(0: routing activity/inactivity event to INT1 disabled;<br>1: routing activity/inactivity event to INT1 enabled)                                        |
| INT1_SINGLE_TAP                  | Routing single-tap recognition event to INT1. Default: 0<br>(0: routing single-tap event to INT1 disabled;<br>1: routing single-tap event to INT1 enabled)                                                                   |
| INT1_WU                          | Routing wake-up event to INT1. Default value: 0<br>(0: routing wake-up event to INT1 disabled;<br>1: routing wake-up event to INT1 enabled)                                                                                  |
| INT1_FF                          | Routing free-fall event to INT1. Default value: 0<br>(0: routing free-fall event to INT1 disabled;<br>1: routing free-fall event to INT1 enabled)                                                                            |
| INT1_DOUBLE_TAP                  | Routing tap event to INT1. Default value: 0<br>(0: routing double-tap event to INT1 disabled;<br>1: routing double-tap event to INT1 enabled)                                                                                |
| INT1_6D                          | Routing 6D event to INT1. Default value: 0<br>(0: routing 6D event to INT1 disabled;<br>1: routing 6D event to INT1 enabled)                                                                                                 |
| INT1_EMB_FUNC                    | Routing embedded functions event to INT1. Default value: 0<br>(0: routing embedded functions event to INT1 disabled;<br>1: routing embedded functions event to INT1 enabled)                                                 |
| INT1_SHUB                        | Routing sensor hub communication concluded event to INT1. Default value: 0<br>(0: routing sensor hub communication concluded event to INT1 disabled;<br>1: routing sensor hub communication concluded event to INT1 enabled) |

1. Activity/inactivity interrupt mode (sleep change or sleep status) depends on the **SLEEP\_STATUS\_ON\_INT** bit in the **INACTIVITY\_DUR (54h)** register.





## 9.63

### MD2\_CFG (5Fh)

Functions routing to INT2 pin register (R/W). Each bit in this register enables a signal to be carried over the INT2 pin. The output of the pin is the OR combination of the signals selected here and in the **INT2\_CTRL (0Eh)** register.

**Table 168. MD2\_CFG register**

| INT2_SLEEP_CHANGE | INT2_SINGLE_TAP | INT2_WU | INT2_FF | INT2_DOUBLE_TAP | INT2_6D | INT2_EMB_FUNC | INT2_TIMESTAMP |
|-------------------|-----------------|---------|---------|-----------------|---------|---------------|----------------|
|-------------------|-----------------|---------|---------|-----------------|---------|---------------|----------------|

**Table 169. MD2\_CFG register description**

|                                  |                                                                                                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT2_SLEEP_CHANGE <sup>(1)</sup> | Routing activity/inactivity recognition event to INT2. Default: 0<br>(0: routing activity/inactivity event to INT2 disabled;<br>1: routing activity/inactivity event to INT2 enabled) |
| INT2_SINGLE_TAP                  | Single-tap recognition routing to INT2. Default: 0<br>(0: routing single-tap event to INT2 disabled;<br>1: routing single-tap event to INT2 enabled)                                  |
| INT2_WU                          | Routing wake-up event to INT2. Default value: 0<br>(0: routing wake-up event to INT2 disabled;<br>1: routing wake-up event to INT2 enabled)                                           |
| INT2_FF                          | Routing free-fall event to INT2. Default value: 0<br>(0: routing free-fall event to INT2 disabled;<br>1: routing free-fall event to INT2 enabled)                                     |
| INT2_DOUBLE_TAP                  | Routing tap event to INT2. Default value: 0<br>(0: routing double-tap event to INT2 disabled;<br>1: routing double-tap event to INT2 enabled)                                         |
| INT2_6D                          | Routing 6D event to INT2. Default value: 0<br>(0: routing 6D event to INT2 disabled;<br>1: routing 6D event to INT2 enabled)                                                          |
| INT2_EMB_FUNC                    | Routing embedded functions event to INT2. Default value: 0<br>(0: routing embedded functions event to INT2 disabled;<br>1: routing embedded functions event to INT2 enabled)          |
| INT2_TIMESTAMP                   | Enables routing the alert for timestamp overflow within 5.6 ms to the INT2 pin.                                                                                                       |

1. Activity/inactivity interrupt mode (sleep change or sleep status) depends on the **SLEEP\_STATUS\_ON\_INT** bit in the **INACTIVITY\_DUR (54h)** register.

## 9.64

### HAODR\_CFG (62h)

HAODR data rate configuration register (R/W)

**Table 170. HAODR\_CFG register**

|                  |                  |                  |                  |                  |                  |             |             |
|------------------|------------------|------------------|------------------|------------------|------------------|-------------|-------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | HAODR_SEL_1 | HAODR_SEL_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 171. HAODR\_CFG register description**

|                 |                                                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HAODR_SEL_[1:0] | Selects the ODR set supported when high-accuracy ODR (HAODR) mode is enabled (see Table 19. Accelerometer and gyroscope ODR selection in high-accuracy ODR mode).<br>Default: 00 |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 9.65

### EMB\_FUNC\_CFG (63h)

Embedded functions configuration register (R/W)

**Table 172. EMB\_FUNC\_CFG register**

|                        |                  |                           |                            |                  |                  |                  |                  |
|------------------------|------------------|---------------------------|----------------------------|------------------|------------------|------------------|------------------|
| XL_DualC_BATCH_FROM_IF | 0 <sup>(1)</sup> | EMB_FUNC_IRQ_MASK_G_SETTL | EMB_FUNC_IRQ_MASK_XL_SETTL | EMB_FUNC_DISABLE | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|------------------------|------------------|---------------------------|----------------------------|------------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 173. EMB\_FUNC\_CFG register description**

|                            |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XL_DualC_BATCH_FROM_IF     | When dual-channel mode is enabled, this bit enables batching the accelerometer channel 2 in FIFO. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                                                                                                               |
| EMB_FUNC_IRQ_MASK_G_SETTL  | Enables / masks execution trigger of the embedded functions when gyroscope data are settling. Default value: 0<br>(0: disabled;<br>1: masks execution trigger of the embedded functions until gyroscope filter settling ends)<br><i>Note: Refer to the product application note for the details regarding operating/power mode configurations, settings, turn-on/off time and on-the-fly changes.</i>         |
| EMB_FUNC_IRQ_MASK_XL_SETTL | Enables / masks execution trigger of the embedded functions when accelerometer data are settling. Default value: 0<br>(0: disabled;<br>1: masks execution trigger of the embedded functions until accelerometer filter settling ends)<br><i>Note: Refer to the product application note for the details regarding operating/power mode configurations, settings, turn-on/off time and on-the-fly changes.</i> |
| EMB_FUNC_DISABLE           | Disables execution of the embedded functions. Default value: 0<br>(0: disabled;<br>1: embedded functions execution trigger is not generated anymore and all initialization procedures are forced when this bit is set back to 0).                                                                                                                                                                             |


**9.66**
**CTRL\_EIS (6Bh)**

Gyroscope EIS channel control register (R/W)

**Table 174. CTRL\_EIS register**

|             |             |                  |              |                  |            |            |            |
|-------------|-------------|------------------|--------------|------------------|------------|------------|------------|
| ODR_G_EIS_1 | ODR_G_EIS_0 | 0 <sup>(1)</sup> | LPF_G_EIS_BW | 0 <sup>(1)</sup> | FS_G_EIS_2 | FS_G_EIS_1 | FS_G_EIS_0 |
|-------------|-------------|------------------|--------------|------------------|------------|------------|------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 175. CTRL\_EIS register description**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ODR_G_EIS_[1:0] | Enables and selects the ODR of the gyroscope EIS channel.<br>(00: EIS channel is off (default);<br>01: 1.92 kHz;<br>10: 960 Hz;<br>11: reserved)                                                                                                                                                                                                                                                                                                                                                         |
| LPF_G_EIS_BW    | Gyroscope digital LPF_EIS filter bandwidth selection. Refer to Table 176.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| FS_G_EIS_[2:0]  | Gyroscope full-scale selection for EIS channel. If the FS_G_[3:0] bits in CTRL6 (15h) are equal to 1100 (±4000 dps), FS_G_EIS_[2:0] must be set to "100" in order to have ±4000 dps full scale on both UI and EIS channels. If the FS_G_3 bit in register CTRL6 (15h) is equal to 0, the EIS channel full scale can be selected as follows:<br>(000: ±125 dps (default);<br>001: ±250 dps;<br>010: ±500 dps;<br>011: ±1000 dps;<br>100: ±2000 dps;<br>101: reserved;<br>110: reserved;<br>111: reserved) |

**Table 176. Gyroscope EIS chain digital LPF\_EIS filter bandwidth selection**

| ODR_G_EIS_[1:0] | Gyroscope EIS ODR [Hz] | LPF_G_EIS_BW | Cutoff [Hz] | Phase @ 20 Hz [°] |
|-----------------|------------------------|--------------|-------------|-------------------|
| 01              | 1.92 kHz               | 0            | 153 Hz      | -13.5°            |
|                 |                        | 1            | 203 Hz      | -10.8°            |
| 10              | 960                    | 0            | 148 Hz      | -15.4°            |
|                 |                        | 1            | 193 Hz      | -12.7°            |



## 9.67 X\_OFS\_USR (73h)

Accelerometer X-axis user offset correction (R/W). The offset value set in the X\_OFS\_USR offset register is internally added to the acceleration value measured on the X-axis.

**Table 177. X\_OFS\_USR register**

| X_OFS_USR_7 | X_OFS_USR_6 | X_OFS_USR_5 | X_OFS_USR_4 | X_OFS_USR_3 | X_OFS_USR_2 | X_OFS_USR_1 | X_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 178. X\_OFS\_USR register description**

|                |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X_OFS_USR[7:0] | Accelerometer X-axis user offset correction, expressed in two's complement, weight depends on USR_OFF_W in CTRL9 (18h). The offset can be applied to the output registers (see USR_OFF_ON_OUT bit in the CTRL9 (18h) register) or to the wake-up function input data (see USR_OFF_ON_WU bit in the WAKE_UP_THS (5Bh) register). The value must be in the range [-127 127]. |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.68 Y\_OFS\_USR (74h)

Accelerometer Y-axis user offset correction (R/W). The offset value set in the Y\_OFS\_USR offset register is internally added to the acceleration value measured on the Y-axis.

**Table 179. Y\_OFS\_USR register**

| Y_OFS_USR_7 | Y_OFS_USR_6 | Y_OFS_USR_5 | Y_OFS_USR_4 | Y_OFS_USR_3 | Y_OFS_USR_2 | Y_OFS_USR_1 | Y_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 180. Y\_OFS\_USR register description**

|                |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y_OFS_USR[7:0] | Accelerometer Y-axis user offset correction, expressed in two's complement, weight depends on USR_OFF_W in CTRL9 (18h). The offset can be applied to the output registers (see USR_OFF_ON_OUT bit in the CTRL9 (18h) register) or to the wake-up function input data (see USR_OFF_ON_WU bit in the WAKE_UP_THS (5Bh) register). The value must be in the range [-127 127]. |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.69 Z\_OFS\_USR (75h)

Accelerometer Z-axis user offset correction (R/W). The offset value set in the Z\_OFS\_USR offset register is internally added to the acceleration value measured on the Z-axis.

**Table 181. Z\_OFS\_USR register**

| Z_OFS_USR_7 | Z_OFS_USR_6 | Z_OFS_USR_5 | Z_OFS_USR_4 | Z_OFS_USR_3 | Z_OFS_USR_2 | Z_OFS_USR_1 | Z_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 182. Z\_OFS\_USR register description**

|                |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Z_OFS_USR[7:0] | Accelerometer Z-axis user offset correction, expressed in two's complement, weight depends on USR_OFF_W in CTRL9 (18h). The offset can be applied to the output registers (see USR_OFF_ON_OUT bit in the CTRL9 (18h) register) or to the wake-up function input data (see USR_OFF_ON_WU bit in the WAKE_UP_THS (5Bh) register). The value must be in the range [-127 127]. |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 9.70 FIFO\_DATA\_OUT\_TAG (78h)

FIFO tag register (R)

**Table 183. FIFO\_DATA\_OUT\_TAG register**

| TAG_SENSOR_4 | TAG_SENSOR_3 | TAG_SENSOR_2 | TAG_SENSOR_1 | TAG_SENSOR_0 | TAG_CNT_1 | TAG_CNT_0 | - |
|--------------|--------------|--------------|--------------|--------------|-----------|-----------|---|
|--------------|--------------|--------------|--------------|--------------|-----------|-----------|---|

**Table 184. FIFO\_DATA\_OUT\_TAG register description**

|                  |                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAG_SENSOR_[4:0] | FIFO tag. Identifies the sensor in:<br>FIFO_DATA_OUT_X_L (79h) and FIFO_DATA_OUT_X_H (7Ah), FIFO_DATA_OUT_Y_L (7Bh) and FIFO_DATA_OUT_Y_H (7Ch), and FIFO_DATA_OUT_Z_L (7Dh) and FIFO_DATA_OUT_Z_H (7Eh)<br>For details, refer to Table 185. |
| TAG_CNT_[1:0]    | 2-bit counter which identifies sensor time slot                                                                                                                                                                                              |

**Table 185. FIFO tag**

| TAG_SENSOR_[4:0] | Sensor name               |
|------------------|---------------------------|
| 0x00             | FIFO empty                |
| 0x01             | Gyroscope NC              |
| 0x02             | Accelerometer NC          |
| 0x03             | Temperature               |
| 0x04             | Timestamp                 |
| 0x05             | CFG_Change                |
| 0x06             | Accelerometer NC_T_2      |
| 0x07             | Accelerometer NC_T_1      |
| 0x08             | Accelerometer 2xC         |
| 0x09             | Accelerometer 3xC         |
| 0x0A             | Gyroscope NC_T_2          |
| 0x0B             | Gyroscope NC_T_1          |
| 0x0C             | Gyroscope 2xC             |
| 0x0D             | Gyroscope 3xC             |
| 0x0E             | Sensor hub slave 0        |
| 0x0F             | Sensor hub slave 1        |
| 0x10             | Sensor hub slave 2        |
| 0x11             | Sensor hub slave 3        |
| 0x12             | Step counter              |
| 0x13             | SFLP game rotation vector |
| 0x16             | SFLP gyroscope bias       |
| 0x17             | SFLP gravity vector       |
| 0x19             | Sensor hub nack           |
| 0x1D             | Accelerometer dualC       |
| 0x1E             | Enhanced EIS gyroscope    |



### 9.71 FIFO\_DATA\_OUT\_X\_L (79h) and FIFO\_DATA\_OUT\_X\_H (7Ah)

FIFO data output X (R)

**Table 186.** FIFO\_DATA\_OUT\_X\_H and FIFO\_DATA\_OUT\_X\_L registers

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
| D7  | D6  | D5  | D4  | D3  | D2  | D1 | D0 |

**Table 187.** FIFO\_DATA\_OUT\_X\_H and FIFO\_DATA\_OUT\_X\_L register description

|         |                    |
|---------|--------------------|
| D[15:0] | FIFO X-axis output |
|---------|--------------------|

### 9.72 FIFO\_DATA\_OUT\_Y\_L (7Bh) and FIFO\_DATA\_OUT\_Y\_H (7Ch)

FIFO data output Y (R)

**Table 188.** FIFO\_DATA\_OUT\_Y\_H and FIFO\_DATA\_OUT\_Y\_L registers

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
| D7  | D6  | D5  | D4  | D3  | D2  | D1 | D0 |

**Table 189.** FIFO\_DATA\_OUT\_Y\_H and FIFO\_DATA\_OUT\_Y\_L register description

|         |                    |
|---------|--------------------|
| D[15:0] | FIFO Y-axis output |
|---------|--------------------|

### 9.73 FIFO\_DATA\_OUT\_Z\_L (7Dh) and FIFO\_DATA\_OUT\_Z\_H (7Eh)

FIFO data output Z (R)

**Table 190.** FIFO\_DATA\_OUT\_Z\_H and FIFO\_DATA\_OUT\_Z\_L registers

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
| D7  | D6  | D5  | D4  | D3  | D2  | D1 | D0 |

**Table 191.** FIFO\_DATA\_OUT\_Z\_H and FIFO\_DATA\_OUT\_Z\_L register description

|         |                    |
|---------|--------------------|
| D[15:0] | FIFO Z-axis output |
|---------|--------------------|



## 10 Embedded functions register mapping

The table given below provides a list of the registers for the embedded functions available in the device and the corresponding addresses. Embedded functions registers are accessible when EMB\_FUNC\_REG\_ACCESS is set to 1 in FUNC\_CFG\_ACCESS (01h).

**Table 192. Register address map - embedded functions**

| Name                 | Type | Register address |          | Default  | Comment |
|----------------------|------|------------------|----------|----------|---------|
|                      |      | Hex              | Binary   |          |         |
| PAGE_SEL             | R/W  | 02               | 00000010 | 00000001 |         |
| EMB_FUNC_EN_A        | R/W  | 04               | 00000100 | 00000000 |         |
| EMB_FUNC_EN_B        | R/W  | 05               | 00000101 | 00000000 |         |
| EMB_FUNC_EXEC_STATUS | R    | 07               | 00000111 | output   |         |
| PAGE_ADDRESS         | R/W  | 08               | 00001000 | 00000000 |         |
| PAGE_VALUE           | R/W  | 09               | 00001001 | 00000000 |         |
| EMB_FUNC_INT1        | R/W  | 0A               | 00001010 | 00000000 |         |
| FSM_INT1             | R/W  | 0B               | 00001011 | 00000000 |         |
| RESERVED             | -    | 0C-0D            |          |          |         |
| EMB_FUNC_INT2        | R/W  | 0E               | 00001110 | 00000000 |         |
| FSM_INT2             | R/W  | 0F               | 00001111 | 00000000 |         |
| RESERVED             | -    | 10-11            |          |          |         |
| EMB_FUNC_STATUS      | R    | 12               | 00010010 | output   |         |
| FSM_STATUS           | R    | 13               | 00010011 | output   |         |
| RESERVED             | -    | 14-15            |          |          |         |
| PAGE_RW              | R/W  | 17               | 00010111 | 00000000 |         |
| RESERVED             | -    | 18-43            |          |          |         |
| EMB_FUNC_FIFO_EN_A   | R/W  | 44               | 01000100 | 00000000 |         |
| RESERVED             | -    | 45               |          |          |         |
| FSM_ENABLE           | R/W  | 46               | 01000110 | 00000000 |         |
| RESERVED             | -    | 47               |          |          |         |
| FSM_LONG_COUNTER_L   | R/W  | 48               | 01001000 | 00000000 |         |
| FSM_LONG_COUNTER_H   | R/W  | 49               | 01001001 | 00000000 |         |
| RESERVED             | -    | 4A               |          |          |         |
| INT_ACK_MASK         | R/W  | 4B               | 01001011 | 00000000 |         |
| FSM_OUTS1            | R    | 4C               | 01001100 | output   |         |
| FSM_OUTS2            | R    | 4D               | 01001101 | output   |         |
| FSM_OUTS3            | R    | 4E               | 01001110 | output   |         |
| FSM_OUTS4            | R    | 4F               | 01001111 | output   |         |
| FSM_OUTS5            | R    | 50               | 01010000 | output   |         |
| FSM_OUTS6            | R    | 51               | 01010001 | output   |         |
| FSM_OUTS7            | R    | 52               | 01010010 | output   |         |
| FSM_OUTS8            | R    | 53               | 01010011 | output   |         |
| RESERVED             | -    | 54- 5D           |          |          |         |



| Name            | Type | Register address |          | Default  | Comment |
|-----------------|------|------------------|----------|----------|---------|
|                 |      | Hex              | Binary   |          |         |
| SFLP_ODR        | R/W  | 5E               | 01011110 | 01011011 |         |
| FSM_ODR         | R/W  | 5F               | 01011111 | 01001011 |         |
| RESERVED        | -    | 60               |          |          |         |
| STEP_COUNTER_L  | R    | 62               | 01100010 | output   |         |
| STEP_COUNTER_H  | R    | 63               | 01100011 | output   |         |
| EMB_FUNC_SRC    | R/W  | 64               | 01100100 | output   |         |
| EMB_FUNC_INIT_A | R/W  | 66               | 01100110 | 00000000 |         |
| EMB_FUNC_INIT_B | R/W  | 67               | 01100111 | 00000000 |         |

Reserved registers must not be changed. Writing to those registers may cause permanent damage to the device. The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.





## 11 Embedded functions register description

### 11.1 PAGE\_SEL (02h)

Enable advanced features dedicated page (R/W)

Table 193. PAGE\_SEL register

|           |           |           |           |                  |                  |                  |                  |
|-----------|-----------|-----------|-----------|------------------|------------------|------------------|------------------|
| PAGE_SEL3 | PAGE_SEL2 | PAGE_SEL1 | PAGE_SEL0 | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 1 <sup>(2)</sup> |
|-----------|-----------|-----------|-----------|------------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.
2. This bit must be set to 1 for the correct operation of the device.

Table 194. PAGE\_SEL register description

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| PAGE_SEL[3:0] | Selects the advanced features dedicated page. Default value: 0000 |
|---------------|-------------------------------------------------------------------|

### 11.2 EMB\_FUNC\_EN\_A (04h)

Enable embedded functions register (R/W)

Table 195. EMB\_FUNC\_EN\_A register

|                  |                  |                |         |         |                  |              |                  |
|------------------|------------------|----------------|---------|---------|------------------|--------------|------------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | SIGN_MOTION_EN | TILT_EN | PEDO_EN | 0 <sup>(1)</sup> | SFLP_GAME_EN | 0 <sup>(1)</sup> |
|------------------|------------------|----------------|---------|---------|------------------|--------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

Table 196. EMB\_FUNC\_EN\_A register description

|                |                                                                                                                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SIGN_MOTION_EN | Enables significant motion detection function. Default value: 0<br>(0: significant motion detection function disabled;<br>1: significant motion detection function enabled)                                                                                                       |
| TILT_EN        | Enables tilt calculation. Default value: 0<br>(0: tilt algorithm disabled;<br>1: tilt algorithm enabled)                                                                                                                                                                          |
| PEDO_EN        | Enables pedometer algorithm. Default value: 0<br>(0: pedometer algorithm disabled;<br>1: pedometer algorithm enabled)                                                                                                                                                             |
| SFLP_GAME_EN   | Enables sensor fusion low-power algorithm for 6-axis (accelerometer + gyroscope) game rotation vector. Default value: 0<br>(0: sensor fusion algorithm for 6-axis accelerometer + gyroscope disabled;<br>1: sensor fusion algorithm for 6-axis accelerometer + gyroscope enabled) |



## 11.3

**EMB\_FUNC\_EN\_B (05h)**

Enable embedded functions register (R/W)

**Table 197. EMB\_FUNC\_EN\_B register**

|                  |                  |                  |             |               |                  |                  |        |
|------------------|------------------|------------------|-------------|---------------|------------------|------------------|--------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | PEDO_ADV_EN | FIFO_COMPR_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | FSM_EN |
|------------------|------------------|------------------|-------------|---------------|------------------|------------------|--------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 198. EMB\_FUNC\_EN\_B register description**

|                              |                                                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| PEDO_ADV_EN                  | Enables pedometer advanced features. Default value: 0<br>(0: pedometer advanced features block disabled;<br>1: pedometer advanced features block enabled) |
| FIFO_COMPR_EN <sup>(1)</sup> | Enables FIFO compression function. Default value: 0<br>(0: FIFO compression function disabled;<br>1: FIFO compression function enabled)                   |
| FSM_EN                       | Enables finite state machine (FSM) function. Default value: 0<br>(0: FSM function disabled; 1: FSM function enabled)                                      |

1. This bit is activated if the FIFO\_COMPR\_RT\_EN bit of FIFO\_CTRL2 (08h) is set to 1.

## 11.4

**EMB\_FUNC\_EXEC\_STATUS (07h)**

Embedded functions execution status register (R)

**Table 199. EMB\_FUNC\_EXEC\_STATUS register**

|   |   |   |   |   |   |                   |                |
|---|---|---|---|---|---|-------------------|----------------|
| 0 | 0 | 0 | 0 | 0 | 0 | EMB_FUNC_EXEC_OVR | EMB_FUNC_ENDOP |
|---|---|---|---|---|---|-------------------|----------------|

**Table 200. EMB\_FUNC\_EXEC\_STATUS register description**

|                   |                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMB_FUNC_EXEC_OVR | This bit is set to 1 when the execution of the embedded functions program exceeds maximum time (new data are generated before the end of the algorithms). Default value: 0 |
| EMB_FUNC_ENDOP    | When this bit is set to 1, no embedded function is running. Default value: 0                                                                                               |



## 11.5 PAGE\_ADDRESS (08h)

Page address register (R/W)

**Table 201. PAGE\_ADDRESS register**

| PAGE_ADDR7 | PAGE_ADDR6 | PAGE_ADDR5 | PAGE_ADDR4 | PAGE_ADDR3 | PAGE_ADDR2 | PAGE_ADDR1 | PAGE_ADDR0 |
|------------|------------|------------|------------|------------|------------|------------|------------|
|------------|------------|------------|------------|------------|------------|------------|------------|

**Table 202. PAGE\_ADDRESS register description**

|                |                                                                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PAGE_ADDR[7:0] | After setting the bit PAGE_WRITE / PAGE_READ in register <a href="#">PAGE_RW (17h)</a> , this register is used to set the address of the register to be written/read in the advanced features page selected through the bits PAGE_SEL[3:0] in register <a href="#">PAGE_SEL (02h)</a> . |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 11.6 PAGE\_VALUE (09h)

Page value register (R/W)

**Table 203. PAGE\_VALUE register**

| PAGE_VALUE7 | PAGE_VALUE6 | PAGE_VALUE5 | PAGE_VALUE4 | PAGE_VALUE3 | PAGE_VALUE2 | PAGE_VALUE1 | PAGE_VALUE0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 204. PAGE\_VALUE register description**

|                 |                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PAGE_VALUE[7:0] | These bits are used to write (if the bit PAGE_WRITE = 1 in register <a href="#">PAGE_RW (17h)</a> ) or read (if the bit PAGE_READ = 1 in register <a href="#">PAGE_RW (17h)</a> ) the data at the address PAGE_ADDR[7:0] of the selected advanced features page. |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 11.7 EMB\_FUNC\_INT1 (0Ah)

INT1 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT1. The pin's output supplies the OR combination of the selected signals.

**Table 205. EMB\_FUNC\_INT1 register**

|             |                  |              |           |                    |                  |                  |                  |
|-------------|------------------|--------------|-----------|--------------------|------------------|------------------|------------------|
| INT1_FSM_LC | 0 <sup>(1)</sup> | INT1_SIG_MOT | INT1_TILT | INT1_STEP_DETECTOR | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|-------------|------------------|--------------|-----------|--------------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 206. EMB\_FUNC\_INT1 register description**

|                                   |                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| INT1_FSM_LC <sup>(1)</sup>        | Routing FSM long counter timeout interrupt event to INT1. Default value: 0 (0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_SIG_MOT <sup>(1)</sup>       | Routing significant motion event to INT1. Default value: 0 (0: routing to INT1 disabled; 1: routing to INT1 enabled)                 |
| INT1_TILT <sup>(1)</sup>          | Routing tilt event to INT1. Default value: 0 (0: routing to INT1 disabled; 1: routing to INT1 enabled)                               |
| INT1_STEP_DETECTOR <sup>(1)</sup> | Routing pedometer step recognition event to INT1. Default value: 0 (0: routing to INT1 disabled; 1: routing to INT1 enabled)         |

1. This bit is activated if the INT1\_EMB\_FUNC bit of [MD1\\_CFG \(5Eh\)](#) is set to 1.



## 11.8

**FSM\_INT1 (0Bh)**

INT1 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT1. The pin's output supplies the OR combination of the selected signals.

**Table 207. FSM\_INT1 register**

| INT1_FSM8 | INT1_FSM7 | INT1_FSM6 | INT1_FSM5 | INT1_FSM4 | INT1_FSM3 | INT1_FSM2 | INT1_FSM1 |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|

**Table 208. FSM\_INT1 register description**

|                          |                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| INT1_FSM8 <sup>(1)</sup> | Routing FSM8 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM7 <sup>(1)</sup> | Routing FSM7 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM6 <sup>(1)</sup> | Routing FSM6 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM5 <sup>(1)</sup> | Routing FSM5 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM4 <sup>(1)</sup> | Routing FSM4 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM3 <sup>(1)</sup> | Routing FSM3 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM2 <sup>(1)</sup> | Routing FSM2 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |
| INT1_FSM1 <sup>(1)</sup> | Routing FSM1 interrupt event to INT1. Default value: 0<br>(0: routing to INT1 disabled; 1: routing to INT1 enabled) |

1. This bit is activated if the INT1\_EMB\_FUNC bit of MD1\_CFG (5Eh) is set to 1.



## 11.9

**EMB\_FUNC\_INT2 (0Eh)**

INT2 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT2. The pin's output supplies the OR combination of the selected signals.

**Table 209. EMB\_FUNC\_INT2 register**

|             |                  |              |           |                    |                  |                  |                  |
|-------------|------------------|--------------|-----------|--------------------|------------------|------------------|------------------|
| INT2_FSM_LC | 0 <sup>(1)</sup> | INT2_SIG_MOT | INT2_TILT | INT2_STEP_DETECTOR | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|-------------|------------------|--------------|-----------|--------------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 210. EMB\_FUNC\_INT2 register description**

|                                   |                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| INT2_FSM_LC <sup>(1)</sup>        | Routing FSM long counter timeout interrupt event to INT2. Default value: 0 (0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_SIG_MOT <sup>(1)</sup>       | Routing significant motion event to INT2. Default value: 0 (0: routing to INT2 disabled; 1: routing to INT2 enabled)                 |
| INT2_TILT <sup>(1)</sup>          | Routing tilt event to INT2. Default value: 0 (0: routing to INT2 disabled; 1: routing to INT2 enabled)                               |
| INT2_STEP_DETECTOR <sup>(1)</sup> | Routing pedometer step recognition event to INT2. Default value: 0 (0: routing to INT2 disabled; 1: routing to INT2 enabled)         |

1. This bit is activated if the INT2\_EMB\_FUNC bit of MD2\_CFG (5Fh) is set to 1.



## 11.10 FSM\_INT2 (0Fh)

INT2 pin control register (R/W)

Each bit in this register enables a signal to be carried over INT2. The pin's output supplies the OR combination of the selected signals.

**Table 211. FSM\_INT2 register**

| INT2_FSM8 | INT2_FSM7 | INT2_FSM6 | INT2_FSM5 | INT2_FSM4 | INT2_FSM3 | INT2_FSM2 | INT2_FSM1 |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|

**Table 212. FSM\_INT2 register description**

|                          |                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| INT2_FSM8 <sup>(1)</sup> | Routing FSM8 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM7 <sup>(1)</sup> | Routing FSM7 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM6 <sup>(1)</sup> | Routing FSM6 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM5 <sup>(1)</sup> | Routing FSM5 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM4 <sup>(1)</sup> | Routing FSM4 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM3 <sup>(1)</sup> | Routing FSM3 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM2 <sup>(1)</sup> | Routing FSM2 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |
| INT2_FSM1 <sup>(1)</sup> | Routing FSM1 interrupt event to INT2. Default value: 0<br>(0: routing to INT2 disabled; 1: routing to INT2 enabled) |

1. This bit is activated if the INT2\_EMB\_FUNC bit of MD2\_CFG (5Fh) is set to 1.

## 11.11 EMB\_FUNC\_STATUS (12h)

Embedded function status register (R)

**Table 213. EMB\_FUNC\_STATUS register**

|           |   |           |         |             |   |   |   |
|-----------|---|-----------|---------|-------------|---|---|---|
| IS_FSM_LC | 0 | IS_SIGMOT | IS_TILT | IS_STEP_DET | 0 | 0 | 0 |
|-----------|---|-----------|---------|-------------|---|---|---|

**Table 214. EMB\_FUNC\_STATUS register description**

|             |                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------|
| IS_FSM_LC   | Interrupt status bit for FSM long counter timeout interrupt event.<br>(1: interrupt detected; 0: no interrupt) |
| IS_SIGMOT   | Interrupt status bit for significant motion detection<br>(1: interrupt detected; 0: no interrupt)              |
| IS_TILT     | Interrupt status bit for tilt detection<br>(1: interrupt detected; 0: no interrupt)                            |
| IS_STEP_DET | Interrupt status bit for step detection<br>(1: interrupt detected; 0: no interrupt)                            |



## 11.12 FSM\_STATUS (13h)

Finite state machine status register (R)

**Table 215. FSM\_STATUS register**

| IS_FSM8 | IS_FSM7 | IS_FSM6 | IS_FSM5 | IS_FSM4 | IS_FSM3 | IS_FSM2 | IS_FSM1 |
|---------|---------|---------|---------|---------|---------|---------|---------|
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 216. FSM\_STATUS register description**

|         |                                                                                         |
|---------|-----------------------------------------------------------------------------------------|
| IS_FSM8 | Interrupt status bit for FSM8 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM7 | Interrupt status bit for FSM7 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM6 | Interrupt status bit for FSM6 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM5 | Interrupt status bit for FSM5 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM4 | Interrupt status bit for FSM4 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM3 | Interrupt status bit for FSM3 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM2 | Interrupt status bit for FSM2 interrupt event. (1: interrupt detected; 0: no interrupt) |
| IS_FSM1 | Interrupt status bit for FSM1 interrupt event. (1: interrupt detected; 0: no interrupt) |

## 11.13 PAGE\_RW (17h)

Enable read and write mode of advanced features dedicated page (R/W)

**Table 217. PAGE\_RW register**

| EMB_FUNC_LIR | PAGE_WRITE | PAGE_READ | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|--------------|------------|-----------|------------------|------------------|------------------|------------------|------------------|
|--------------|------------|-----------|------------------|------------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 218. PAGE\_RW register description**

|              |                                                                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMB_FUNC_LIR | Latched interrupt mode for embedded functions. Default value: 0<br>(0: embedded functions interrupt request not latched;<br>1: embedded functions interrupt request latched) |
| PAGE_WRITE   | Enables writes to the selected advanced features dedicated page. <sup>(1)</sup> Default value: 0<br>(1: enable; 0: disable)                                                  |
| PAGE_READ    | Enables reads from the selected advanced features dedicated page. <sup>(1)</sup> Default value: 0<br>(1: enable; 0: disable)                                                 |

1. Page selected by PAGE\_SEL[3:0] in PAGE\_SEL (02h) register.



## 11.14 EMB\_FUNC\_FIFO\_EN\_A (44h)

Embedded functions FIFO configuration register A (R/W)

**Table 219. EMB\_FUNC\_FIFO\_EN\_A register**

|                  |                      |                    |                      |                  |                  |                   |                  |
|------------------|----------------------|--------------------|----------------------|------------------|------------------|-------------------|------------------|
| 0 <sup>(1)</sup> | STEP_COUNTER_FIFO_EN | SFLP_GBIAS_FIFO_EN | SFLP_GRAVITY_FIFO_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | SFLP_GAME_FIFO_EN | 0 <sup>(1)</sup> |
|------------------|----------------------|--------------------|----------------------|------------------|------------------|-------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 220. EMB\_FUNC\_FIFO\_EN\_A register description**

|                      |                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STEP_COUNTER_FIFO_EN | Enables batching the step counter values in the FIFO buffer. Default value: 0 (0: disabled; 1: enabled)                                                     |
| SFLP_GBIAS_FIFO_EN   | Enables batching the gyroscope bias values computed by the SFLP algorithm in the FIFO buffer. Default value: 0 (0: disabled; 1: enabled)                    |
| SFLP_GRAVITY_FIFO_EN | Enables batching the gravity values computed by the SFLP algorithm in the FIFO buffer. Default value: 0 (0: disabled; 1: enabled)                           |
| SFLP_GAME_FIFO_EN    | Enables batching the game rotation vector (quaternion) values computed by the SFLP algorithm in the FIFO buffer. Default value: 0 (0: disabled; 1: enabled) |

## 11.15 FSM\_ENABLE (46h)

Enable FSM register (R/W)

**Table 221. FSM\_ENABLE register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| FSM8_EN | FSM7_EN | FSM6_EN | FSM5_EN | FSM4_EN | FSM3_EN | FSM2_EN | FSM1_EN |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 222. FSM\_ENABLE register description**

|         |                                                                    |
|---------|--------------------------------------------------------------------|
| FSM8_EN | Enables FSM8. Default value: 0 (0: FSM8 disabled; 1: FSM8 enabled) |
| FSM7_EN | Enables FSM7. Default value: 0 (0: FSM7 disabled; 1: FSM7 enabled) |
| FSM6_EN | Enables FSM6. Default value: 0 (0: FSM6 disabled; 1: FSM6 enabled) |
| FSM5_EN | Enables FSM5. Default value: 0 (0: FSM5 disabled; 1: FSM5 enabled) |
| FSM4_EN | Enables FSM4. Default value: 0 (0: FSM4 disabled; 1: FSM4 enabled) |
| FSM3_EN | Enables FSM3. Default value: 0 (0: FSM3 disabled; 1: FSM3 enabled) |
| FSM2_EN | Enables FSM2. Default value: 0 (0: FSM2 disabled; 1: FSM2 enabled) |
| FSM1_EN | Enables FSM1. Default value: 0 (0: FSM1 disabled; 1: FSM1 enabled) |





## 11.16 FSM\_LONG\_COUNTER\_L (48h) and FSM\_LONG\_COUNTER\_H (49h)

FSM long counter status register (R/W)

The long counter value is an unsigned integer value (16-bit format).

**Table 223. FSM\_LONG\_COUNTER\_L register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| FSM_LC_7 | FSM_LC_6 | FSM_LC_5 | FSM_LC_4 | FSM_LC_3 | FSM_LC_2 | FSM_LC_1 | FSM_LC_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 224. FSM\_LONG\_COUNTER\_L register description**

|              |                                                              |
|--------------|--------------------------------------------------------------|
| FSM_LC_[7:0] | Long counter current value (LSbyte). Default value: 00000000 |
|--------------|--------------------------------------------------------------|

**Table 225. FSM\_LONG\_COUNTER\_H register**

|           |           |           |           |           |           |          |          |
|-----------|-----------|-----------|-----------|-----------|-----------|----------|----------|
| FSM_LC_15 | FSM_LC_14 | FSM_LC_13 | FSM_LC_12 | FSM_LC_11 | FSM_LC_10 | FSM_LC_9 | FSM_LC_8 |
|-----------|-----------|-----------|-----------|-----------|-----------|----------|----------|

**Table 226. FSM\_LONG\_COUNTER\_H register description**

|               |                                                              |
|---------------|--------------------------------------------------------------|
| FSM_LC_[15:8] | Long counter current value (MSbyte). Default value: 00000000 |
|---------------|--------------------------------------------------------------|



## 11.17 INT\_ACK\_MASK (4Bh)

Reset status register (R/W)

**Table 227. INT\_ACK\_MASK register**

| IACK_MASK7 | IACK_MASK6 | IACK_MASK5 | IACK_MASK4 | IACK_MASK3 | IACK_MASK2 | IACK_MASK1 | IACK_MASK0 |
|------------|------------|------------|------------|------------|------------|------------|------------|
|------------|------------|------------|------------|------------|------------|------------|------------|

**Table 228. INT\_ACK\_MASK register description**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IACK_MASK7 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 7 of the status register is not reset. When this bit is set to 0, bit 7 of the status register is reset. Default value: 0 |
| IACK_MASK6 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 6 of the status register is not reset. When this bit is set to 0, bit 6 of the status register is reset. Default value: 0 |
| IACK_MASK5 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 5 of the status register is not reset. When this bit is set to 0, bit 5 of the status register is reset. Default value: 0 |
| IACK_MASK4 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 4 of the status register is not reset. When this bit is set to 0, bit 4 of the status register is reset. Default value: 0 |
| IACK_MASK3 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 3 of the status register is not reset. When this bit is set to 0, bit 3 of the status register is reset. Default value: 0 |
| IACK_MASK2 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 2 of the status register is not reset. When this bit is set to 0, bit 2 of the status register is reset. Default value: 0 |
| IACK_MASK1 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 1 of the status register is not reset. When this bit is set to 0, bit 1 of the status register is reset. Default value: 0 |
| IACK_MASK0 | If set to 1, when reading the <a href="#">EMB_FUNC_STATUS (12h)</a> / <a href="#">EMB_FUNC_STATUS_MAINPAGE (49h)</a> and <a href="#">FSM_STATUS (13h)</a> / <a href="#">FSM_STATUS_MAINPAGE (4Ah)</a> registers in latched mode (when the <a href="#">EMB_FUNC_LIR</a> bit is set to 1 in the <a href="#">PAGE_RW (17h)</a> register), bit 0 of the status register is not reset. When this bit is set to 0, bit 0 of the status register is reset. Default value: 0 |



## 11.18 FSM\_OUTS1 (4Ch)

FSM1 output register (R)

**Table 229. FSM\_OUTS1 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 230. FSM\_OUTS1 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM1 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM1 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM1 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM1 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM1 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM1 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM1 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM1 output: negative event detected on the vector. (0: event not detected; 1: event detected) |

## 11.19 FSM\_OUTS2 (4Dh)

FSM2 output register (R)

**Table 231. FSM\_OUTS2 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 232. FSM\_OUTS2 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM2 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM2 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM2 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM2 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM2 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM2 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM2 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM2 output: negative event detected on the vector. (0: event not detected; 1: event detected) |



## 11.20

**FSM\_OUTS3 (4Eh)**

FSM3 output register (R)

**Table 233. FSM\_OUTS3 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 234. FSM\_OUTS3 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM3 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM3 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM3 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM3 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM3 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM3 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM3 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM3 output: negative event detected on the vector. (0: event not detected; 1: event detected) |

## 11.21

**FSM\_OUTS4 (4Fh)**

FSM4 output register (R)

**Table 235. FSM\_OUTS4 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 236. FSM\_OUTS4 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM4 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM4 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM4 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM4 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM4 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM4 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM4 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM4 output: negative event detected on the vector. (0: event not detected; 1: event detected) |



## 11.22 FSM\_OUTS5 (50h)

FSM5 output register (R)

**Table 237. FSM\_OUTS5 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 238. FSM\_OUTS5 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM5 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM5 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM5 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM5 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM5 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM5 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM5 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM5 output: negative event detected on the vector. (0: event not detected; 1: event detected) |

## 11.23 FSM\_OUTS6 (51h)

FSM6 output register (R)

**Table 239. FSM\_OUTS6 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 240. FSM\_OUTS6 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM6 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM6 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM6 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM6 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM6 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM6 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM6 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM6 output: negative event detected on the vector. (0: event not detected; 1: event detected) |



## 11.24 FSM\_OUTS7 (52h)

FSM7 output register (R)

**Table 241. FSM\_OUTS7 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 242. FSM\_OUTS7 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM7 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM7 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM7 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM7 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM7 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM7 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM7 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM7 output: negative event detected on the vector. (0: event not detected; 1: event detected) |

## 11.25 FSM\_OUTS8 (53h)

FSM8 output register (R)

**Table 243. FSM\_OUTS8 register**

| P_X | N_X | P_Y | N_Y | P_Z | N_Z | P_V | N_V |
|-----|-----|-----|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|-----|-----|-----|

**Table 244. FSM\_OUTS8 register description**

|     |                                                                                                |
|-----|------------------------------------------------------------------------------------------------|
| P_X | FSM8 output: positive event detected on the X-axis. (0: event not detected; 1: event detected) |
| N_X | FSM8 output: negative event detected on the X-axis. (0: event not detected; 1: event detected) |
| P_Y | FSM8 output: positive event detected on the Y-axis. (0: event not detected; 1: event detected) |
| N_Y | FSM8 output: negative event detected on the Y-axis. (0: event not detected; 1: event detected) |
| P_Z | FSM8 output: positive event detected on the Z-axis. (0: event not detected; 1: event detected) |
| N_Z | FSM8 output: negative event detected on the Z-axis. (0: event not detected; 1: event detected) |
| P_V | FSM8 output: positive event detected on the vector. (0: event not detected; 1: event detected) |
| N_V | FSM8 output: negative event detected on the vector. (0: event not detected; 1: event detected) |



## 11.26

**SFLP\_ODR (5Eh)**

Sensor fusion low-power output data rate configuration register (R/W)

**Table 245. SFLP\_ODR register**

|                  |                  |                 |                 |                 |                  |                  |                  |
|------------------|------------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|
| 0 <sup>(1)</sup> | 1 <sup>(2)</sup> | SFLP_GAME_ODR_2 | SFLP_GAME_ODR_1 | SFLP_GAME_ODR_0 | 0 <sup>(1)</sup> | 1 <sup>(2)</sup> | 1 <sup>(2)</sup> |
|------------------|------------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.
2. This bit must be set to 1 for the correct operation of the device.

**Table 246. SFLP\_ODR register description**

|                     |                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SFLP_GAME_ODR_[2:0] | <p>ODR configuration of the SFLP game algorithm:</p> <p>(000: 15 Hz;<br/>           001: 30 Hz;<br/>           010: 60 Hz;<br/>           011: 120 Hz (default);<br/>           100: 240 Hz;<br/>           101: 480 Hz)</p> |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 11.27

**FSM\_ODR (5Fh)**

Finite state machine output data rate configuration register (R/W)

**Table 247. FSM\_ODR register**

|                  |                  |           |           |           |                  |                  |                  |
|------------------|------------------|-----------|-----------|-----------|------------------|------------------|------------------|
| 0 <sup>(1)</sup> | 1 <sup>(2)</sup> | FSM_ODR_2 | FSM_ODR_1 | FSM_ODR_0 | 0 <sup>(1)</sup> | 1 <sup>(2)</sup> | 1 <sup>(2)</sup> |
|------------------|------------------|-----------|-----------|-----------|------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.
2. This bit must be set to 1 for the correct operation of the device.

**Table 248. FSM\_ODR register description**

|               |                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSM_ODR_[2:0] | Finite state machine ODR configuration:<br>(000: 15 Hz;<br>001: 30 Hz (default);<br>010: 60 Hz;<br>011: 120 Hz;<br>100: 240 Hz;<br>101: 480 Hz;<br>110: 960 Hz) |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 11.28

**STEP\_COUNTER\_L (62h) and STEP\_COUNTER\_H (63h)**

Step counter output register (R)

**Table 249. STEP\_COUNTER\_L register**

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| STEP_7 | STEP_6 | STEP_5 | STEP_4 | STEP_3 | STEP_2 | STEP_1 | STEP_0 |
|--------|--------|--------|--------|--------|--------|--------|--------|

**Table 250. STEP\_COUNTER\_L register description**

|            |                              |
|------------|------------------------------|
| STEP_[7:0] | Step counter output (LSbyte) |
|------------|------------------------------|

**Table 251. STEP\_COUNTER\_H register**

|         |         |         |         |         |         |        |        |
|---------|---------|---------|---------|---------|---------|--------|--------|
| STEP_15 | STEP_14 | STEP_13 | STEP_12 | STEP_11 | STEP_10 | STEP_9 | STEP_8 |
|---------|---------|---------|---------|---------|---------|--------|--------|

**Table 252. STEP\_COUNTER\_H register description**

|             |                              |
|-------------|------------------------------|
| STEP_[15:8] | Step counter output (MSbyte) |
|-------------|------------------------------|





## 11.29

**EMB\_FUNC\_SRC (64h)**

Embedded function source register (R/W)

**Table 253. EMB\_FUNC\_SRC register**

|               |                  |               |                     |               |                       |                  |                  |
|---------------|------------------|---------------|---------------------|---------------|-----------------------|------------------|------------------|
| PEDO_RST_STEP | 0 <sup>(1)</sup> | STEP_DETECTED | STEP_COUNT_DELTA_IA | STEP_OVERFLOW | STEPS_COUNTER_BIT_SET | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|---------------|------------------|---------------|---------------------|---------------|-----------------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 254. EMB\_FUNC\_SRC register description**

|                       |                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PEDO_RST_STEP         | Reset pedometer step counter. Read/write bit.<br>(0: disabled; 1: enabled)                                                                                                                                                            |
| STEP_DETECTED         | Step detector event detection status. Read-only bit.<br>(0: step detection event not detected; 1: step detection event detected)                                                                                                      |
| STEP_COUNT_DELTA_IA   | Pedometer step recognition on delta time status. Read-only bit.<br>(0: no step recognized during delta time;<br>1: at least one step recognized during delta time)                                                                    |
| STEP_OVERFLOW         | Step counter overflow status. Read-only bit.<br>(0: step counter value < 2 <sup>16</sup> ; 1: step counter value reached 2 <sup>16</sup> )                                                                                            |
| STEPS_COUNTER_BIT_SET | This bit is equal to 1 when the step count is increased. If a timer period is programmed in PEDO_SC_DELTAT_L (D0h) and PEDO_SC_DELTAT_H (D1h) embedded advanced features (page 1) registers, this bit is kept at 0.<br>Read-only bit. |



## 11.30

**EMB\_FUNC\_INIT\_A (66h)**

Embedded functions initialization register (R/W)

**Table 255. EMB\_FUNC\_INIT\_A register**

|                  |                  |              |           |               |                  |                |                  |
|------------------|------------------|--------------|-----------|---------------|------------------|----------------|------------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | SIG_MOT_INIT | TILT_INIT | STEP_DET_INIT | 0 <sup>(1)</sup> | SFLP_GAME_INIT | 0 <sup>(1)</sup> |
|------------------|------------------|--------------|-----------|---------------|------------------|----------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 256. EMB\_FUNC\_INIT\_A register description**

|                |                                                                                    |
|----------------|------------------------------------------------------------------------------------|
| SIG_MOT_INIT   | Significant motion detection algorithm initialization request. Default value: 0    |
| TILT_INIT      | Tilt algorithm initialization request. Default value: 0                            |
| STEP_DET_INIT  | Pedometer step counter/detector algorithm initialization request. Default value: 0 |
| SFLP_GAME_INIT | SFLP game algorithm initialization request. Default value: 0                       |

## 11.31

**EMB\_FUNC\_INIT\_B (67h)**

Embedded functions initialization register (R/W)

**Table 257. EMB\_FUNC\_INIT\_B register**

|                  |                  |                  |                  |                 |                  |                  |          |
|------------------|------------------|------------------|------------------|-----------------|------------------|------------------|----------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | FIFO_COMPR_INIT | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | FSM_INIT |
|------------------|------------------|------------------|------------------|-----------------|------------------|------------------|----------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 258. EMB\_FUNC\_INIT\_B register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| FIFO_COMPR_INIT | FIFO compression feature initialization request. Default value: 0 |
| FSM_INIT        | FSM initialization request. Default value: 0                      |



## 12 Embedded advanced features pages

The table given below provides a list of the registers for the embedded advanced features page 0. These registers are accessible when PAGE\_SEL[3:0] are set to 0000 in [PAGE\\_SEL \(02h\)](#).

Note:

*External sensor offset compensation registers and transformation matrix correction registers affect FSM data only. When these registers are set with their default values, no compensation is applied.*

**Table 259. Register address map - embedded advanced features page 0**

| Name                  | Type | Register address |          | Default  | Comment |
|-----------------------|------|------------------|----------|----------|---------|
|                       |      | Hex              | Binary   |          |         |
| SFLP_GAME_GBIASX_L    | R/W  | 6E               | 01101110 | 00000000 |         |
| SFLP_GAME_GBIASX_H    | R/W  | 6F               | 01101111 | 00000000 |         |
| SFLP_GAME_GBIASY_L    | R/W  | 70               | 01110000 | 00000000 |         |
| SFLP_GAME_GBIASY_H    | R/W  | 71               | 01110001 | 00000000 |         |
| SFLP_GAME_GBIASZ_L    | R/W  | 72               | 01110010 | 00000000 |         |
| SFLP_GAME_GBIASZ_H    | R/W  | 73               | 01110011 | 00000000 |         |
| FSM_EXT_SENSITIVITY_L | R/W  | BA               | 10111010 | 00100100 |         |
| FSM_EXT_SENSITIVITY_H | R/W  | BB               | 10111011 | 00010110 |         |
| FSM_EXT_OFFX_L        | R/W  | C0               | 11000000 | 00000000 |         |
| FSM_EXT_OFFX_H        | R/W  | C1               | 11000001 | 00000000 |         |
| FSM_EXT_OFFY_L        | R/W  | C2               | 11000010 | 00000000 |         |
| FSM_EXT_OFFY_H        | R/W  | C3               | 11000011 | 00000000 |         |
| FSM_EXT_OFFZ_L        | R/W  | C4               | 11000100 | 00000000 |         |
| FSM_EXT_OFFZ_H        | R/W  | C5               | 11000101 | 00000000 |         |
| FSM_EXT_MATRIX_XX_L   | R/W  | C6               | 11000110 | 00000000 |         |
| FSM_EXT_MATRIX_XX_H   | R/W  | C7               | 11000111 | 00111100 |         |
| FSM_EXT_MATRIX_XY_L   | R/W  | C8               | 11001000 | 00000000 |         |
| FSM_EXT_MATRIX_XY_H   | R/W  | C9               | 11001001 | 00000000 |         |
| FSM_EXT_MATRIX_XZ_L   | R/W  | CA               | 11001010 | 00000000 |         |
| FSM_EXT_MATRIX_XZ_H   | R/W  | CB               | 11001011 | 00000000 |         |
| FSM_EXT_MATRIX_YY_L   | R/W  | CC               | 11001100 | 00000000 |         |
| FSM_EXT_MATRIX_YY_H   | R/W  | CD               | 11001101 | 00111100 |         |
| FSM_EXT_MATRIX_YZ_L   | R/W  | CE               | 11001110 | 00000000 |         |
| FSM_EXT_MATRIX_YZ_H   | R/W  | CF               | 11001111 | 00000000 |         |
| FSM_EXT_MATRIX_ZZ_L   | R/W  | D0               | 11010000 | 00000000 |         |
| FSM_EXT_MATRIX_ZZ_H   | R/W  | D1               | 11010001 | 00111100 |         |
| EXT_CFG_A             | R/W  | D4               | 11010100 | 00000101 |         |
| EXT_CFG_B             | R/W  | D5               | 11010101 | 00000010 |         |



The following table provides a list of the registers for the embedded advanced features page 1. These registers are accessible when PAGE\_SEL[3:0] are set to 0001 in [PAGE\\_SEL \(02h\)](#).

**Table 260. Register address map - embedded advanced features page 1**

| Name                | Type | Register address |          | Default  | Comment |
|---------------------|------|------------------|----------|----------|---------|
|                     |      | Hex              | Binary   |          |         |
| FSM_LC_TIMEOUT_L    | R/W  | 7A               | 01111010 | 00000000 |         |
| FSM_LC_TIMEOUT_H    | R/W  | 7B               | 01111011 | 00000000 |         |
| FSM_PROGRAMS        | R/W  | 7C               | 01111100 | 00000000 |         |
| FSM_START_ADD_L     | R/W  | 7E               | 01111110 | 00000000 |         |
| FSM_START_ADD_H     | R/W  | 7F               | 01111111 | 00000000 |         |
| PEDO_CMD_REG        | R/W  | 83               | 10000011 | 00000000 |         |
| PEDO_DEB_STEPS_CONF | R/W  | 84               | 10000100 | 00001010 |         |
| PEDO_SC_DELTAT_L    | R/W  | D0               | 11010000 | 00000000 |         |
| PEDO_SC_DELTAT_H    | R/W  | D1               | 11010001 | 00000000 |         |

The following table provides a list of the registers for the embedded advanced features page 2. These registers are accessible when PAGE\_SEL[3:0] are set to 0010 in [PAGE\\_SEL \(02h\)](#).

**Table 261. Register address map - embedded advanced features page 2**

| Name                    | Type | Register address |          | Default  | Comment |
|-------------------------|------|------------------|----------|----------|---------|
|                         |      | Hex              | Binary   |          |         |
| EXT_FORMAT              | R/W  | 00               | 00000000 | 00000000 |         |
| EXT_3BYTE_SENSITIVITY_L | R/W  | 02               | 00000010 | 00000000 |         |
| EXT_3BYTE_SENSITIVITY_H | R/W  | 03               | 00000011 | 00001100 |         |
| EXT_3BYTE_OFFSET_XL     | R/W  | 06               | 00000110 | 00000000 |         |
| EXT_3BYTE_OFFSET_L      | R/W  | 07               | 00000111 | 01010100 |         |
| EXT_3BYTE_OFFSET_H      | R/W  | 08               | 00001000 | 00111111 |         |

Reserved registers must not be changed. Writing to those registers may cause permanent damage to the device. The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.

**Write procedure example:** write value 06h in register at address 84h (PEDO\_DEB\_STEPS\_CONF) in Page 1

1. Write bit EMB\_FUNC\_REG\_ACCESS = 1 in FUNC\_CFG\_ACCESS (01h) // Enable access to embedded functions registers
2. Write bit PAGE\_WRITE = 1 in PAGE\_RW (17h) register // Select write operation mode
3. Write 0001 in PAGE\_SEL[3:0] field of register PAGE\_SEL (02h) // Select page 1
4. Write 84h in PAGE\_ADDR register (08h) // Set address
5. Write 06h in PAGE\_DATA register (09h) // Set value to be written
6. Write bit PAGE\_WRITE = 0 in PAGE\_RW (17h) register // Write operation disabled
7. Write bit EMB\_FUNC\_REG\_ACCESS = 0 in FUNC\_CFG\_ACCESS (01h) // Disable access to embedded functions registers



**Read procedure example:** read value of register at address 84h (PEDO\_DEB\_STEPS\_CONF) in Page 1

1. Write bit EMB\_FUNC\_REG\_ACCESS = 1 in FUNC\_CFG\_ACCESS (01h) // Enable access to embedded functions registers
2. Write bit PAGE\_READ = 1 in PAGE\_RW (17h) register // Select read operation mode
3. Write 0001 in PAGE\_SEL[3:0] field of register PAGE\_SEL (02h) // Select page 1
4. Write 84h in PAGE\_ADDR register (08h) // Set address
5. Read value of PAGE\_DATA register (09h) // Get register value
6. Write bit PAGE\_READ = 0 in PAGE\_RW (17h) register // Read operation disabled
7. Write bit EMB\_FUNC\_REG\_ACCESS = 0 in FUNC\_CFG\_ACCESS (01h) // Disable access to embedded functions registers

**Note:**

*Steps 1 and 2 of both procedures are intended to be performed at the beginning of the procedure. Steps 6 and 7 of both procedures are intended to be performed at the end of the procedure. If the procedure involves multiple operations, only steps 3, 4 and 5 must be repeated for each operation. If, in particular, the multiple operations involve consecutive registers, only step 5 can be performed.*



## 13 Embedded advanced features register description

### 13.1 Page 0 - embedded advanced features registers

#### 13.1.1 SFLP\_GAME\_GBIASX\_L (6Eh) and SFLP\_GAME\_GBIASX\_H (6Fh)

SFLP game algorithm X-axis gyroscope bias register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEFFFFFFFFF (S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 262. SFLP\_GAME\_GBIASX\_L register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| GAME_GBIASX_7 | GAME_GBIASX_6 | GAME_GBIASX_5 | GAME_GBIASX_4 | GAME_GBIASX_3 | GAME_GBIASX_2 | GAME_GBIASX_1 | GAME_GBIASX_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 263. SFLP\_GAME\_GBIASX\_L register description**

|                   |                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASX_[7:0] | SFLP game algorithm X-axis gbias: temporary register for gbias setting procedure (LSbyte).<br>Default value: 00000000 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|

**Table 264. SFLP\_GAME\_GBIASX\_H register**

|                |                |                |                |                |                |               |               |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|
| GAME_GBIASX_15 | GAME_GBIASX_14 | GAME_GBIASX_13 | GAME_GBIASX_12 | GAME_GBIASX_11 | GAME_GBIASX_10 | GAME_GBIASX_9 | GAME_GBIASX_8 |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|

**Table 265. SFLP\_GAME\_GBIASX\_H register description**

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASX_[15:8] | SFLP game algorithm X-axis gbias: temporary register for gbias setting procedure (MSbyte).<br>Default value: 00000000 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|

#### 13.1.2 SFLP\_GAME\_GBIASY\_L (70h) and SFLP\_GAME\_GBIASY\_H (71h)

SFLP game algorithm Y-axis gyroscope bias register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEFFFFFFFFF (S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 266. SFLP\_GAME\_GBIASY\_L register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| GAME_GBIASY_7 | GAME_GBIASY_6 | GAME_GBIASY_5 | GAME_GBIASY_4 | GAME_GBIASY_3 | GAME_GBIASY_2 | GAME_GBIASY_1 | GAME_GBIASY_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 267. SFLP\_GAME\_GBIASY\_L register description**

|                   |                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASY_[7:0] | SFLP game algorithm Y-axis gbias: temporary register for gbias setting procedure (LSbyte).<br>Default value: 00000000 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|

**Table 268. SFLP\_GAME\_GBIASY\_H register**

|                |                |                |                |                |                |               |               |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|
| GAME_GBIASY_15 | GAME_GBIASY_14 | GAME_GBIASY_13 | GAME_GBIASY_12 | GAME_GBIASY_11 | GAME_GBIASY_10 | GAME_GBIASY_9 | GAME_GBIASY_8 |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|

**Table 269. SFLP\_GAME\_GBIASY\_H register description**

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASY_[15:8] | SFLP game algorithm Y-axis gbias: temporary register for gbias setting procedure (MSbyte).<br>Default value: 00000000 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|



## 13.1.3

**SFLP\_GAME\_GBIASZ\_L (72h) and SFLP\_GAME\_GBIASZ\_H (73h)**

SFLP game algorithm Z-axis gyroscope bias register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEE (S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

Table 270. SFLP\_GAME\_GBIASZ\_L register

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| GAME_GBIASZ_7 | GAME_GBIASZ_6 | GAME_GBIASZ_5 | GAME_GBIASZ_4 | GAME_GBIASZ_3 | GAME_GBIASZ_2 | GAME_GBIASZ_1 | GAME_GBIASZ_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

Table 271. SFLP\_GAME\_GBIASZ\_L register description

|                   |                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASZ_[7:0] | SFLP game algorithm Z-axis gbias: temporary register for gbias setting procedure (LSbyte). Default value: 00000000 |
|-------------------|--------------------------------------------------------------------------------------------------------------------|

Table 272. SFLP\_GAME\_GBIASZ\_H register

|                |                |                |                |                |                |               |               |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|
| GAME_GBIASZ_15 | GAME_GBIASZ_14 | GAME_GBIASZ_13 | GAME_GBIASZ_12 | GAME_GBIASZ_11 | GAME_GBIASZ_10 | GAME_GBIASZ_9 | GAME_GBIASZ_8 |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|

Table 273. SFLP\_GAME\_GBIASZ\_H register description

|                    |                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| GAME_GBIASZ_[15:8] | SFLP game algorithm Z-axis gbias: temporary register for gbias setting procedure (MSbyte). Default value: 00000000 |
|--------------------|--------------------------------------------------------------------------------------------------------------------|

## 13.1.4

**FSM\_EXT\_SENSITIVITY\_L (BAh) and FSM\_EXT\_SENSITIVITY\_H (BBh)**

External sensor sensitivity value register for the finite state machine (R/W)

This register corresponds to the conversion value of the external sensor. The register value is expressed as half-precision floating-point format: SEEEEEEEEEEEE (S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

Default value of FSM\_EXT\_S\_[15:0] bits is 0x1624 (when using an external magnetometer this value corresponds to 0.0015 gauss/LSB).

Table 274. FSM\_EXT\_SENSITIVITY\_L register

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| FSM_EXT_S_7 | FSM_EXT_S_6 | FSM_EXT_S_5 | FSM_EXT_S_4 | FSM_EXT_S_3 | FSM_EXT_S_2 | FSM_EXT_S_1 | FSM_EXT_S_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

Table 275. FSM\_EXT\_SENSITIVITY\_L register description

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| FSM_EXT_S_[7:0] | External sensor sensitivity (LSbyte). Default value: 00100100 |
|-----------------|---------------------------------------------------------------|

Table 276. FSM\_EXT\_SENSITIVITY\_H register

|              |              |              |              |              |              |             |             |
|--------------|--------------|--------------|--------------|--------------|--------------|-------------|-------------|
| FSM_EXT_S_15 | FSM_EXT_S_14 | FSM_EXT_S_13 | FSM_EXT_S_12 | FSM_EXT_S_11 | FSM_EXT_S_10 | FSM_EXT_S_9 | FSM_EXT_S_8 |
|--------------|--------------|--------------|--------------|--------------|--------------|-------------|-------------|

Table 277. FSM\_EXT\_SENSITIVITY\_H register description

|                  |                                                   |
|------------------|---------------------------------------------------|
| FSM_EXT_S_[15:8] | External sensor (MSbyte). Default value: 00010110 |
|------------------|---------------------------------------------------|



### 13.1.5 FSM\_EXT\_OFFX\_L (C0h) and FSM\_EXT\_OFFX\_H (C1h)

External sensor X-axis offset

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 278. FSM\_EXT\_OFFX\_L register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| FSM_EXT_OFFX_7 | FSM_EXT_OFFX_6 | FSM_EXT_OFFX_5 | FSM_EXT_OFFX_4 | FSM_EXT_OFFX_3 | FSM_EXT_OFFX_2 | FSM_EXT_OFFX_1 | FSM_EXT_OFFX_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 279. FSM\_EXT\_OFFX\_L register description**

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFX[7:0] | External sensor X-axis offset (LSbyte). Default value: 00000000 |
|-------------------|-----------------------------------------------------------------|

**Table 280. FSM\_EXT\_OFFX\_H register**

|                 |                 |                 |                 |                 |                 |                |                |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|
| FSM_EXT_OFFX_15 | FSM_EXT_OFFX_14 | FSM_EXT_OFFX_13 | FSM_EXT_OFFX_12 | FSM_EXT_OFFX_11 | FSM_EXT_OFFX_10 | FSM_EXT_OFFX_9 | FSM_EXT_OFFX_8 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|

**Table 281. FSM\_EXT\_OFFX\_H register description**

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFX[15:8] | External sensor X-axis offset (MSbyte). Default value: 00000000 |
|--------------------|-----------------------------------------------------------------|

### 13.1.6 FSM\_EXT\_OFFY\_L (C2h) and FSM\_EXT\_OFFY\_H (C3h)

External sensor Y-axis offset

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 282. FSM\_EXT\_OFFY\_L register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| FSM_EXT_OFFY_7 | FSM_EXT_OFFY_6 | FSM_EXT_OFFY_5 | FSM_EXT_OFFY_4 | FSM_EXT_OFFY_3 | FSM_EXT_OFFY_2 | FSM_EXT_OFFY_1 | FSM_EXT_OFFY_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 283. FSM\_EXT\_OFFY\_L register description**

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFY[7:0] | External sensor Y-axis offset (LSbyte). Default value: 00000000 |
|-------------------|-----------------------------------------------------------------|

**Table 284. FSM\_EXT\_OFFY\_H register**

|                 |                 |                 |                 |                 |                 |                |                |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|
| FSM_EXT_OFFY_15 | FSM_EXT_OFFY_14 | FSM_EXT_OFFY_13 | FSM_EXT_OFFY_12 | FSM_EXT_OFFY_11 | FSM_EXT_OFFY_10 | FSM_EXT_OFFY_9 | FSM_EXT_OFFY_8 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|

**Table 285. FSM\_EXT\_OFFY\_H register description**

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFY[15:8] | External sensor Y-axis offset (MSbyte). Default value: 00000000 |
|--------------------|-----------------------------------------------------------------|





### 13.1.7 FSM\_EXT\_OFFZ\_L (C4h) and FSM\_EXT\_OFFZ\_H (C5h)

External sensor Z-axis offset register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEEE

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 286. FSM\_EXT\_OFFZ\_L register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| FSM_EXT_OFFZ_7 | FSM_EXT_OFFZ_6 | FSM_EXT_OFFZ_5 | FSM_EXT_OFFZ_4 | FSM_EXT_OFFZ_3 | FSM_EXT_OFFZ_2 | FSM_EXT_OFFZ_1 | FSM_EXT_OFFZ_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 287. FSM\_EXT\_OFFZ\_L register description**

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFZ_[7:0] | External sensor Z-axis offset (LSbyte). Default value: 00000000 |
|--------------------|-----------------------------------------------------------------|

**Table 288. FSM\_EXT\_OFFZ\_H register**

|                 |                 |                 |                 |                 |                 |                |                |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|
| FSM_EXT_OFFZ_15 | FSM_EXT_OFFZ_14 | FSM_EXT_OFFZ_13 | FSM_EXT_OFFZ_12 | FSM_EXT_OFFZ_11 | FSM_EXT_OFFZ_10 | FSM_EXT_OFFZ_9 | FSM_EXT_OFFZ_8 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|

**Table 289. FSM\_EXT\_OFFZ\_H register description**

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| FSM_EXT_OFFZ_[15:8] | External sensor Z-axis offset (MSbyte). Default value: 00000000 |
|---------------------|-----------------------------------------------------------------|

### 13.1.8 FSM\_EXT\_MATRIX\_XX\_L (C6h) and FSM\_EXT\_MATRIX\_XX\_H (C7h)

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEEE

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 290. FSM\_EXT\_MATRIX\_XX\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_XX_7 | FSM_EXT_MAT_XX_6 | FSM_EXT_MAT_XX_5 | FSM_EXT_MAT_XX_4 | FSM_EXT_MAT_XX_3 | FSM_EXT_MAT_XX_2 | FSM_EXT_MAT_XX_1 | FSM_EXT_MAT_XX_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 291. FSM\_EXT\_MATRIX\_XX\_L register description**

|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_XX_[7:0] | Transformation matrix row1 col1 coefficient (LSbyte). Default value: 00000000 |
|----------------------|-------------------------------------------------------------------------------|

**Table 292. FSM\_EXT\_MATRIX\_XX\_H register**

|                   |                   |                   |                   |                   |                   |                  |                  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|
| FSM_EXT_MAT_XX_15 | FSM_EXT_MAT_XX_14 | FSM_EXT_MAT_XX_13 | FSM_EXT_MAT_XX_12 | FSM_EXT_MAT_XX_11 | FSM_EXT_MAT_XX_10 | FSM_EXT_MAT_XX_9 | FSM_EXT_MAT_XX_8 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|

**Table 293. FSM\_EXT\_MATRIX\_XX\_H register description**

|                       |                                                                               |
|-----------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_XX_[15:8] | Transformation matrix row1 col1 coefficient (MSbyte). Default value: 00111100 |
|-----------------------|-------------------------------------------------------------------------------|



## 13.1.9

**FSM\_EXT\_MATRIX\_XY\_L (C8h) and FSM\_EXT\_MATRIX\_XY\_H (C9h)**

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 294. FSM\_EXT\_MATRIX\_XY\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_XY_7 | FSM_EXT_MAT_XY_6 | FSM_EXT_MAT_XY_5 | FSM_EXT_MAT_XY_4 | FSM_EXT_MAT_XY_3 | FSM_EXT_MAT_XY_2 | FSM_EXT_MAT_XY_1 | FSM_EXT_MAT_XY_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 295. FSM\_EXT\_MATRIX\_XY\_L register description**

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_XY_[7:0] | Transformation matrix row1 col2 (and row2 col1) coefficient (LSbyte). Default value: 00000000 |
|----------------------|-----------------------------------------------------------------------------------------------|

**Table 296. FSM\_EXT\_MATRIX\_XY\_H register**

|               |               |               |               |               |               |              |              |
|---------------|---------------|---------------|---------------|---------------|---------------|--------------|--------------|
| FSM_EXT_XY_15 | FSM_EXT_XY_14 | FSM_EXT_XY_13 | FSM_EXT_XY_12 | FSM_EXT_XY_11 | FSM_EXT_XY_10 | FSM_EXT_XY_9 | FSM_EXT_XY_8 |
|---------------|---------------|---------------|---------------|---------------|---------------|--------------|--------------|

**Table 297. FSM\_EXT\_MATRIX\_XY\_H register description**

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_XY_[15:8] | Transformation matrix row1 col2 (and row2 col1) coefficient (MSbyte). Default value: 00000000 |
|-----------------------|-----------------------------------------------------------------------------------------------|

## 13.1.10

**FSM\_EXT\_MATRIX\_XZ\_L (CAh) and FSM\_EXT\_MATRIX\_XZ\_H (CBh)**

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 298. FSM\_EXT\_MATRIX\_XZ\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_XZ_7 | FSM_EXT_MAT_XZ_6 | FSM_EXT_MAT_XZ_5 | FSM_EXT_MAT_XZ_4 | FSM_EXT_MAT_XZ_3 | FSM_EXT_MAT_XZ_2 | FSM_EXT_MAT_XZ_1 | FSM_EXT_MAT_XZ_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 299. FSM\_EXT\_MATRIX\_XZ\_L register description**

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_XZ_[7:0] | Transformation matrix row1 col3 (and row3 col1) coefficient (LSbyte). Default value: 00000000 |
|----------------------|-----------------------------------------------------------------------------------------------|

**Table 300. FSM\_EXT\_MATRIX\_XZ\_H register**

|                   |                   |                   |                   |                   |                   |                  |                  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|
| FSM_EXT_MAT_XZ_15 | FSM_EXT_MAT_XZ_14 | FSM_EXT_MAT_XZ_13 | FSM_EXT_MAT_XZ_12 | FSM_EXT_MAT_XZ_11 | FSM_EXT_MAT_XZ_10 | FSM_EXT_MAT_XZ_9 | FSM_EXT_MAT_XZ_8 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|

**Table 301. FSM\_EXT\_MATRIX\_XZ\_H register description**

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_XZ_[15:8] | Transformation matrix row1 col3 (and row3 col1) coefficient (MSbyte). Default value: 00000000 |
|-----------------------|-----------------------------------------------------------------------------------------------|



### 13.1.11 FSM\_EXT\_MATRIX\_YY\_L (CCh) and FSM\_EXT\_MATRIX\_YY\_H (CDh)

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 302. FSM\_EXT\_MATRIX\_YY\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_YY_7 | FSM_EXT_MAT_YY_6 | FSM_EXT_MAT_YY_5 | FSM_EXT_MAT_YY_4 | FSM_EXT_MAT_YY_3 | FSM_EXT_MAT_YY_2 | FSM_EXT_MAT_YY_1 | FSM_EXT_MAT_YY_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 303. FSM\_EXT\_MATRIX\_YY\_L register description**

|                     |                                                                               |
|---------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_YY[7:0] | Transformation matrix row2 col2 coefficient (LSbyte). Default value: 00000000 |
|---------------------|-------------------------------------------------------------------------------|

**Table 304. FSM\_EXT\_MATRIX\_YY\_H register**

|                   |                   |                   |                   |                   |                   |                  |                  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|
| FSM_EXT_MAT_YY_15 | FSM_EXT_MAT_YY_14 | FSM_EXT_MAT_YY_13 | FSM_EXT_MAT_YY_12 | FSM_EXT_MAT_YY_11 | FSM_EXT_MAT_YY_10 | FSM_EXT_MAT_YY_9 | FSM_EXT_MAT_YY_8 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|

**Table 305. FSM\_EXT\_MATRIX\_YY\_H register description**

|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_YY[15:8] | Transformation matrix row2 col2 coefficient (MSbyte). Default value: 00111100 |
|----------------------|-------------------------------------------------------------------------------|

### 13.1.12 FSM\_EXT\_MATRIX\_YZ\_L (CEh) and FSM\_EXT\_MATRIX\_YZ\_H (CFh)

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 306. FSM\_EXT\_MATRIX\_YZ\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_YZ_7 | FSM_EXT_MAT_YZ_6 | FSM_EXT_MAT_YZ_5 | FSM_EXT_MAT_YZ_4 | FSM_EXT_MAT_YZ_3 | FSM_EXT_MAT_YZ_2 | FSM_EXT_MAT_YZ_1 | FSM_EXT_MAT_YZ_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 307. FSM\_EXT\_MATRIX\_YZ\_L register description**

|                     |                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_YZ[7:0] | Transformation matrix row2 col3 (and row3 col2) coefficient (LSbyte).<br>Default value: 00000000 |
|---------------------|--------------------------------------------------------------------------------------------------|

**Table 308. FSM\_EXT\_MATRIX\_YZ\_H register**

|                   |                   |                   |                   |                   |                   |                  |                  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|
| FSM_EXT_MAT_YZ_15 | FSM_EXT_MAT_YZ_14 | FSM_EXT_MAT_YZ_13 | FSM_EXT_MAT_YZ_12 | FSM_EXT_MAT_YZ_11 | FSM_EXT_MAT_YZ_10 | FSM_EXT_MAT_YZ_9 | FSM_EXT_MAT_YZ_8 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|

**Table 309. FSM\_EXT\_MATRIX\_YZ\_H register description**

|                      |                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------|
| FSM_EXT_MAT_YZ[15:8] | Transformation matrix row2 col3 (and row3 col2) coefficient (MSbyte).<br>Default value: 00000000 |
|----------------------|--------------------------------------------------------------------------------------------------|



### 13.1.13 FSM\_EXT\_MATRIX\_ZZ\_L (D0h) and FSM\_EXT\_MATRIX\_ZZ\_H (D1h)

External sensor transformation matrix register (R/W)

The value is expressed as half-precision floating-point format: SEEEEEEEEEEEEFFFF

(S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

**Table 310. FSM\_EXT\_MATRIX\_ZZ\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FSM_EXT_MAT_ZZ_7 | FSM_EXT_MAT_ZZ_6 | FSM_EXT_MAT_ZZ_5 | FSM_EXT_MAT_ZZ_4 | FSM_EXT_MAT_ZZ_3 | FSM_EXT_MAT_ZZ_2 | FSM_EXT_MAT_ZZ_1 | FSM_EXT_MAT_ZZ_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 311. FSM\_EXT\_MATRIX\_ZZ\_L register description**

|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_ZZ_[7:0] | Transformation matrix row3 col3 coefficient (LSbyte). Default value: 00000000 |
|----------------------|-------------------------------------------------------------------------------|

**Table 312. FSM\_EXT\_MATRIX\_ZZ\_H register**

|                   |                   |                   |                   |                   |                   |                  |                  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|
| FSM_EXT_MAT_ZZ_15 | FSM_EXT_MAT_ZZ_14 | FSM_EXT_MAT_ZZ_13 | FSM_EXT_MAT_ZZ_12 | FSM_EXT_MAT_ZZ_11 | FSM_EXT_MAT_ZZ_10 | FSM_EXT_MAT_ZZ_9 | FSM_EXT_MAT_ZZ_8 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|------------------|

**Table 313. FSM\_EXT\_MATRIX\_ZZ\_H register description**

|                       |                                                                               |
|-----------------------|-------------------------------------------------------------------------------|
| FSM_EXT_MAT_ZZ_[15:8] | Transformation matrix row3 col3 coefficient (MSbyte). Default value: 00111100 |
|-----------------------|-------------------------------------------------------------------------------|



### 13.1.14 EXT\_CFG\_A (D4h)

External sensor coordinates (Z and Y axes) rotation register (r/w).

**Table 314. EXT\_CFG\_A register**

|                  |             |             |             |                  |             |             |             |
|------------------|-------------|-------------|-------------|------------------|-------------|-------------|-------------|
| 0 <sup>(1)</sup> | EXT_Y_AXIS2 | EXT_Y_AXIS1 | EXT_Y_AXIS0 | 0 <sup>(1)</sup> | EXT_Z_AXIS2 | EXT_Z_AXIS1 | EXT_Z_AXIS0 |
|------------------|-------------|-------------|-------------|------------------|-------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 315. EXT\_CFG\_A description**

|                 |                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXT_Y_AXIS[2:0] | External sensor Y-axis coordinates rotation (to be aligned to accelerometer/gyroscope axes orientation)<br>(000: Y = Y; (default)<br>001: Y = -Y;<br>010: Y = X;<br>011: Y = -X;<br>100: Y = -Z;<br>101: Y = Z;<br>Others: Y = Y) |
| EXT_Z_AXIS[2:0] | External sensor Z-axis coordinates rotation (to be aligned to accelerometer/gyroscope axes orientation)<br>(000: Z = Y;<br>001: Z = -Y;<br>010: Z = X;<br>011: Z = -X;<br>100: Z = -Z;<br>101: Z = Z; (default)<br>Others: Z = Y) |

### 13.1.15 EXT\_CFG\_B (D5h)

External sensor coordinates (X-axis) rotation register (r/w).

**Table 316. EXT\_CFG\_B register**

|                  |                  |                  |                  |                  |             |             |             |
|------------------|------------------|------------------|------------------|------------------|-------------|-------------|-------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | EXT_X_AXIS2 | EXT_X_AXIS1 | EXT_X_AXIS0 |
|------------------|------------------|------------------|------------------|------------------|-------------|-------------|-------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 317. EXT\_CFG\_B description**

|                 |                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXT_X_AXIS[2:0] | External sensor X-axis coordinates rotation (to be aligned to accelerometer/gyroscope axes orientation)<br>(000: X = Y;<br>001: X = -Y;<br>010: X = X; (default)<br>011: X = -X;<br>100: X = -Z;<br>101: X = Z;<br>Others: X = Y) |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 13.2 Page 1 - embedded advanced features registers

### 13.2.1 FSM\_LC\_TIMEOUT\_L (7Ah) and FSM\_LC\_TIMEOUT\_H (7Bh)

FSM long counter timeout register (R/W)

The long counter timeout value is an unsigned integer value (16-bit format). When the long counter value reached this value, the FSM generates an interrupt.

**Table 318. FSM\_LC\_TIMEOUT\_L register**

|                 |                 |                 |                 |                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| FSM_LC_TIMEOUT7 | FSM_LC_TIMEOUT6 | FSM_LC_TIMEOUT5 | FSM_LC_TIMEOUT4 | FSM_LC_TIMEOUT3 | FSM_LC_TIMEOUT2 | FSM_LC_TIMEOUT1 | FSM_LC_TIMEOUT0 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|

**Table 319. FSM\_LC\_TIMEOUT\_L register description**

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| FSM_LC_TIMEOUT[7:0] | FSM long counter timeout value (LSbyte). Default value: 00000000 |
|---------------------|------------------------------------------------------------------|

**Table 320. FSM\_LC\_TIMEOUT\_H register**

|                  |                  |                  |                  |                  |                  |                 |                 |
|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|-----------------|
| FSM_LC_TIMEOUT15 | FSM_LC_TIMEOUT14 | FSM_LC_TIMEOUT13 | FSM_LC_TIMEOUT12 | FSM_LC_TIMEOUT11 | FSM_LC_TIMEOUT10 | FSM_LC_TIMEOUT9 | FSM_LC_TIMEOUT8 |
|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|-----------------|

**Table 321. FSM\_LC\_TIMEOUT\_H register description**

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| FSM_LC_TIMEOUT[15:8] | FSM long counter timeout value (MSbyte). Default value: 00000000 |
|----------------------|------------------------------------------------------------------|

### 13.2.2 FSM\_PROGRAMS (7Ch)

FSM number of programs register (R/W)

**Table 322. FSM\_PROGRAMS register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| FSM_N_PROG7 | FSM_N_PROG6 | FSM_N_PROG5 | FSM_N_PROG4 | FSM_N_PROG3 | FSM_N_PROG2 | FSM_N_PROG1 | FSM_N_PROG0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 323. FSM\_PROGRAMS register description**

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| FSM_N_PROG[7:0] | Number of FSM programs; must be less than or equal to 8.<br>Default value: 00000000 |
|-----------------|-------------------------------------------------------------------------------------|



### 13.2.3 FSM\_START\_ADD\_L (7Eh) and FSM\_START\_ADD\_H (7Fh)

FSM start address register (R/W). First available address is 0x35C.

**Table 324. FSM\_START\_ADD\_L register**

| FSM_START7 | FSM_START6 | FSM_START5 | FSM_START4 | FSM_START3 | FSM_START2 | FSM_START1 | FSM_START0 |
|------------|------------|------------|------------|------------|------------|------------|------------|
|------------|------------|------------|------------|------------|------------|------------|------------|

**Table 325. FSM\_START\_ADD\_L register description**

|                |                                                           |
|----------------|-----------------------------------------------------------|
| FSM_START[7:0] | FSM start address value (LSbyte). Default value: 00000000 |
|----------------|-----------------------------------------------------------|

**Table 326. FSM\_START\_ADD\_H register**

| FSM_START15 | FSM_START14 | FSM_START13 | FSM_START12 | FSM_START11 | FSM_START10 | FSM_START9 | FSM_START8 |
|-------------|-------------|-------------|-------------|-------------|-------------|------------|------------|
|-------------|-------------|-------------|-------------|-------------|-------------|------------|------------|

**Table 327. FSM\_START\_ADD\_H register description**

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| FSM_START[15:8] | FSM start address value (MSbyte). Default value: 00000000 |
|-----------------|-----------------------------------------------------------|

### 13.2.4 PEDO\_CMD\_REG (83h)

Pedometer configuration register (R/W)

**Table 328. PEDO\_CMD\_REG register**

|                  |                  |                  |                  |                |                 |                  |                  |
|------------------|------------------|------------------|------------------|----------------|-----------------|------------------|------------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | CARRY_COUNT_EN | FP_REJECTION_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|------------------|------------------|------------------|------------------|----------------|-----------------|------------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 329. PEDO\_CMD\_REG register description**

|                                |                                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| CARRY_COUNT_EN                 | Set when user wants to generate interrupt only on count overflow event. |
| FP_REJECTION_EN <sup>(1)</sup> | Enables the false-positive rejection feature.                           |

1. This bit is activated if the PEDO\_ADV\_EN bit of the [EMB\\_FUNC\\_EN\\_B \(05h\)](#) register is set to 1.



### 13.2.5 PEDO\_DEB\_STEPS\_CONF (84h)

Pedometer debounce configuration register (R/W)

**Table 330. PEDO\_DEB\_STEPS\_CONF register**

|           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| DEB_STEP7 | DEB_STEP6 | DEB_STEP5 | DEB_STEP4 | DEB_STEP3 | DEB_STEP2 | DEB_STEP1 | DEB_STEP0 |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|

**Table 331. PEDO\_DEB\_STEPS\_CONF register description**

|               |                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------|
| DEB_STEP[7:0] | Debounce threshold. Minimum number of steps to increment the step counter (debounce).<br>Default value: 00001010 |
|---------------|------------------------------------------------------------------------------------------------------------------|

### 13.2.6 PEDO\_SC\_DELTAT\_L (D0h) and PEDO\_SC\_DELTAT\_H (D1h)

Time period register for step detection on delta time (R/W)

**Table 332. PEDO\_SC\_DELTAT\_L register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| PD_SC_7 | PD_SC_6 | PD_SC_5 | PD_SC_4 | PD_SC_3 | PD_SC_2 | PD_SC_1 | PD_SC_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 333. PEDO\_SC\_DELTAT\_H register**

|          |          |          |          |          |          |         |         |
|----------|----------|----------|----------|----------|----------|---------|---------|
| PD_SC_15 | PD_SC_14 | PD_SC_13 | PD_SC_12 | PD_SC_11 | PD_SC_10 | PD_SC_9 | PD_SC_8 |
|----------|----------|----------|----------|----------|----------|---------|---------|

**Table 334. PEDO\_SC\_DELTAT\_H/L register description**

|              |                                   |
|--------------|-----------------------------------|
| PD_SC_[15:0] | Time period value (1LSB = 6.4 ms) |
|--------------|-----------------------------------|





## 13.3 Page 2 - embedded advanced features registers

### 13.3.1 EXT\_FORMAT (00h)

External sensor data format (2-byte or 3-byte) for the finite state machine (R/W)

**Table 335. EXT\_FORMAT register**

|                  |                  |                  |                  |                  |                  |                |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|----------------|------------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | EXT_FORMAT_SEL | 0 <sup>(1)</sup> |
|------------------|------------------|------------------|------------------|------------------|------------------|----------------|------------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 336. EXT\_FORMAT register description**

|                |                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| EXT_FORMAT_SEL | Selects the format of the external sensor data for FSM processing. Default value: 0<br>(0: 2-byte format; 1: 3-byte format) |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|

### 13.3.2 EXT\_3BYTE\_SENSITIVITY\_L (02h) and EXT\_3BYTE\_SENSITIVITY\_H (03h)

External sensor (3-byte output data) sensitivity value register for the finite state machine (R/W)

This register corresponds to the conversion value of the external sensor having 3-byte output data. The register value is expressed as half-precision floating-point format: S: 1 sign bit; E: 5 exponent bits; F: 10 fraction bits).

The default value of EXT\_3BYTE\_S [15:0] is 0x0C00 (when using an external pressure sensor this value corresponds to 2.441e-04 hPa/LSB).

**Table 337. EXT\_3BYTE\_SENSITIVITY\_L register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| EXT_3BYTE_S_7 | EXT_3BYTE_S_6 | EXT_3BYTE_S_5 | EXT_3BYTE_S_4 | EXT_3BYTE_S_3 | EXT_3BYTE_S_2 | EXT_3BYTE_S_1 | EXT_3BYTE_S_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 338. EXT\_3BYTE\_SENSITIVITY\_L register description**

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| EXT_3BYTE_S [7:0] | External sensor (3-byte output data) sensitivity (LSbyte). Default value: 00000000 |
|-------------------|------------------------------------------------------------------------------------|

**Table 339. EXT\_3BYTE\_SENSITIVITY\_H register**

|                |                |                |                |                |                |               |               |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|
| EXT_3BYTE_S_15 | EXT_3BYTE_S_14 | EXT_3BYTE_S_13 | EXT_3BYTE_S_12 | EXT_3BYTE_S_11 | EXT_3BYTE_S_10 | EXT_3BYTE_S_9 | EXT_3BYTE_S_8 |
|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|

**Table 340. EXT\_3BYTE\_SENSITIVITY\_H register description**

|                    |                                                                                    |
|--------------------|------------------------------------------------------------------------------------|
| EXT_3BYTE_S [15:8] | External sensor (3-byte output data) sensitivity (MSbyte). Default value: 00001100 |
|--------------------|------------------------------------------------------------------------------------|



## 13.3.3

**EXT\_3BYTE\_OFFSET\_XL (06h), EXT\_3BYTE\_OFFSET\_L (07h) and EXT\_3BYTE\_OFFSET\_H (08h)**

External sensor (3-byte output data) offset value register for the finite state machine (R/W)

This register corresponds to the offset value applied to external sensor output data (3-byte) before being processed in the FSM. The value is expressed as a 24-bit word in two's complement.

The default value of EXT\_3BYTE\_OFF\_[23:0] is 0x3F5400, which corresponds to 4150272 LSB. This offset is subtracted from the external output data (in LSB) before the sensitivity (configured in the EXT\_3BYTE\_SENSITIVITY\_L (02h) and EXT\_3BYTE\_SENSITIVITY\_H (03h) registers) is internally applied.

**Table 341. EXT\_3BYTE\_OFFSET\_XL register**

| EXT_3BYTE_OFF_7 | EXT_3BYTE_OFF_6 | EXT_3BYTE_OFF_5 | EXT_3BYTE_OFF_4 | EXT_3BYTE_OFF_3 | EXT_3BYTE_OFF_2 | EXT_3BYTE_OFF_1 | EXT_3BYTE_OFF_0 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|

**Table 342. EXT\_3BYTE\_OFFSET\_XL register description**

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| EXT_3BYTE_OFF_[7:0] | External sensor (3-byte output data) offset (low byte). Default value: 00000000 |
|---------------------|---------------------------------------------------------------------------------|

**Table 343. EXT\_3BYTE\_OFFSET\_L register**

| EXT_3BYTE_OFF_15 | EXT_3BYTE_OFF_14 | EXT_3BYTE_OFF_13 | EXT_3BYTE_OFF_12 | EXT_3BYTE_OFF_11 | EXT_3BYTE_OFF_10 | EXT_3BYTE_OFF_9 | EXT_3BYTE_OFF_8 |
|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|-----------------|
|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|-----------------|

**Table 344. EXT\_3BYTE\_OFFSET\_L register description**

|                      |                                                                                 |
|----------------------|---------------------------------------------------------------------------------|
| EXT_3BYTE_OFF_[15:8] | External sensor (3-byte output data) offset (mid byte). Default value: 01010100 |
|----------------------|---------------------------------------------------------------------------------|

**Table 345. EXT\_3BYTE\_OFFSET\_H register**

| EXT_3BYTE_OFF_23 | EXT_3BYTE_OFF_22 | EXT_3BYTE_OFF_21 | EXT_3BYTE_OFF_20 | EXT_3BYTE_OFF_19 | EXT_3BYTE_OFF_18 | EXT_3BYTE_OFF_17 | EXT_3BYTE_OFF_16 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 346. EXT\_3BYTE\_OFFSET\_H register description**

|                       |                                                                                  |
|-----------------------|----------------------------------------------------------------------------------|
| EXT_3BYTE_OFF_[23:16] | External sensor (3-byte output data) offset (high byte). Default value: 00111111 |
|-----------------------|----------------------------------------------------------------------------------|



## 14 Sensor hub register mapping

The table given below provides a list of the registers for the sensor hub functions available in the device and the corresponding addresses. The sensor hub registers are accessible when bit SHUB\_REG\_ACCESS is set to 1 in FUNC\_CFG\_ACCESS (01h).

**Table 347. Register address map - sensor hub registers**

| Name           | Type | Register address |          | Default  | Comment |
|----------------|------|------------------|----------|----------|---------|
|                |      | Hex              | Binary   |          |         |
| SENSOR_HUB_1   | R    | 02               | 00000010 | output   |         |
| SENSOR_HUB_2   | R    | 03               | 00000011 | output   |         |
| SENSOR_HUB_3   | R    | 04               | 00000100 | output   |         |
| SENSOR_HUB_4   | R    | 05               | 00000101 | output   |         |
| SENSOR_HUB_5   | R    | 06               | 00000110 | output   |         |
| SENSOR_HUB_6   | R    | 07               | 00000111 | output   |         |
| SENSOR_HUB_7   | R    | 08               | 00001000 | output   |         |
| SENSOR_HUB_8   | R    | 09               | 00001001 | output   |         |
| SENSOR_HUB_9   | R    | 0A               | 00001010 | output   |         |
| SENSOR_HUB_10  | R    | 0B               | 00001011 | output   |         |
| SENSOR_HUB_11  | R    | 0C               | 00001100 | output   |         |
| SENSOR_HUB_12  | R    | 0D               | 00001101 | output   |         |
| SENSOR_HUB_13  | R    | 0E               | 00001110 | output   |         |
| SENSOR_HUB_14  | R    | 0F               | 00001111 | output   |         |
| SENSOR_HUB_15  | R    | 10               | 00010000 | output   |         |
| SENSOR_HUB_16  | R    | 11               | 00010001 | output   |         |
| SENSOR_HUB_17  | R    | 12               | 00010010 | output   |         |
| SENSOR_HUB_18  | R    | 13               | 00010011 | output   |         |
| MASTER_CONFIG  | R/W  | 14               | 00010100 | 00000000 |         |
| SLV0_ADD       | R/W  | 15               | 00010101 | 00000000 |         |
| SLV0_SUBADD    | R/W  | 16               | 00010110 | 00000000 |         |
| SLV0_CONFIG    | R/W  | 17               | 00010111 | 10000000 |         |
| SLV1_ADD       | R/W  | 18               | 00011000 | 00000000 |         |
| SLV1_SUBADD    | R/W  | 19               | 00011001 | 00000000 |         |
| SLV1_CONFIG    | R/W  | 1A               | 00011010 | 00010000 |         |
| SLV2_ADD       | R/W  | 1B               | 00011011 | 00000000 |         |
| SLV2_SUBADD    | R/W  | 1C               | 00011100 | 00000000 |         |
| SLV2_CONFIG    | R/W  | 1D               | 00011101 | 00000000 |         |
| SLV3_ADD       | R/W  | 1E               | 00011110 | 00000000 |         |
| SLV3_SUBADD    | R/W  | 1F               | 00011111 | 00000000 |         |
| SLV3_CONFIG    | R/W  | 20               | 00100000 | 00000000 |         |
| DATAWRITE_SLV0 | R/W  | 21               | 00100001 | 00000000 |         |
| STATUS_MASTER  | R    | 22               | 00100010 | output   |         |

Reserved registers must not be changed. Writing to those registers may cause permanent damage to the device. The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.



## 15 Sensor hub register description

### 15.1 SENSOR\_HUB\_1 (02h)

Sensor hub output register (R)

First byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 348. SENSOR\_HUB\_1 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub1_7 | Sensor Hub1_6 | Sensor Hub1_5 | Sensor Hub1_4 | Sensor Hub1_3 | Sensor Hub1_2 | Sensor Hub1_1 | Sensor Hub1_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 349. SENSOR\_HUB\_1 register description**

|                  |                                           |
|------------------|-------------------------------------------|
| SensorHub1_[7:0] | First byte associated to external sensors |
|------------------|-------------------------------------------|

### 15.2 SENSOR\_HUB\_2 (03h)

Sensor hub output register (R)

Second byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 350. SENSOR\_HUB\_2 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub2_7 | Sensor Hub2_6 | Sensor Hub2_5 | Sensor Hub2_4 | Sensor Hub2_3 | Sensor Hub2_2 | Sensor Hub2_1 | Sensor Hub2_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 351. SENSOR\_HUB\_2 register description**

|                  |                                            |
|------------------|--------------------------------------------|
| SensorHub2_[7:0] | Second byte associated to external sensors |
|------------------|--------------------------------------------|

### 15.3 SENSOR\_HUB\_3 (04h)

Sensor hub output register (R)

Third byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 352. SENSOR\_HUB\_3 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub3_7 | Sensor Hub3_6 | Sensor Hub3_5 | Sensor Hub3_4 | Sensor Hub3_3 | Sensor Hub3_2 | Sensor Hub3_1 | Sensor Hub3_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 353. SENSOR\_HUB\_3 register description**

|                  |                                           |
|------------------|-------------------------------------------|
| SensorHub3_[7:0] | Third byte associated to external sensors |
|------------------|-------------------------------------------|



## 15.4 SENSOR\_HUB\_4 (05h)

Sensor hub output register (R)

Fourth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 354. SENSOR\_HUB\_4 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub4_7 | Sensor Hub4_6 | Sensor Hub4_5 | Sensor Hub4_4 | Sensor Hub4_3 | Sensor Hub4_2 | Sensor Hub4_1 | Sensor Hub4_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 355. SENSOR\_HUB\_4 register description**

|                  |                                            |
|------------------|--------------------------------------------|
| SensorHub4_[7:0] | Fourth byte associated to external sensors |
|------------------|--------------------------------------------|

## 15.5 SENSOR\_HUB\_5 (06h)

Sensor hub output register (R)

Fifth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 356. SENSOR\_HUB\_5 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub5_7 | Sensor Hub5_6 | Sensor Hub5_5 | Sensor Hub5_4 | Sensor Hub5_3 | Sensor Hub5_2 | Sensor Hub5_1 | Sensor Hub5_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 357. SENSOR\_HUB\_5 register description**

|                  |                                           |
|------------------|-------------------------------------------|
| SensorHub5_[7:0] | Fifth byte associated to external sensors |
|------------------|-------------------------------------------|

## 15.6 SENSOR\_HUB\_6 (07h)

Sensor hub output register (R)

Sixth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 358. SENSOR\_HUB\_6 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub6_7 | Sensor Hub6_6 | Sensor Hub6_5 | Sensor Hub6_4 | Sensor Hub6_3 | Sensor Hub6_2 | Sensor Hub6_1 | Sensor Hub6_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 359. SENSOR\_HUB\_6 register description**

|                  |                                           |
|------------------|-------------------------------------------|
| SensorHub6_[7:0] | Sixth byte associated to external sensors |
|------------------|-------------------------------------------|



## 15.7 SENSOR\_HUB\_7 (08h)

Sensor hub output register (R)

Seventh byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 360. SENSOR\_HUB\_7 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub7_7 | Sensor Hub7_6 | Sensor Hub7_5 | Sensor Hub7_4 | Sensor Hub7_3 | Sensor Hub7_2 | Sensor Hub7_1 | Sensor Hub7_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 361. SENSOR\_HUB\_7 register description**

|                  |                                             |
|------------------|---------------------------------------------|
| SensorHub7_[7:0] | Seventh byte associated to external sensors |
|------------------|---------------------------------------------|

## 15.8 SENSOR\_HUB\_8 (09h)

Sensor hub output register (R)

Eighth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 362. SENSOR\_HUB\_8 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub8_7 | Sensor Hub8_6 | Sensor Hub8_5 | Sensor Hub8_4 | Sensor Hub8_3 | Sensor Hub8_2 | Sensor Hub8_1 | Sensor Hub8_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 363. SENSOR\_HUB\_8 register description**

|                  |                                            |
|------------------|--------------------------------------------|
| SensorHub8_[7:0] | Eighth byte associated to external sensors |
|------------------|--------------------------------------------|

## 15.9 SENSOR\_HUB\_9 (0Ah)

Sensor hub output register (R)

Ninth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 364. SENSOR\_HUB\_9 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Sensor Hub9_7 | Sensor Hub9_6 | Sensor Hub9_5 | Sensor Hub9_4 | Sensor Hub9_3 | Sensor Hub9_2 | Sensor Hub9_1 | Sensor Hub9_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 365. SENSOR\_HUB\_9 register description**

|                  |                                           |
|------------------|-------------------------------------------|
| SensorHub9_[7:0] | Ninth byte associated to external sensors |
|------------------|-------------------------------------------|



## 15.10 SENSOR\_HUB\_10 (0Bh)

Sensor hub output register (R)

Tenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 366. SENSOR\_HUB\_10 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub10_7 | Sensor Hub10_6 | Sensor Hub10_5 | Sensor Hub10_4 | Sensor Hub10_3 | Sensor Hub10_2 | Sensor Hub10_1 | Sensor Hub10_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 367. SENSOR\_HUB\_10 register description**

|                   |                                           |
|-------------------|-------------------------------------------|
| SensorHub10_[7:0] | Tenth byte associated to external sensors |
|-------------------|-------------------------------------------|

## 15.11 SENSOR\_HUB\_11 (0Ch)

Sensor hub output register (R)

Eleventh byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 368. SENSOR\_HUB\_11 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub11_7 | Sensor Hub11_6 | Sensor Hub11_5 | Sensor Hub11_4 | Sensor Hub11_3 | Sensor Hub11_2 | Sensor Hub11_1 | Sensor Hub11_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 369. SENSOR\_HUB\_11 register description**

|                   |                                              |
|-------------------|----------------------------------------------|
| SensorHub11_[7:0] | Eleventh byte associated to external sensors |
|-------------------|----------------------------------------------|

## 15.12 SENSOR\_HUB\_12 (0Dh)

Sensor hub output register (R)

Twelfth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 370. SENSOR\_HUB\_12 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub12_7 | Sensor Hub12_6 | Sensor Hub12_5 | Sensor Hub12_4 | Sensor Hub12_3 | Sensor Hub12_2 | Sensor Hub12_1 | Sensor Hub12_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 371. SENSOR\_HUB\_12 register description**

|                   |                                             |
|-------------------|---------------------------------------------|
| SensorHub12_[7:0] | Twelfth byte associated to external sensors |
|-------------------|---------------------------------------------|



### 15.13 SENSOR\_HUB\_13 (0Eh)

Sensor hub output register (R)

Thirteenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 372. SENSOR\_HUB\_13 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub13_7 | Sensor Hub13_6 | Sensor Hub13_5 | Sensor Hub13_4 | Sensor Hub13_3 | Sensor Hub13_2 | Sensor Hub13_1 | Sensor Hub13_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 373. SENSOR\_HUB\_13 register description**

|                   |                                                |
|-------------------|------------------------------------------------|
| SensorHub13_[7:0] | Thirteenth byte associated to external sensors |
|-------------------|------------------------------------------------|

### 15.14 SENSOR\_HUB\_14 (0Fh)

Sensor hub output register (R)

Fourteenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 374. SENSOR\_HUB\_14 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub14_7 | Sensor Hub14_6 | Sensor Hub14_5 | Sensor Hub14_4 | Sensor Hub14_3 | Sensor Hub14_2 | Sensor Hub14_1 | Sensor Hub14_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 375. SENSOR\_HUB\_14 register description**

|                   |                                                |
|-------------------|------------------------------------------------|
| SensorHub14_[7:0] | Fourteenth byte associated to external sensors |
|-------------------|------------------------------------------------|

### 15.15 SENSOR\_HUB\_15 (10h)

Sensor hub output register (R)

Fifteenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 376. SENSOR\_HUB\_15 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub15_7 | Sensor Hub15_6 | Sensor Hub15_5 | Sensor Hub15_4 | Sensor Hub15_3 | Sensor Hub15_2 | Sensor Hub15_1 | Sensor Hub15_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 377. SENSOR\_HUB\_15 register description**

|                   |                                               |
|-------------------|-----------------------------------------------|
| SensorHub15_[7:0] | Fifteenth byte associated to external sensors |
|-------------------|-----------------------------------------------|





## 15.16 SENSOR\_HUB\_16 (11h)

Sensor hub output register (R)

Sixteenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 378. SENSOR\_HUB\_16 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub16_7 | Sensor Hub16_6 | Sensor Hub16_5 | Sensor Hub16_4 | Sensor Hub16_3 | Sensor Hub16_2 | Sensor Hub16_1 | Sensor Hub16_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 379. SENSOR\_HUB\_16 register description**

|                   |                                               |
|-------------------|-----------------------------------------------|
| SensorHub16_[7:0] | Sixteenth byte associated to external sensors |
|-------------------|-----------------------------------------------|

## 15.17 SENSOR\_HUB\_17 (12h)

Sensor hub output register (R)

Seventeenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 380. SENSOR\_HUB\_17 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub17_7 | Sensor Hub17_6 | Sensor Hub17_5 | Sensor Hub17_4 | Sensor Hub17_3 | Sensor Hub17_2 | Sensor Hub17_1 | Sensor Hub17_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 381. SENSOR\_HUB\_17 register description**

|                   |                                                 |
|-------------------|-------------------------------------------------|
| SensorHub17_[7:0] | Seventeenth byte associated to external sensors |
|-------------------|-------------------------------------------------|

## 15.18 SENSOR\_HUB\_18 (13h)

Sensor hub output register (R)

Eighteenth byte associated to external sensors. The content of the register is consistent with the SLVx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 382. SENSOR\_HUB\_17 register**

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Sensor Hub18_7 | Sensor Hub18_6 | Sensor Hub18_5 | Sensor Hub18_4 | Sensor Hub18_3 | Sensor Hub18_2 | Sensor Hub18_1 | Sensor Hub18_0 |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|

**Table 383. SENSOR\_HUB\_17 register description**

|                   |                                                |
|-------------------|------------------------------------------------|
| SensorHub18_[7:0] | Eighteenth byte associated to external sensors |
|-------------------|------------------------------------------------|



## 15.19

**MASTER\_CONFIG (14h)**

Master configuration register (R/W)

**Table 384. MASTER\_CONFIG register**

|                 |            |              |                   |                  |           |              |              |
|-----------------|------------|--------------|-------------------|------------------|-----------|--------------|--------------|
| RST_MASTER_REGS | WRITE_ONCE | START_CONFIG | PASS_THROUGH_MODE | 0 <sup>(1)</sup> | MASTER_ON | AUX_SENS_ON1 | AUX_SENS_ON0 |
|-----------------|------------|--------------|-------------------|------------------|-----------|--------------|--------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 385. MASTER\_CONFIG register description**

|                   |                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RST_MASTER_REGS   | Resets master logic and output registers. Must be set to 1 and then set to 0. Default value: 0                                                                                                            |
| WRITE_ONCE        | Slave 0 write operation is performed only at the first sensor hub cycle.<br>Default value: 0<br>(0: write operation for each sensor hub cycle;<br>1: write operation only for the first sensor hub cycle) |
| START_CONFIG      | Sensor hub trigger signal selection. Default value: 0<br>(0: sensor hub trigger signal is the accelerometer/gyro data-ready;<br>1: sensor hub trigger signal external from INT2 pin)                      |
| PASS_THROUGH_MODE | I <sup>2</sup> C interface pass-through. Default value: 0<br>(0: pass-through disabled;<br>1: pass-through enabled, primary I <sup>2</sup> C line is short-circuited with the sensor hub line)            |
| MASTER_ON         | Enables sensor hub I <sup>2</sup> C master. Default: 0<br>(0: master I <sup>2</sup> C of sensor hub disabled; 1: master I <sup>2</sup> C of sensor hub enabled)                                           |
| AUX_SENS_ON[1:0]  | Number of external sensors to be read by the sensor hub.<br>(00: one sensor (default);<br>01: two sensors;<br>10: three sensors;<br>11: four sensors)                                                     |



## 15.20 SLV0\_ADD (15h)

I<sup>2</sup>C slave address of the first external sensor (sensor 0) register (R/W)

**Table 386. SLV0\_ADD register**

|             |             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|
| slave0_add6 | slave0_add5 | slave0_add4 | slave0_add3 | slave0_add2 | slave0_add1 | slave0_add0 | rw_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|

**Table 387. SLV\_ADD register description**

|                 |                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------|
| slave0_add[6:0] | I <sup>2</sup> C slave address of sensor 0 that can be read by the sensor hub.<br>Default value: 0000000 |
| rw_0            | Read/write operation on sensor 0. Default value: 0<br>(0: write operation; 1: read operation)            |

## 15.21 SLV0\_SUBADD (16h)

Address of register on the first external sensor (sensor 0) register (R/W)

**Table 388. SLV0\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| slave0_reg7 | slave0_reg6 | slave0_reg5 | slave0_reg4 | slave0_reg3 | slave0_reg2 | slave0_reg1 | slave0_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 389. SLV0\_SUBADD register description**

|                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| slave0_reg[7:0] | Address of register on sensor 0 that has to be read/written according to the rw_0 bit value in <a href="#">SLV0_ADD (15h)</a> . Default value: 00000000 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 15.22 SLV0\_CONFIG (17h)

First external sensor (sensor 0) configuration and sensor hub settings register (R/W)

**Table 390. SLV0\_CONFIG register**

|            |            |            |                  |                     |               |               |               |
|------------|------------|------------|------------------|---------------------|---------------|---------------|---------------|
| SHUB_ODR_2 | SHUB_ODR_1 | SHUB_ODR_0 | 0 <sup>(1)</sup> | BATCH_EXT_SENS_0_EN | Slave0_numop2 | Slave0_numop1 | Slave0_numop0 |
|------------|------------|------------|------------------|---------------------|---------------|---------------|---------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 391. SLV0\_CONFIG register description**

|                     |                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SHUB_ODR_2[2:0]     | Rate at which the master communicates.<br>(000: 1.875 Hz;<br>001: 15 Hz;<br>010: 30 Hz;<br>011: 60 Hz;<br>100: 120 Hz (default);<br>101: 240 Hz;<br>110: 480 Hz;<br>111: reserved |
| BATCH_EXT_SENS_0_EN | Enables FIFO data batching of first slave. Default value: 0                                                                                                                       |
| Slave0_numop[2:0]   | Number of read operations on sensor 0. Default value: 000                                                                                                                         |



### 15.23 SLV1\_ADD (18h)

I<sup>2</sup>C slave address of the second external sensor (sensor 1) register (R/W)

**Table 392. SLV1\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave1_add6 | Slave1_add5 | Slave1_add4 | Slave1_add3 | Slave1_add2 | Slave1_add1 | Slave1_add0 | r_1 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 393. SLV1\_ADD register description**

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Slave1_add[6:0] | I <sup>2</sup> C slave address of sensor 1 that can be read by the sensor hub.<br>Default value: 0000000        |
| r_1             | Enables read operation on sensor 1. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

### 15.24 SLV1\_SUBADD (19h)

Address of register on the second external sensor (sensor 1) register (R/W)

**Table 394. SLV1\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave1_reg7 | Slave1_reg6 | Slave1_reg5 | Slave1_reg4 | Slave1_reg3 | Slave1_reg2 | Slave1_reg1 | Slave1_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 395. SLV1\_SUBADD register description**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Slave1_reg[7:0] | Address of register on sensor 1 that has to be read/written according to the r_1 bit value in SLV1_ADD (18h). |
|-----------------|---------------------------------------------------------------------------------------------------------------|

### 15.25 SLV1\_CONFIG (1Ah)

Second external sensor (sensor 2) configuration register (R/W)

**Table 396. SLV1\_CONFIG register**

|                  |                  |                  |                  |                     |               |               |               |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 1 <sup>(2)</sup> | BATCH_EXT_SENS_1_EN | Slave1_numop2 | Slave1_numop1 | Slave1_numop0 |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|

1. This bit must be set to 0 for the correct operation of the device.
2. This bit must be set to 1 for the correct operation of the device.

**Table 397. SLV1\_CONFIG register description**

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| BATCH_EXT_SENS_1_EN | Enables FIFO data batching of second slave. Default value: 0 |
| Slave1_numop[2:0]   | Number of read operations on sensor 2. Default value: 000    |



## 15.26 SLV2\_ADD (1Bh)

I<sup>2</sup>C slave address of the third external sensor (sensor 2) register (R/W)

**Table 398. SLV2\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave2_add6 | Slave2_add5 | Slave2_add4 | Slave2_add3 | Slave2_add2 | Slave2_add1 | Slave2_add0 | r_2 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 399. SLV2\_ADD register description**

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Slave2_add[6:0] | I <sup>2</sup> C slave address of sensor 2 that can be read by the sensor hub.                                  |
| r_2             | Enables read operation on sensor 2. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

## 15.27 SLV2\_SUBADD (1Ch)

Address of register on the third external sensor (sensor 2) register (R/W)

**Table 400. SLV2\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave2_reg7 | Slave2_reg6 | Slave2_reg5 | Slave2_reg4 | Slave2_reg3 | Slave2_reg2 | Slave2_reg1 | Slave2_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 401. SLV2\_SUBADD register description**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Slave2_reg[7:0] | Address of register on sensor 2 that has to be read/written according to the r_2 bit value in SLV2_ADD (1Bh). |
|-----------------|---------------------------------------------------------------------------------------------------------------|

## 15.28 SLV2\_CONFIG (1Dh)

Third external sensor (sensor 2) configuration register (R/W)

**Table 402. SLV2\_CONFIG register**

|                  |                  |                  |                  |                     |               |               |               |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | BATCH_EXT_SENS_2_EN | Slave2_numop2 | Slave2_numop1 | Slave2_numop0 |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 403. SLV2\_CONFIG register description**

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| BATCH_EXT_SENS_2_EN | Enables FIFO data batching of third slave. Default value: 0 |
| Slave2_numop[2:0]   | Number of read operations on sensor 2. Default value: 000   |



## 15.29 SLV3\_ADD (1Eh)

I<sup>2</sup>C slave address of the fourth external sensor (sensor 3) register (R/W)

**Table 404. SLV3\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave3_add6 | Slave3_add5 | Slave3_add4 | Slave3_add3 | Slave3_add2 | Slave3_add1 | Slave3_add0 | r_3 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 405. SLV3\_ADD register description**

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Slave3_add[6:0] | I <sup>2</sup> C slave address of sensor 3 that can be read by the sensor hub.                                  |
| r_3             | Enables read operation on sensor 3. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

## 15.30 SLV3\_SUBADD (1Fh)

Address of register on the fourth external sensor (sensor 3) register (R/W)

**Table 406. SLV3\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave3_reg7 | Slave3_reg6 | Slave3_reg5 | Slave3_reg4 | Slave3_reg3 | Slave3_reg2 | Slave3_reg1 | Slave3_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 407. SLV3\_SUBADD register description**

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| Slave3_reg[7:0] | Address of register on sensor 3 that has to be read according to the r_3 bit value in SLV3_ADD (1Eh). |
|-----------------|-------------------------------------------------------------------------------------------------------|

## 15.31 SLV3\_CONFIG (20h)

Fourth external sensor (sensor 3) configuration register (R/W)

**Table 408. SLV3\_CONFIG register**

|                  |                  |                  |                  |                     |               |               |               |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | BATCH_EXT_SENS_3_EN | Slave3_numop2 | Slave3_numop1 | Slave3_numop0 |
|------------------|------------------|------------------|------------------|---------------------|---------------|---------------|---------------|

1. This bit must be set to 0 for the correct operation of the device.

**Table 409. SLV3\_CONFIG register description**

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| BATCH_EXT_SENS_3_EN | Enables FIFO data batching of fourth slave. Default value: 0 |
| Slave3_numop[2:0]   | Number of read operations on sensor 3. Default value: 000    |



### 15.32 DATAWRITE\_SLV0 (21h)

Data to be written into the slave device register (R/W)

**Table 410. DATAWRITE\_SLV0 register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Slave0_dataw7 | Slave0_dataw6 | Slave0_dataw5 | Slave0_dataw4 | Slave0_dataw3 | Slave0_dataw2 | Slave0_dataw1 | Slave0_dataw0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 411. DATAWRITE\_SLV0 register description**

|                   |                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Slave0_dataw[7:0] | Data to be written into the slave 0 device according to the rw_0 bit in register <a href="#">SLV0_ADD (15h)</a> .<br>Default value: 00000000 |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

### 15.33 STATUS\_MASTER (22h)

Sensor hub source register (R)

**Table 412. STATUS\_MASTER register**

|              |             |             |             |             |   |   |                |
|--------------|-------------|-------------|-------------|-------------|---|---|----------------|
| WR_ONCE_DONE | SLAVE3_NACK | SLAVE2_NACK | SLAVE1_NACK | SLAVE0_NACK | 0 | 0 | SENS_HUB_ENDOP |
|--------------|-------------|-------------|-------------|-------------|---|---|----------------|

**Table 413. STATUS\_MASTER register description**

|                |                                                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WR_ONCE_DONE   | When the bit WRITE_ONCE in <a href="#">MASTER_CONFIG (14h)</a> is configured as 1, this bit is set to 1 when the write operation on slave 0 has been performed and completed. Default value: 0 |
| SLAVE3_NACK    | This bit is set to 1 if Not acknowledge occurs on slave 3 communication. Default value: 0                                                                                                      |
| SLAVE2_NACK    | This bit is set to 1 if Not acknowledge occurs on slave 2 communication. Default value: 0                                                                                                      |
| SLAVE1_NACK    | This bit is set to 1 if Not acknowledge occurs on slave 1 communication. Default value: 0                                                                                                      |
| SLAVE0_NACK    | This bit is set to 1 if Not acknowledge occurs on slave 0 communication. Default value: 0                                                                                                      |
| SENS_HUB_ENDOP | Sensor hub communication status. Default value: 0<br>(0: sensor hub communication not concluded;<br>1: sensor hub communication concluded)                                                     |



## 16 Soldering information

The LGA package is compliant with the [ECOPACK](#) and RoHS standard.

It is qualified for soldering heat resistance according to JEDEC J-STD-020.

For land pattern and soldering recommendations, consult technical note [TN0018](#) available on [www.st.com](http://www.st.com).

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DRAFT



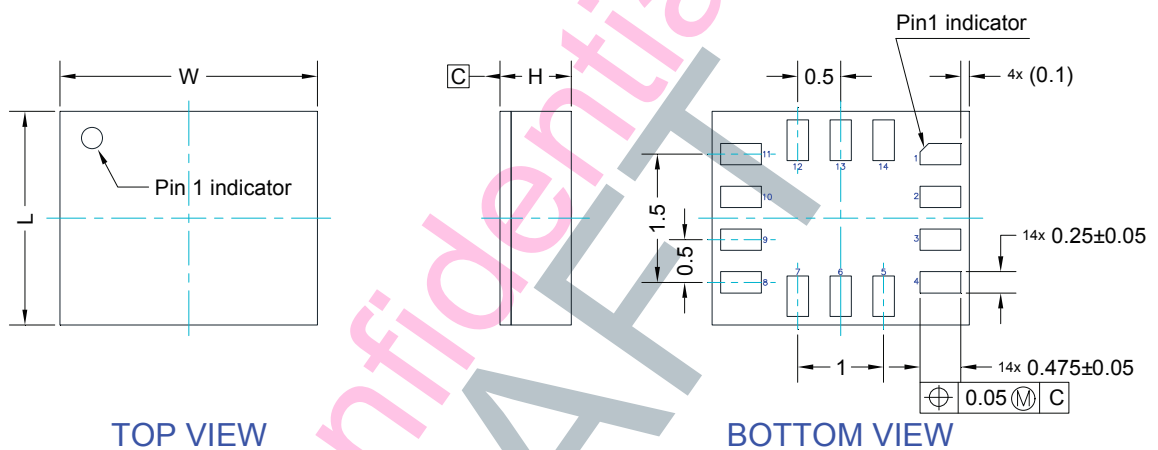


## 17 Package information

To meet environmental requirements, ST offers these devices in different grades of **ECOPACK** packages, depending on their level of environmental compliance. ECOPACK specifications, grade definitions, and product status are available at: [www.st.com](http://www.st.com). ECOPACK is an ST trademark.

### 17.1 LGA-14L package information

**Figure 25. LGA-14L 2.5 x 3.0 x 0.86 mm package outline and mechanical data**



Dimensions are in millimeter unless otherwise specified  
 General tolerance is  $\pm 0.1$  mm unless otherwise specified

#### OUTER DIMENSIONS

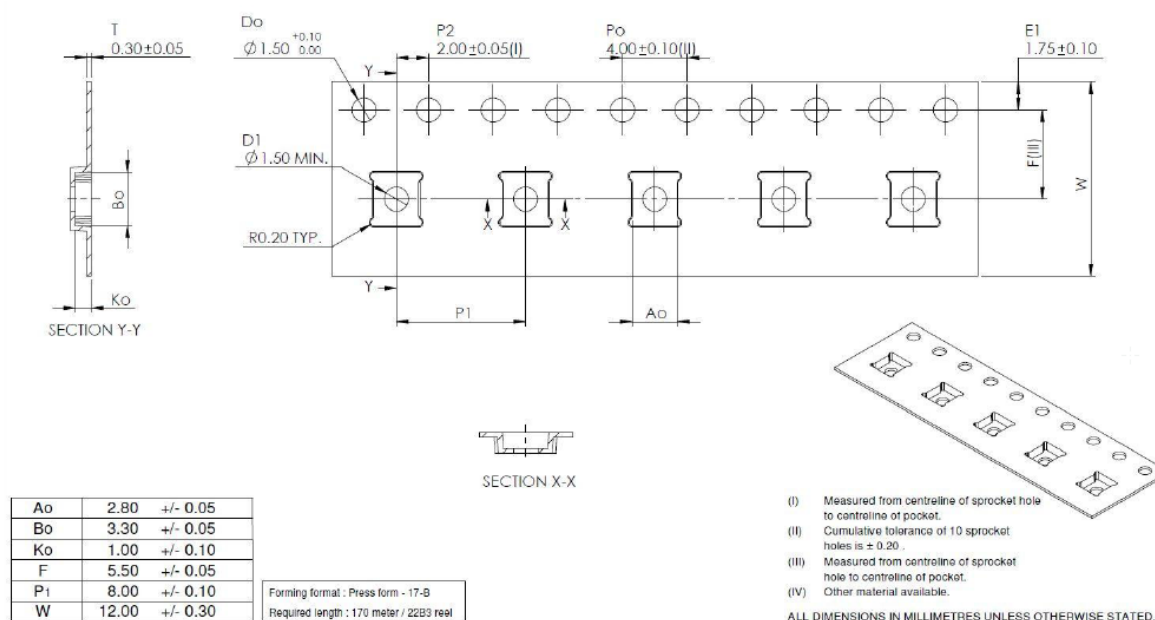
| ITEM       | DIMENSION [mm] | TOLERANCE [mm] |
|------------|----------------|----------------|
| Length [L] | 2.50           | $\pm 0.1$      |
| Width [W]  | 3.00           | $\pm 0.1$      |
| Height [H] | 0.86           | MAX            |

DM00249496\_5

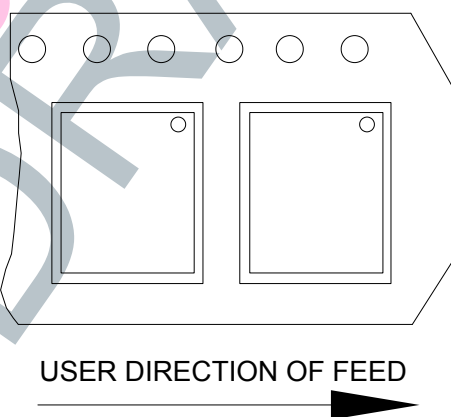


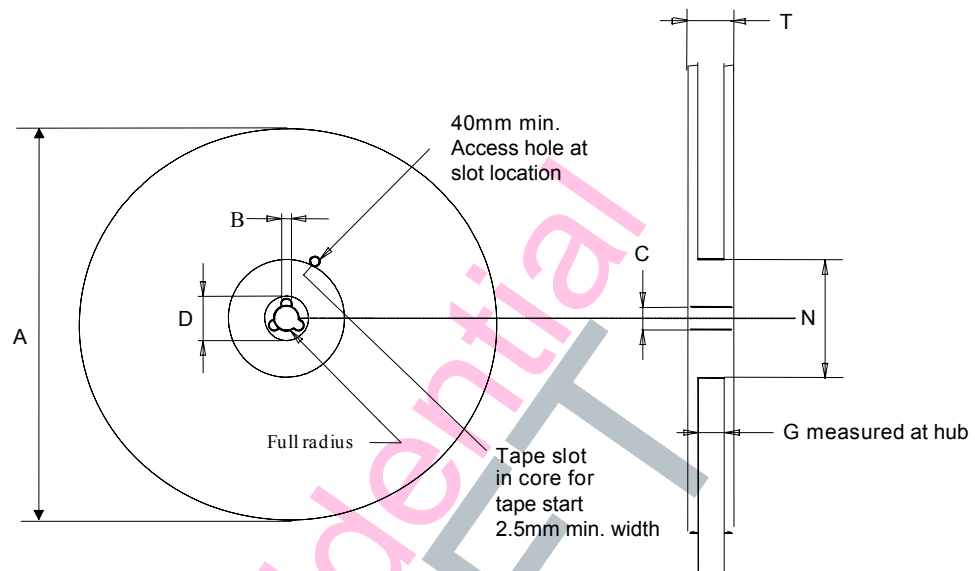
## 17.2 LGA-14L packing information

**Figure 26. Carrier tape information for LGA-14L package**



**Figure 27. LGA-14L package orientation in carrier tape**




**Figure 28. Reel information for carrier tape of LGA-14L package**

**Table 414. Reel dimensions for carrier tape of LGA-14L package**

| Reel dimensions (mm) |            |
|----------------------|------------|
| A (max)              | 330        |
| B (min)              | 1.5        |
| C                    | 13 ±0.25   |
| D (min)              | 20.2       |
| N (min)              | 60         |
| G                    | 12.4 +2/-0 |
| T (max)              | 18.4       |



## Revision history

**Table 415. Document revision history**

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03-Mar-2023 | 1        | Initial release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 17-May-2023 | 2        | Updated Section 2.1 Pedometer functions: step detector and step counters<br>Updated footnotes of Table 3. Mechanical characteristics<br>Updated Section 6.5 High-accuracy ODR mode<br>Updated registers in Section 9 Register description                                                                                                                                                                                                                                                                                                                                   |
| 09-Aug-2023 | 3        | Updated Section 2.2: Pedometer algorithm<br>Updated EMB_FUNC_EN_B (05h) and PEDO_CMD_REG (83h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 22-Jan-2025 | 4        | Updated footnote <sup>(7)</sup> and <sup>(8)</sup> of Table 3. Mechanical characteristics<br>Added Table 7. SPI slave timing values (Vdd_IO at 1.2 V)<br>Added footnote <sup>(3)</sup> to maximum capacitive load in Table 8. I <sup>2</sup> C slave timing values<br>Updated Section 6.5: High-accuracy ODR mode<br>Updated bits 0 and 1 in PIN_CTRL (02h)<br>Updated description of BUS_ACT_SEL_[1:0] in CTRL5 (14h)<br>Updated bit resolution in TIMESTAMP0 (40h), TIMESTAMP1 (41h),<br>TIMESTAMP2 (42h), and TIMESTAMP3 (43h)<br>Updated Table 335. EXT_FORMAT register |
| 25-Mar-2025 | 5        | Updated independent I/O supply in Features<br>Updated test conditions (V <sub>IH</sub> , V <sub>IL</sub> , V <sub>OH</sub> , V <sub>OL</sub> ) in Table 4. Electrical characteristics<br>Updated default value for GETMRL in Table 17. MIPI I3C <sup>®</sup> CCC commands<br>Updated Section 6.9: FIFO<br>Updated description of SLEEP_DUR_[3:0] in WAKE_UP_DUR (5Ch)<br>Updated X_OFS_USR (73h), Y_OFS_USR (74h), Z_OFS_USR (75h)                                                                                                                                          |



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